

NEKpress Research

Psychology Statistics Practice Materials with Jamovi

Model Choice, Worked Solutions, and Scientific Evidence

Problems and Worked Solutions - Version 1.0

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github.com/nicholaskarlson/data

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Repository, source files, synthetic datasets, dictionaries, and version history:

<https://github.com/nicholaskarlson/data>.

Questions, corrections, teaching ideas, and suggested improvements are welcome in GitHub Discussions.

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Introduction: From Statistical Results to Scientific Evidence

Causality, cumulative inquiry, and how to use these materials

Statistics is sometimes taught as a collection of procedures. Identify the number of groups, click the corresponding menu, find a test statistic and a p value, and write a conclusion. Those skills matter, but they are not the purpose of statistical analysis. Statistics serves scientific questions. A method is useful only when it helps researchers learn something defensible about a question they care about.

These materials therefore begin one step before calculation. They ask what the researchers want to know, what was observed, how the observations were obtained, what comparison the model estimates, and what conclusion the design can support. The numerical answer matters. So does the scientific argument in which that answer has a place.

Science rarely establishes an important claim through one decisive study. It progressively constrains credible explanations through carefully designed studies, critical testing, replication, triangulation, and synthesis.

That proposition is the resource's scientific contract. Every plausible scenario, problem, and worked solution should help you distinguish a statistical result from a scientific conclusion and one study from a cumulative body of evidence.

Statistics Serves Scientific Questions

A software table does not know the research question. It does not know whether the rows represent independent people, repeated observations from the same people, schools containing students, or observations selected because an earlier result was extreme. It does not know whether conditions were randomly assigned, whether a scale measured the intended construct, or whether a result will generalize to another population.

Researchers supply those meanings through theory, design, measurement, and documentation. Statistical analysis then helps them describe patterns, estimate quantities, represent uncertainty, compare explanations, and discover where the evidence remains weak. The analysis is part of the scientific argument, not a replacement for it.

Consider a before-and-after comparison. A paired-samples analysis can estimate the average change among the people who were measured twice. It can quantify how variable the changes were and how precisely the mean change was estimated.

Pause and Decide

A university counseling center measures mindfulness in 72 students before a six-week program and again afterward. Scores rise by an average of 6.09 points, 95% CI [4.91, 7.26]. The center's report concludes: *the program produced a significant improvement in student mindfulness*.

Before reading on, write down three things: whether that conclusion is supported, one reason, and how confident you are from 1 to 5.

Every unit in these materials asks you to commit like this before the explanation arrives. Committing is what makes the explanation useful.

Why That Conclusion Fails

The estimate and the interval are right, and they answer the statistical question the data structure poses. The conclusion is not supported, because a paired analysis cannot, by itself, show that a program caused the change. History, maturation, testing effects, attrition, changing measurement conditions, and regression toward the mean can also produce a before-and-after difference. Nothing in the arithmetic distinguishes them.

Notice what did the work. Not the p value, not the interval width, not the sample size, but the absence of a comparison condition — a fact about the design, visible before any analysis ran.

The correct calculation answers the statistical question posed by the data structure. The design determines whether that statistical answer also identifies a causal effect.

What a Statistical Result Can—and Cannot—Do

Three layers of reasoning should remain distinct.

Layer	Central question	Principal contribution
Statistical analysis	What pattern was estimated, and how uncertain is it?	Estimates, intervals, diagnostics, and model-based comparisons
Design and identification	What does the comparison mean, and could it represent a causal effect?	Assignment, counterfactual reasoning, assumptions, and alternative explanations
Cumulative science	How does this result fit with other evidence?	Replication, triangulation, mechanism testing, synthesis, and generalization

A confidence interval can show which effect sizes remain compatible with a model and dataset. A p value can summarize how surprising a test statistic would be under a stated null model. An effect size can put a difference or relationship on a useful scale. None of these automatically supplies random assignment, a representative sample, valid measurement, faithful implementation, or a correct causal structure.

This is why statistical significance and scientific importance are not the same. A precisely estimated trivial difference may have a very small p value. A practically important estimate may remain uncertain in a small study. An interval containing zero does not prove that the effect is exactly zero. A small p value does not prove that the favored explanation is the only possible explanation.

The strongest conclusion authorized by the complete evidence record is the study's **claim ceiling**. A favorable numerical result never raises that ceiling. If the assignment mechanism does not identify a causal contrast, a small p value cannot manufacture one. If recruitment does not support population inference, a narrow interval cannot make the sample representative.

Science as Cumulative, Self-Correcting Inquiry

The phrase *the scientific method* can suggest a rigid recipe: pose a hypothesis, conduct one experiment, calculate a result, and announce a conclusion. Real inquiry is more iterative. Questions are refined, measures are improved, explanations are challenged, and new studies are designed in response to the limitations of earlier studies.

A useful cycle is:

1. Observe a problem and formulate a question.
2. Propose explanations that make distinguishable predictions.
3. Define the constructs, population, cases, and quantity to be estimated.
4. Choose a design capable of separating plausible explanations.
5. Collect and analyze evidence with transparent procedures.
6. Examine uncertainty, assumptions, and competing interpretations.
7. Replicate, vary, triangulate, and synthesize.
8. Revise the claim and identify the next informative study.

The final step leads back to the first. A study can resolve one uncertainty and expose another. That is progress, not failure.

Science can be described as converging toward truth, but convergence is not automatic. Ten studies can reproduce the same bias. A literature can appear consistent because unfavorable results were not reported,

outcome definitions changed, or every research group used the same imperfect measure. The more defensible goal is **convergence toward increasingly well-supported, precisely bounded, and openly revisable claims**.

Direct replication asks whether a result reappears when important features are held similar. Conceptual replication varies an operational definition or setting while testing the same underlying explanation.

Triangulation compares methods with different weaknesses. Mechanism studies test processes that a causal explanation predicts. Multisite studies reveal variation that one setting cannot show. Meta-analysis can synthesize comparable estimates, but it cannot make biased or incompatible studies unbiased and compatible.

The point is not to collect significant results as votes. It is to build a research program in which different forms of evidence make the remaining explanations increasingly difficult to sustain.

What Causation Means

A causal question concerns what would happen to an outcome if a condition were changed, compared with an appropriate alternative, under stated circumstances. It therefore contains a counterfactual comparison. For the same case at the same time, researchers cannot ordinarily observe both the outcome under the condition and the outcome under its alternative. Design and assumptions must make a comparison among observed outcomes informative about that missing counterfactual.

The causal quantity of interest must be stated clearly. It might be the average effect of offering a program, the effect among people who comply with an assignment, or the effect in a particular target population. These are not interchangeable. The **estimand** names the quantity the study is intended to estimate. Model choice should follow from the estimand and design rather than from whichever software output looks most favorable.

Random assignment can make treatment groups comparable in expectation and thereby support a causal contrast when implementation, attrition, interference, measurement, and analysis are also handled defensibly. Random sampling serves a different purpose: it can support generalization from a sample to a population. Neither procedure guarantees the other. A randomized experiment conducted with a narrow convenience sample may have strong internal validity and limited population reach. A representative observational survey may describe a population well while leaving causal alternatives unresolved.

Regression adjustment does not automatically recreate an experiment. Whether adjustment reduces or introduces bias depends on the causal role, timing, and measurement of the variables included. The fitted coefficient is always conditional on a model; causal interpretation requires an identification argument beyond the existence of that coefficient.

Responsiveness, Consistency, and Mechanism

Mosteller and Tukey (1977) described three complementary forms of support for causal reasoning: responsiveness, consistency, and mechanism.

Responsiveness asks whether the outcome responds when the proposed cause is deliberately changed. Random assignment and credible implementation can provide especially strong responsiveness evidence because the change is introduced by design rather than merely observed. A comparison of existing groups is not responsiveness evidence simply because the groups differ.

Consistency asks whether a coherent relationship recurs across studies, settings, populations, measures, time periods, and reasonable analyses. Exact numerical equality is not required. Genuine effects can vary. What matters is whether the pattern and its variation make sense across informative changes in the evidence.

Mechanism asks whether a defensible process connects the proposed cause to the outcome and makes additional predictions that can be tested. A mechanism is more than a plausible story told after the result. It should imply observable intermediate changes, timing, boundary conditions, or failures that could count against the explanation.

These forms of support are not a two-out-of-three scorecard. Consistency cannot repair a bias shared by every study. A plausible mechanism cannot rescue a comparison that does not identify the proposed effect. Responsiveness in one setting does not establish broad generality. Mechanism evidence can validate one link without proving the complete causal chain.

Modern causal identification provides the logical bridge connecting design and assumptions to the causal estimand. Identification is not a fourth box beside responsiveness, consistency, and mechanism. It asks whether the observed data, given the design and stated assumptions, can recover the causal comparison of interest. jamovi can estimate the specified model. It cannot decide whether the identification argument is true.

Why Repeated Evidence Can Still Mislead

Repeated results deserve confidence only to the extent that the research program exposes rather than reproduces its weaknesses. Several recurring problems can produce false convergence.

Shared bias. If every study recruits the same narrow population, uses the same confounded comparison, or repeats the same measurement defect, repetition can make the biased answer more precise without making it more accurate.

Selective evidence. Publication bias, outcome switching, unreported model choices, and stopping rules can make the visible literature look more coherent than the underlying evidence.

Measurement non-equivalence. A score may not represent the same construct across groups, occasions, or settings. Comparing means then risks attaching one name to quantities that differ in meaning.

Incompatible estimands. Studies described with the same topic label may estimate different populations, interventions, outcomes, time horizons, or causal contrasts. Pooling them can conceal rather than resolve the scientific question.

Significance counting. Treating significant and nonsignificant results as opposite votes discards effect magnitude and precision. Two estimates can be similar even when one has $p < .05$ and the other does not; estimates can differ substantially even when both cross the same threshold.

Defensible cumulative reasoning therefore examines the estimates, uncertainty, designs, measures, sources of bias, and patterns of variation. It also treats contradiction as information. A result that fails to replicate can reveal a boundary condition, an unstable procedure, an exaggerated original estimate, or a mistaken explanation.

No Study Stands Alone

A study should be evaluated both as an answer to its immediate question and as a contribution to a larger evidence program. Ask five questions:

1. What did this study estimate?
2. What does its design permit researchers to claim?
3. Does it principally add responsiveness, consistency, or mechanism evidence?
4. What important explanation or generalization can it not settle?
5. What next study would most reduce the remaining uncertainty?

A one-group pre-post study can estimate change among measured participants. It may reveal feasibility, measurement problems, or the range of individual responses. Without a comparison condition and an identification argument, it cannot uniquely attribute change to the program. The next informative study might introduce randomized assignment, measure a proposed mechanism, or repeat the work in a contrasting population.

A scientifically successful study is therefore not necessarily one producing $p < .05$. Success can mean obtaining an informative interval around a small effect, ruling out an explanation, discovering that a finding does not generalize, detecting a measurement failure, contradicting an earlier result, or identifying the next experiment capable of discriminating among competing explanations.

These materials distinguish scientific success from a favorable result pattern. A favorable pattern is the direction, magnitude, and precision the hypothetical researchers hope to see. Scientific success is a credible answer with honest uncertainty and limitations, whether favorable, unfavorable, or inconclusive.

How These Materials Teach Scientific Reasoning

These materials are designed for practice rather than as a second conceptual textbook. They teach by requiring decisions and then making the reasoning behind those decisions visible.

Each substantive unit moves through the same six levels: recall and recognize, predict before clicking, calculate and explain, read jamovi output, choose the model, and report and critique. They run in that order for a reason. Each level gives you something to be right or wrong about before the next one arrives, so that vocabulary is in place before you predict, a prediction is on record before you calculate, and a model has been chosen deliberately before you are asked to defend a claim built on it. How to Use These Materials sets out what each level asks of you in practice.

The materials use synthetic data so that every exercise can be reproduced and verified without implying access to real participants. Plausible scenarios add scientific context while stating clearly that the studies are fictional. Worked solutions provide corrective feedback and model expert reasoning. Rejected alternatives teach selection by comparison rather than by a memorized flowchart. Inconclusive cases prevent scientific usefulness from becoming synonymous with favorable findings. Repeated and mixed practice develops transfer beyond the cue supplied by a chapter title.

Throughout the solutions, a recurring **Study in the Larger Evidence Program** box explains what the represented study adds, what it cannot settle, which form of causal support it can contribute, and what evidence should come next. Those questions connect one statistical exercise to the cumulative work of science.

The Reader's Evidence Checklist

Before accepting a statistical conclusion, ask:

1. What is the scientific question?
2. What are the cases or independent units?
3. How were the constructs measured, and are the scores comparable?
4. How were observations sampled, assigned, exposed, and retained?
5. What estimand does the analysis target?
6. Why does the selected model match the design and dependence structure?
7. What estimate and interval answer the question?
8. Which assumptions and diagnostic evidence matter for that answer?
9. What alternative explanations remain?
10. What is the claim ceiling?
11. What does the study contribute to responsiveness, consistency, or mechanism?
12. What next evidence would most reduce the remaining uncertainty?

Use this checklist before opening jamovi, while reading its output, and again when writing the conclusion. The purpose is not to make every claim timid. It is to make each claim as strong as the evidence permits—and no stronger.

How to Use These Materials

Read the introduction first. It explains why every analysis in these materials is treated as one part of a scientific argument and a larger evidence program. This section describes the practical study routine.

Statistics becomes usable through decisions, not recognition alone. Reading a clear explanation can make a method feel familiar; solving a new case reveals whether the reasoning can be retrieved and transferred.

Each unit therefore moves through six levels. Begin without looking at the solutions. Commit to an answer, mark your confidence, and then compare your reasoning with the worked response. When an answer is wrong, record the cue you missed rather than merely copying the correction.

The six levels

1. **Recall and recognize:** retrieve the vocabulary and design features.
2. **Predict before clicking:** state what pattern you expect before seeing software output.
3. **Calculate and explain:** connect a small calculation to the model's logic.
4. **Read jamovi output:** locate, translate, and integrate the relevant evidence.
5. **Choose the model:** select one analysis and reject plausible alternatives.
6. **Report and critique:** communicate the result and repair reasoning that exceeds the evidence.

A useful practice loop

For each problem, write four brief notes: **answer**, **reason**, **confidence**, and **revision**. Reattempt missed problems after a delay. A correct answer that depends on remembering the page is less valuable than a correct answer reached from the design and data structure.

When a solution includes **Study in the Larger Evidence Program**, record what the study adds, what it cannot settle, and which next study would be most informative. This keeps one statistical result in its scientific context.

Companion Text, Data, and Feedback

For fuller conceptual and jamovi workflow explanations, use Psychology Research Methods and Statistics by Design with Jamovi. Use these materials for additional attempts, contrasting cases, worked solutions, and model-choice challenges.

The public companion files include study_02_mindfulness_paired.csv, study_03_feedback_two_groups.csv, and study_04_practice_spacing_anova.csv, together with data dictionaries. Browse them at <https://nicholaskarlson.github.io/data/>; use [Discussions](#) for questions and improvements.

Optional Books for Deeper Study

This open resource is complete on its own. The following published books are optional purchases for readers who want fuller conceptual instruction, reproducible code workflows, or guidance for planning real research and consulting relationships.

Psychology Research Methods and Statistics by Design with Jamovi

This 27-chapter, beginner-friendly guide is the closest conceptual companion to these practice materials. Use it when a problem reveals that you need a fuller explanation of research design, a named jamovi Desktop workflow, APA-style reporting, or the limitation that keeps a result defensible. It uses the same public synthetic-data ecosystem and works well before or alongside the practice units. [View on Amazon](#).

Psychological Statistics by Design

This advanced, code-first psychology guide follows evidence from a research question and data layout through Python analysis, independent R verification, parity checks, and reviewer packets. Use it after these materials when you want to see model choice, estimands, claim ceilings, and reproducibility expressed as inspectable code and handoff artifacts. [View on Amazon](#).

Applied Statistics with Python and R

This accessible Python-first guide develops a broader applied workflow, using R for optional independent verification and PyStatsV1 as a bridge to a proof-first evidence package. Use it when you want to turn the reasoning practiced here into a reproducible, cross-language workflow for common tests, analysis of variance, correlation, regression, and APA-style reporting. [View on Amazon](#).

Before You Hire a Statistician

This guide explains what statistical consulting can and cannot do, what to prepare before contact, how scope, fees, timelines, deliverables, and revisions work, and how privacy, authorship, ethics, and responsibility should be handled. Use it before adapting these exercises to a real study, sharing data, or hiring statistical help. [View on Amazon](#).

PART I

Questions, data, and description

Unit 1: From Research Questions to Statistical Questions

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Unit Purpose

Most analyses that go wrong went wrong before any software opened. A research question is a sentence in ordinary language; a statistical question names a population, a set of cases, a measured variable, a quantity to be estimated, and a claim the design could support. Translating one into the other is a set of decisions, and every decision you skip gets made for you by whichever analysis you happen to run.

Learning Objectives

After completing this unit, a learner should be able to:

- identify the population, cases, variables, and estimand implied by a question;
- distinguish a research question from the statistical question that stands in for it;
- state what quantity an analysis estimates, in words, before naming a method;
- recognise when a question names no comparison and say what follows;
- reject a statistical question that answers something easier than what was asked; and
- write the claim a design could support, before any data exist.

Decision Map

Ask these questions in order:

1. Who or what is the question about, and who is actually represented?
2. What is the case—the independent unit that contributes information?
3. What is measured, and on what scale?
4. What quantity would answer the question if you could compute it perfectly?
5. Is a comparison named, and if so, to what?
6. What would count as an answer, and what would count as a useful non-answer?
7. What claim could this design support even in the best case?

Unit Study Files

Study	Data structure	Used here for	Unit role
Study 02	Pre and post scores for 72 synthetic participants	Identifying cases, variables, and the estimand from a file	Translating a vague question into a stated quantity

Plausible Research Scenario

The scenario creates decisions rather than documenting real research. At the **Pause and Decide**, stop reading and write an answer, reason, and confidence rating. The explanation that follows is feedback on that commitment.

Plausible Research Scenario: “Does Mindfulness Help Our Students?”

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

A university counseling centre's director asks her fictional research team a reasonable question: *does mindfulness help our students?* She wants an answer before funding a six-week program.

The sentence is a research question, not yet a statistical one, and the gap between them is where the study gets designed. Five things are unspecified.

Which students. Everyone enrolled? Those who seek help with stress? Those who would volunteer for a six-week commitment? Different populations, and only the last is one a voluntary pilot could represent.

Help with what. Distress, sleep, academic performance, self-reported mindfulness? Each implies a different instrument, and one measuring mindfulness cannot report on distress however encouraging its numbers.

Compared to what. “Help” is comparative and no comparison is named. To not participating? To a waiting list? To how the same students were six weeks earlier? Only the last is available in a one-group pilot, and it is the weakest.

How much. Without a magnitude that would matter, any result reads as support.

What the case is. One participant, or one questionnaire? The file will hold 144 scores from 72 people, and there are 72 independent cases.

The team must replace the question with one the study can answer, and be explicit that it is a replacement. The honest statistical question is narrower than the director's, and saying so is what makes the eventual answer trustworthy.

Pause and Decide

Write the statistical question you would put in the protocol. Name the population, the case, the outcome, and the quantity to be estimated. Then state plainly what the director asked that your question does not answer.

Why This Model?

There is no model yet, and that is the point. The estimand comes first: the mean post-minus-pre change in mindfulness score among the represented participants. A method is chosen to estimate that quantity, not the other way round.

Two habits are rejected. Choosing a test from the file's shape—two columns, therefore a two-sample test—selects a method before a question exists. And choosing the question from what the data support most impressively answers whatever is easiest rather than what was asked.

What Would a Favorable Result Look Like?

Nothing to favour yet. The study succeeds if the question it answers is stated clearly enough that a reader can see the distance between it and the director's question.

Best Defensible Claim

Statistical evidence: None yet. The unit produces a stated estimand and a described population, not a result.

Causal identification: The design implied by the question—one group, measured twice—identifies nothing causal, and this is knowable before any data are collected.

Claim ceiling: Whatever the eventual numbers, the strongest available claim is a described change among participants who volunteered at one centre.

What Must Be True?

The population must be describable in advance, the case must be the independent unit rather than the measurement, and the outcome must be fixed before results are seen. An estimand changed after inspecting the data answers a question the data selected.

Defensibility Procedures

Write the statistical question before collecting anything; name the population and the case; define the outcome and its scale; state the estimand in words before naming a method; record the comparison the design does and does not provide; and state what magnitude would matter.

Next evidence priority: Decide what design could support a stronger claim, which is a question about assignment rather than about measurement.

Claim Audit

Repair this claim: “We will find out whether mindfulness helps students.” Name the population it implies, the comparison it assumes, and the claim ceiling the available design permits.

Problems

Do not consult the solutions until you have recorded an answer, reason, and confidence rating for every attempted item.

Level 1: Recall and Recognize

Problem 1.1 — Name the parts

List the components a statistical question must name before a method can be chosen. For each, give one sentence saying what goes wrong if it is left out.

Problem 1.2 — Population and represented sample

Distinguish the population a question is about from the participants a study represents. Name one situation in which the two differ so much that the estimate answers a different question.

Problem 1.3 — Define an estimand

Define “estimand” without naming any statistical method. Then state the estimand for a study measuring the same students before and after an interval.

Level 2: Predict Before Clicking

Problem 1.4 — Predict what changes with the population

The same six-week program is evaluated once among students who sought help with stress, and once among all enrolled students. Predict which quantities in the statistical question change and which stay the same.

Problem 1.5 — Predict what the file cannot answer

Study 02 contains a pre score, a post score, and a practice log for each of 72 participants. Predict which of these two questions it can address and which it cannot: *how much did scores change?* and *did the program change scores more than no program would have?*

Level 4: Read jamovi Output

Problem 1.6 — Identify the cases from output

A jamovi Descriptives table for Study 02 reports $N = 72$ for mindfulness before and $N = 72$ for mindfulness after. A colleague concludes the study has 144 observations. State what the table shows and what it does not.

Problem 1.7 — Locate the outcome

The same table reports a mean of 43.293 for one variable and 49.379 for another. Explain what you would need to know before deciding which is the outcome, and where in the study record that information lives.

Problem 1.8 — Read a table against a question

The director asks whether students who practised more improved more. State whether a Descriptives table alone can address that question, and what kind of output would.

Level 5: Choose the Model

Problem 1.9 — Choose the question the design can answer

Given a one-group pre-post design, choose between two candidate statistical questions—one about change and one about effect—and justify the choice by what the design supplies.

Problem 1.10 — Reject the question chosen by the file

A colleague proposes comparing the pre and post columns as two independent groups because “the file has two columns of scores.” Identify the step that went wrong and name what should have come first.

Problem 1.11 — Reject the substituted outcome

The team cannot measure academic performance, so a draft proposes reporting the mindfulness score and describing it as evidence about academic outcomes. Reject the substitution and say what may be claimed instead.

Problem 1.12 — Reject the question chosen after the result

A draft protocol says the team will “examine the data and report the clearest finding.” Explain what this does to any quantity eventually reported.

Level 6: Report and Critique

Problem 1.13 — Write the statistical question

Write the statistical question for the mindfulness pilot in two sentences: one naming population, case, outcome and estimand, and one naming what the design cannot address.

Problem 1.14 — Repair a question with no comparison

Repair this question: “Does the mindfulness program improve wellbeing?” Name every unspecified element and supply a version a one-group pilot could answer.

Problem 1.15 — Repair a population claim

Repair this sentence: “This study tells us how mindfulness affects university students.” Identify the population actually represented and rewrite the claim.

Problem 1.16 — Repair an estimand that changed

A report states that the team originally planned to examine mindfulness change but “focused on practice minutes because the relationship was stronger.” Explain what this does to the reported result and how the report should be written.

Worked Solutions

Each solution identifies the decisive cue, reasoning path, relevant evidence, rejected alternative, claim boundary, and a transfer cue.

Level 1 Solutions

Solution 1.1

A statistical question names the **population** (leave it out and the result describes nobody in particular), the **case** or independent unit (leave it out and the amount of information is miscounted), the **variables and their scales** (leave them out and no summary is interpretable), the **estimand** (leave it out and the method chooses the question), the **comparison, if any** (leave it out and comparative words creep into a non-comparative study), and the **magnitude that would matter** (leave it out and any result reads as support). **Transfer cue:** every one of these is decided before software opens.

Solution 1.2

The **population** is the group the question is about; the **represented participants** are those the study actually observed. They differ whenever participation is voluntary, recruitment is local, or attrition is selective. A programme evaluated among students who volunteered for a six-week commitment estimates change among people willing to make that commitment, which may be a very different quantity from change among students in general. **Claim ceiling:** describe who was represented, and do not silently promote them to a population.

Solution 1.3

An **estimand** is the quantity you would compute if you could observe the relevant data perfectly—a target, stated in words, independent of any method used to approximate it. For students measured before and after an interval it is the **mean post-minus-pre change in the outcome among the represented participants**. Note that this names a direction, a population, and an outcome, and names no test.

Level 2 Solutions

Solution 1.4

The **population** changes, and so does everything that depends on it: the estimand is now change among all enrolled students, and the claim ceiling refers to a different group. The **case** is unchanged—still one participant. The **outcome and its scale** are unchanged. The **design's causal limits** are unchanged, because they follow from having no comparison condition rather than from who was sampled. **Transfer cue:** changing who you study changes what your number means, even when the arithmetic is identical.

Solution 1.5

It can address **how much did scores change**: every participant has a pre and a post score, so a per-person difference exists. It cannot address **did the program change scores more than no program would have**, because no participant was observed without the program, so the comparison the question requires was never measured. **Decisive cue:** a comparative question needs an observed comparison, and no analysis can manufacture one.

Level 4 Solutions

Solution 1.6

The table shows that **72 participants** each supplied a value on two occasions. It does not show 144 independent observations: the two scores in a row belong to one person and are linked. The N column reports how many values were available for each variable, not how many independent cases the study contains.

Rejected rule: “add the Ns.”

Solution 1.7

You would need the **study's definition of the outcome and the direction of interest**, which live in the protocol, the study-design record and the data dictionary—not in the output. A Descriptives table reports numbers for whatever columns exist; it carries no information about which column the question is about. Here the pre and post scores are the same instrument at two occasions, so neither is the outcome on its own: the outcome is the change between them.

Solution 1.8

A Descriptives table cannot address it. Descriptives summarise variables one at a time and say nothing about how two variables move together. The question asks about a **relationship** between practice minutes and change, which needs a scatterplot and a correlation or regression. Note also that such a relationship would be observational even within an experiment, because practice was not assigned.

Level 5 Solutions

Solution 1.9

Choose the question about **change**: what is the mean post-minus-pre difference among the represented participants? The design supplies two measurements of the same people and no comparison condition, so it can estimate how much scores moved and cannot estimate what would have happened otherwise. Choosing the effect question would commit the study to answering something it cannot, which is worse than answering a narrower question well.

Solution 1.10

The step that went wrong is that a **method was chosen from the file's shape** rather than from the question. What should have come first is the estimand: a per-person change, which the paired structure exists to support. Treating the columns as independent groups discards the participant matching and answers a question about unrelated people that nobody asked. **Transfer cue:** file layout is a storage decision, not a scientific one.

Solution 1.11

Reject it. A mindfulness questionnaire measures self-reported mindfulness; it is not a measure of academic performance, and no result on it licenses a claim about grades. What may be claimed instead is change on the measured construct, with the gap between construct and outcome stated explicitly. **Claim ceiling:** you may report what you measured, using the name of what you measured.

Solution 1.12

It makes the reported quantity a **selected** one. If the estimand is chosen after seeing which comparison looks clearest, the reported result is the largest of several implicitly examined, and every interval and *p* value attached to it understates the uncertainty—correct arithmetic applied to a question the data picked. A protocol should name the primary estimand in advance and label anything else as exploratory.

Study in the Larger Evidence Program: Stating the Mindfulness Question

- **What this study adds:** A stated population, case, outcome and estimand, fixed before data exist, which is what makes every later number in the programme interpretable.
- **What it cannot settle:** Anything empirical. No data have been collected, and the unit produces a question rather than an answer.
- **Causal-support role:** None yet, but it fixes the ceiling: a one-group design cannot supply responsiveness evidence however the numbers come out.
- **Next evidence priority:** Examine what design would raise that ceiling, which is a question about assignment rather than measurement.

Level 6 Solutions

Solution 1.13

“Among students who volunteer for a six-week mindfulness programme at this centre, what is the mean post-minus-pre change in self-reported mindfulness score, with one participant as the independent case? Because every participant receives the programme and no comparison condition is observed, this question concerns change over the interval and cannot address what the programme caused or how students in general would respond.”

Solution 1.14

Unspecified: the **population** (which students), the **outcome** (wellbeing is not an instrument), the **comparison** (“improve” relative to what), the **magnitude** that would matter, and the **case**. A one-group pilot could answer: “Among participants who complete both questionnaires, what is the mean post-minus-pre change in self-reported mindfulness score over six weeks?” **Transfer cue:** every comparative word in a question is a promise the design has to keep.

Solution 1.15

The represented participants are students who sought help with stress at one centre and volunteered for a six-week commitment—not university students. And “affects” asserts causation the design cannot identify. Corrected: “Among the students represented in this pilot, self-reported mindfulness scores changed by the estimated amount over six weeks. The design does not establish what caused the change or how other students would respond.”

Solution 1.16

The reported result is now the outcome of a **search**, not of the planned analysis, so its stated uncertainty is too small and its apparent strength is partly a selection artifact. The report should state the pre-specified estimand and its result first, whatever it was, then present the practice-minutes analysis explicitly as exploratory and unplanned, so a reader can weight it accordingly. **Decisive cue:** what makes a finding exploratory is when the question was chosen, not how interesting it is.

Unit 2: Design, Assignment, and Causal Limits

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Unit Purpose

What a study may claim is fixed by how its observations came to exist, not by how its numbers came out. This unit builds one habit above all others: before reading any result, find the assignment record and ask what comparison the design actually observed. A large, precise, beautifully analysed estimate from a one-group study and a small, noisy one from the same design have exactly the same claim ceiling.

Learning Objectives

After completing this unit, a learner should be able to:

- distinguish experimental, quasi-experimental, and observational evidence by assignment;
- distinguish within-person from between-person comparison structures;
- identify the counterfactual a causal claim requires and say whether it was observed;
- distinguish an undocumented assignment record from a documented absence of assignment;
- explain why a result's size never raises the claim ceiling; and
- name the specific alternative explanations a one-group pre-post design leaves open.

Decision Map

Ask these questions in order:

1. How did each case come to be in the condition it is in?
2. Was that mechanism random, chosen, pre-existing, or simply unrecorded?
3. Is the comparison within the same people or between different people?
4. What would have happened otherwise, and was it observed?
5. Which alternative explanations does the design leave open?
6. What is the strongest claim this evidence record supports?
7. Would that claim change if the numbers came out differently?

Unit Study Files

Study	Data structure	Assignment record	Unit role
Study 02	Pre and post scores for 72 synthetic participants	Not documented in the pinned source assets	An unrecorded mechanism
Study 02C	Pre and post scores for 72 synthetic participants	None; every participant is measured before and after	A documented absence of assignment

Plausible Research Scenario

The scenario creates decisions rather than documenting real research. At the **Pause and Decide**, stop reading and write an answer, reason, and confidence rating. The explanation that follows is feedback on that commitment.

Plausible Research Scenario: What the Assignment Record Actually Says

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

Two files sit in front of the counseling centre's fictional research team. Study 02 and Study 02C have identical structure: 72 synthetic participants, a mindfulness score before a six-week interval and one after. One shows a clear average rise, the other almost none.

A junior colleague proposes the first supports a claim the second does not. Seeing why that is backwards is this unit's work.

Look at what the study-design record says about assignment. Study 02C: *none; every synthetic participant is measured before and after*. A **documented absence** — no assignment happened, because there is one condition. Study 02: *not documented in the pinned source assets*. An **unrecorded mechanism**: the record does not say what happened, which is not the same as saying nothing did.

The distinction is real and does not rescue Study 02. Silence is not evidence, so a reader must assume the weakest defensible reading. Both files sit at the same ceiling: a described change over an interval among the represented participants.

Study 02's rise is large and precise; Study 02C's is near zero and imprecise. Whether that difference in what they found is also a difference in what they may claim is the question below.

What the one-group structure leaves open is a specific list of rival explanations, each of which would produce a pre-post difference with no programme effect.

Pause and Decide

Decide whether Study 02's larger estimate licenses a stronger claim than Study 02C's, and what an unrecorded mechanism permits a reader to assume. Then name two rival explanations a comparison group would rule out, and one it would not.

Why This Model?

The design question comes before the model and is answered from the study record, not the data. Both files support a paired-samples comparison: the measurements are linked within participants. That model estimates change and does not identify an effect, and no other model would change what the design observed.

A between-groups model is rejected because the same people appear at both occasions; a causal model, because no counterfactual was observed.

What Would a Favorable Result Look Like?

A favourable result informs the next design decision, which both do. Study 02 shows the measurement can register change; Study 02C shows change is not guaranteed. Neither says the programme works — and treating the first as though it did is the failure this unit prevents.

Best Defensible Claim

Statistical evidence: Each study estimates the mean post-minus-pre change among its represented participants.

Causal identification: None in either study; no comparison for what would have happened without the programme was observed.

Claim ceiling: Describe change over the represented interval — identical for both files, and immovable by the estimate.

What Must Be True?

Participant matching must be correct across occasions and the measurement must mean the same at both. A causal claim—which neither study supports—would need an observed comparison condition with a recorded assignment mechanism.

Defensibility Procedures

Record the assignment mechanism explicitly, including when there was none; distinguish an absent mechanism from an absent record; document recruitment, timing, exposure and attrition; fix the claim ceiling before results exist; and repeat it whatever the numbers.

Next evidence priority: A comparison condition with a recorded assignment mechanism — the only change that raises the ceiling.

Claim Audit

Repair this claim: “The change was large and highly precise, so the programme clearly works.” Separate an estimate’s size from the design that produced it, and name what must be added to the study rather than the analysis.

Problems

Do not consult the solutions until you have recorded an answer, reason, and confidence rating for every attempted item.

Level 1: Recall and Recognize

Problem 2.1 — Sort designs by assignment

Define experimental, quasi-experimental, and observational evidence in terms of how cases came to be in their conditions. State which of the three a one-group pre-post study is.

Problem 2.2 — Name the counterfactual

State what counterfactual a claim that the programme caused a change would require, and say whether the mindfulness pilot observed it.

Problem 2.3 — Within-person and between-person

Distinguish a within-person comparison from a between-person one. Give one advantage of each that the other does not have.

Level 2: Predict Before Clicking

Problem 2.4 — Predict what a comparison group would change

Suppose the next study adds a randomly assigned waiting-list group. Predict which of these change: the estimand, the claim ceiling, the alternative explanations that remain open, and the statistical model.

Problem 2.5 — Predict the effect of a larger estimate on the ceiling

Suppose the pilot had produced a mean change three times larger, with a much narrower interval. Predict what happens to the claim ceiling, and state the principle behind your prediction.

Level 4: Read jamovi Output

Problem 2.6 — Read a result against its design

jamovi reports a mean difference of 6.086 with a 95% interval of [4.913, 7.259] for Study 02. Write what this output establishes and the one word a careless reader would add that it does not support.

Problem 2.7 — Read two results with one ceiling

Study 02 gives a mean change of 6.086; Study 02C gives 0.400 with an interval of [-0.821, 1.621]. State what differs between the two studies and what does not.

Problem 2.8 — Read output that cannot answer the question

A colleague asks jamovi for every analysis the file supports, hoping one will establish that the programme worked. Explain why no output from this file can, and what the colleague is actually varying.

Level 5: Choose the Model

Problem 2.9 — Choose the comparison the design observed

For the mindfulness pilot, state which comparison is available and which is merely desired. Explain what happens to a report that quietly substitutes the second for the first.

Problem 2.10 — Reject randomisation by assumption

A draft describes Study 02's assignment as "presumably random, since nothing suggests otherwise." Reject the inference and state what an unrecorded mechanism permits a reader to assume.

Problem 2.11 — Reject a causal claim rescued by an analysis

A colleague proposes adjusting for practice minutes so the result "can be read causally." Explain why a covariate cannot supply a missing comparison condition.

Problem 2.12 — Reject a ceiling that moves with the result

A report states its claim ceiling in the methods section and then, having found a large effect, states a stronger conclusion in the discussion. Name the error and say when the ceiling should have been fixed.

Level 6: Report and Critique

Problem 2.13 — Write the design paragraph

Write the two-sentence design statement for the mindfulness pilot: what comparison was observed, and what claim that permits.

Problem 2.14 — Repair a causal conclusion

Repair this sentence: "Mindfulness training increased students' mindfulness scores by about six points." Identify every element that exceeds the design.

Problem 2.15 — Repair a claim about an unrecorded mechanism

Repair this sentence: "Participants were assigned to the programme, so the comparison is fair." Say what the study record supports and what it does not.

Problem 2.16 — Repair a limitations section that does the wrong job

A limitations section reads: "As with all studies, some caution is warranted." Rewrite it to name the specific alternative explanations this design leaves open.

Worked Solutions

Each solution identifies the decisive cue, reasoning path, relevant evidence, rejected alternative, claim boundary, and a transfer cue.

Level 1 Solutions

Solution 2.1

In **experimental** evidence the researcher assigns cases to conditions by a known mechanism, usually randomisation. In **quasi-experimental** evidence cases are in conditions by some non-random but identifiable process the design tries to account for. In **observational** evidence conditions are pre-existing or self-selected and the researcher only observes. A one-group pre-post study is none of these in the comparative sense: it has **one condition** and therefore no assignment at all. Its comparison is across time within the same people.

Solution 2.2

It would require knowing how the **same participants would have changed over the same six weeks without the programme**. The pilot did not observe it: every represented participant received the programme, so the counterfactual is missing by construction. **Decisive cue:** a causal claim names two states of the world and a study must have observed something standing in for both.

Solution 2.3

A **within-person** comparison measures the same case under both conditions, so each person serves as their own control and stable individual differences are removed—this is why the paired analysis is more precise. A **between-person** comparison measures different cases in each condition, so no carryover, practice or fatigue can contaminate the contrast, and the two conditions can be simultaneous rather than sequential. **Transfer cue:** each design removes a different threat, which is why neither is universally better.

Level 2 Solutions

Solution 2.4

The **estimand** changes: from mean change within participants to a difference in change between assigned groups. The **claim ceiling** rises: with random assignment and adequate execution, a causal effect for the represented experimental population becomes defensible. The **alternative explanations** narrow sharply—history, maturation, testing effects and regression to the mean now affect both groups and largely cancel. The **statistical model** changes too, but it is the least important of the four. **Transfer cue:** the ceiling is raised by design, not by analysis.

Solution 2.5

The claim ceiling is **unchanged**. It is determined by what the design observed, and the design observed no comparison condition regardless of how the numbers came out. The principle: if a large estimate licensed a causal claim, the licence could be earned by chance—a researcher with a favourable sample would acquire an entitlement that a researcher with the same design and an unfavourable sample would not. **Rejected rule:** “strong results justify strong claims.”

Level 4 Solutions

Solution 2.6

It establishes that, among the 72 represented participants, mindfulness scores were an estimated 6.086 points higher after the interval than before, with the data compatible with average rises of roughly 4.913 to 7.259 points. The word a careless reader adds is “**because**”—or any of its relatives: *increased by*, *produced*, *led to*. The output describes a change over time and cannot attribute it. **Claim ceiling**: description, however precise the interval.

Solution 2.7

What differs is the **evidence**: the size of the estimated change and how precisely each study pinned it down. What does not differ is the **claim ceiling**, the **design**, the **available comparison**, and the **list of alternative explanations**. Two studies of the same design differ in what they found, not in what they may say. **Transfer cue**: read the design before the result, so the result cannot influence what you think the design permits.

Solution 2.8

No output can, because the limitation is in the **data**, not in the analysis: the file contains no observation of anyone not receiving the programme. What the colleague is varying is the **analysis**, and no analysis creates an observation that was never made. Running many analyses in search of a causal reading also makes whichever result is eventually reported a selected one, so the search weakens the evidence twice over.

Level 5 Solutions

Solution 2.9

The comparison **observed** is between the same participants at two times. The comparison **desired** is between participants who received the programme and participants who did not. A report that substitutes the second for the first is not making an arithmetic error—the numbers are correct—it is describing a study that was not conducted. Readers cannot detect the substitution from the results section, which is why the design paragraph has to state the available comparison explicitly.

Solution 2.10

Reject it. “Nothing suggests otherwise” is an argument from an absent record, and an absent record is not evidence for any particular mechanism. An unrecorded assignment mechanism permits a reader to assume **only the weakest defensible reading**: that the mechanism is unknown and may be related to the outcome. Note the contrast with Study 02C, whose record affirms that no assignment occurred—a documented absence, which is a different and more informative state than silence. **Transfer cue**: silence in a study record is a gap, never a permission.

Solution 2.11

Because a covariate adjusts a comparison that exists; it does not create one. Adjusting for practice minutes changes which participants are being compared with which, within a design that still contains no unexposed participants. There is no set of statistical controls that can stand in for an observation nobody made. Adjustment can make an existing comparison more defensible; it cannot supply the counterfactual. **Claim ceiling**: unchanged by any covariate.

Solution 2.12

The error is a **ceiling that moved with the result**. It should have been fixed in the protocol, before any data existed, and repeated unchanged in the discussion. A ceiling that is stated in the methods and then exceeded in

the discussion is worse than one never stated at all, because the methods section gives a reader false assurance that the boundary was being observed.

Level 6 Solutions

Solution 2.13

“All 72 represented participants received the six-week programme and were measured before and after it, so the only comparison observed is within participants across time. This design supports a description of how scores changed over the interval among these participants, and does not support a claim that the programme caused the change or that other students would respond similarly.”

Solution 2.14

“Mindfulness training increased” asserts causation with no comparison condition. “Students” promotes the represented volunteers to a general population. “By about six points” is the only defensible part, and only as a description. Corrected: “Among the represented participants, mindfulness scores were an estimated 6.086 points higher after the six-week interval than before. Because every participant received the programme, the design does not establish what caused the change.”

Solution 2.15

The record supports neither half. It does not document that participants were assigned—it is silent—and “fair” claims a property that only a known assignment mechanism could establish. Corrected: “The assignment mechanism is not documented in the study record. In the absence of that documentation, the comparison must be treated as observational, and differences between participants cannot be assumed to be balanced.”

Solution 2.16

“As with all studies, some caution is warranted” names nothing and warns of nothing. A limitations section should name the specific rival explanations this design leaves open: **history** (something else happened over the six weeks), **maturation** (participants would have changed anyway), **testing effects** (completing the questionnaire twice), **regression toward the mean** (those scoring lowest at intake tend to score higher later regardless), **measurement drift** (administration conditions differing between occasions), and **selective attrition** (who completed both questionnaires may not be who started). **Transfer cue**: a limitation a reader could not have written themselves is worth printing; one they could is filler.

Study in the Larger Evidence Program: Fixing the Mindfulness Claim Ceiling

- **What this study adds:** An explicit assignment record and a claim ceiling fixed before results exist, which is what stops a favourable number from being read as a stronger finding than the design allows.
- **What it cannot settle:** Anything causal. Both files describe change over an interval with no observed comparison condition.
- **Causal-support role:** It establishes the boundary rather than supplying evidence: neither responsiveness nor mechanism can be argued from a one-group design however the numbers fall.
- **Next evidence priority:** A comparison condition with a recorded assignment mechanism, which is the only change that raises the ceiling.

Unit 3: Measurement, Coding, and Data Quality

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Unit Purpose

Between a construct and a column sits a chain of decisions: what the construct is, how it is operationalised, what scale results, how values are coded, what counts as missing, and what a value outside the permitted range means. Each link can break quietly. A codebook written before collection is what makes the chain inspectable afterwards, and an analysis that cannot say what its numbers measure is not saved by being correct.

Learning Objectives

After completing this unit, a learner should be able to:

- trace a construct to an operational definition and a coded variable;
- assign a measurement level and say what summaries it permits;
- distinguish a variable's role in a design from its data type;
- state a missing-data policy before collection and report against it afterwards;
- reject an analysis that treats a code as a quantity; and
- read a data dictionary as evidence about what a result can mean.

Decision Map

Ask these questions in order:

1. What construct is this variable meant to represent?
2. How was it operationalised, and by what instrument?
3. What is its measurement level, and what summaries does that permit?
4. What role does it play—case identifier, outcome, condition, covariate?
5. What values are permitted, and what happens to values outside them?
6. What is the missing-data policy, and what was actually observed?
7. What can a result on this variable be about?

Unit Study Files

Study	Data structure	Used here for	Unit role
Study 02	Four variables for 72 synthetic participants	Roles, measurement levels, permitted ranges, and a declared missing-data policy	Reading and writing a codebook

Plausible Research Scenario

The scenario creates decisions rather than documenting real research. At the **Pause and Decide**, stop reading and write an answer, reason, and confidence rating. The explanation that follows is feedback on that commitment.

Plausible Research Scenario: Writing the Codebook Before Anyone Is Measured

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

The counseling centre's fictional team has its statistical question and must now specify what will be recorded. The output is a codebook: one entry per variable, written before collection begins.

Four variables are planned. A **participant identifier**, whose role is to link each person's two measurements—a label, not a quantity. A **mindfulness score before** the interval and one **after**, both continuous, both from the same instrument, both outcomes. And **practice minutes**, self-reported, continuous, a covariate rather than an outcome.

Each entry records more than a name. The **role** says what the variable does in the design; the **measurement level**, which summaries are meaningful; the **planned type**, what the file should contain; the **missing-data policy**, what is expected and what will be done; and a **permitted range**, which values are impossible rather than merely surprising.

The team writes the mindfulness variables as complete: a participant supplies both scores or is not a complete pair. Practice minutes get a different policy, `limited_synthetic_missingness`, because a self-reported log is a request rather than a measurement and some participants will not return one.

That distinction is the interesting one. Declaring it in advance makes the eventual gap a documented feature of the design rather than a discovery; a team that declares nothing and later reports a mean for whoever answered has made the same gap invisible.

Pause and Decide

Write the codebook entry for practice minutes: role, measurement level, planned type, missing-data policy, and permitted range. Then state what a summary of that variable will be able to describe, and who it will leave out.

Why This Model?

There is no statistical model here; the deliverable is a specification. What makes it a decision rather than paperwork is that each field constrains later analysis. An identifier cannot be averaged. A declared missing-data policy gives a reportable denominator. A permitted range turns an impossible value into a data-quality finding rather than an outlier.

Two shortcuts are rejected. Letting the file define the variables makes the specification a description of an accident. And deferring the missing-data policy until missingness appears invites a policy chosen to suit the result.

What Would a Favorable Result Look Like?

Nothing here is favourable or unfavourable. The codebook succeeds if a reader who has never seen the study can say what each column means, what summaries it permits, and how many cases stand behind it.

Best Defensible Claim

Statistical evidence: None. This stage produces a specification against which later data can be checked.

Causal identification: Unaffected. Measurement quality constrains what a result can mean but supplies no comparison the design lacks.

Claim ceiling: A later claim reaches the construct only as far as the operational definition supports, and covers only the participants who supplied the values.

What Must Be True?

The operational definition must precede collection; the instrument must be administered comparably across occasions; permitted ranges must come from the instrument rather than from observed data; and the missing-data policy must be fixed before anyone is measured.

Defensibility Procedures

Write one entry per variable before collection; record role, measurement level, planned type, missing policy and permitted range; version the codebook; report observed missing counts against the declared policy; and never let observed data revise a permitted range without recording why.

Next evidence priority: Describe what was actually collected against this specification, which is where the described distributions come in.

Claim Audit

Repair this claim: “Participants reported practising about twenty minutes a day, measured by our mindfulness scale.” Identify the instrument confusion and the population confusion, and say what the sentence supports.

Problems

Do not consult the solutions until you have recorded an answer, reason, and confidence rating for every attempted item.

Level 1: Recall and Recognize

Problem 3.1 — Construct to column

Name the steps between a construct and a coded variable in a data file. State which step an analysis can inspect and which it must take on trust.

Problem 3.2 — Role and type are different

Distinguish a variable's role in a design from its data type. Give one variable whose role and type could each be changed without the other changing.

Problem 3.3 — Measurement level constrains summaries

For each of an identifier, an ordered category, and a continuous score, state one summary that is meaningful and one that is not.

Level 2: Predict Before Clicking

Problem 3.4 — Predict what a missing policy produces

Study 02 declares mindfulness scores complete and practice minutes limited_synthetic_missingness. Predict what you expect to see in the observed missing counts for each, and what it would mean if the mindfulness variables had missing values.

Problem 3.5 — Predict the consequence of a mis-set level

Suppose the participant identifier were recorded as a continuous variable. Predict what software would then permit, and name the specific nonsense that becomes available.

Level 4: Read jamovi Output

Problem 3.6 — Read the Ns against the policy

A Descriptives table reports $N = 72$ for both mindfulness variables and $N = 68$ for practice minutes. State which declared policy each figure is consistent with, and what would have to be reported alongside.

Problem 3.7 — Read a range against a permitted range

The table shows practice minutes running from a minimum of 5.000 to a maximum of 31.700, and mindfulness before running from 23.900 to 64.600. Explain what these ranges can and cannot tell you about data quality.

Problem 3.8 — Read a variable's role from output alone

A colleague, looking only at a Descriptives table showing means of 43.293 and 49.379, asks which variable is the covariate. Explain why the table cannot answer and where the answer lives.

Level 5: Choose the Model

Problem 3.9 — Choose the summary the level permits

For an identifier, a self-reported practice log, and a mindfulness score, choose what may be summarised and how. Justify each from the measurement level.

Problem 3.10 — Reject the inferred codebook

A colleague proposes writing the dictionary after collection, by inspecting what the file contains. Reject the proposal and name two things that can no longer be checked.

Problem 3.11 — Reject a policy set after the fact

Missingness appears in practice minutes, and a draft proposes deciding how to handle it now that its pattern is visible. Explain what this does to the analysis.

Problem 3.12 — Reject a construct substitution

A report describes the mindfulness score as a measure of student wellbeing. Reject the substitution and state the claim the instrument supports.

Level 6: Report and Critique

Problem 3.13 — Write two codebook entries

Write full codebook entries for the participant identifier and for practice minutes, each naming role, measurement level, planned type, missing policy, and permitted range.

Problem 3.14 — Repair a mean over an identifier

A colleague recodes the identifiers P001 to P072 as the numbers 1 to 72 so that jamovi will accept them as continuous, then writes: “The average participant ID was 36.5, confirming a balanced sample.” Name both errors — the recoding and the conclusion — and say what may be reported instead.

Problem 3.15 — Repair an undocumented denominator

Repair this sentence: “Participants practised an average of 20.185 minutes a day.” Supply the missing information and name the policy it should be reported against.

Problem 3.16 — Repair a measurement claim

Repair this sentence: “Because the mindfulness scale is well established, our scores are valid measures of mindfulness in this population.” Identify what validity in one setting does and does not establish in another.

Worked Solutions

Each solution identifies the decisive cue, reasoning path, relevant evidence, rejected alternative, claim boundary, and a transfer cue.

Level 1 Solutions

Solution 3.1

Construct → **operational definition** → **instrument** → **administration** → **raw response** → **coding rule** → **stored value**. An analysis can inspect the stored values and, if a codebook exists, the coding rule and permitted range. It must take the **construct-to-instrument link on trust**, because no property of the numbers reveals whether the instrument measures what it claims. **Transfer cue**: the weakest link in the chain is the one no computation can check.

Solution 3.2

Role is what the variable does in the design—case identifier, outcome, condition, covariate—and comes from the research question. **Type** is what the file stores. Practice minutes is a covariate stored as numeric; it could be promoted to an outcome by a different question with no change to the file, or stored as text with no change to its role. **Decisive cue**: role is a scientific decision, type is a storage decision.

Solution 3.3

An **identifier** may be counted and checked for duplicates; its mean is meaningless. An **ordered category** may be reported as a median or as frequencies; its mean is not generally interpretable, because the spacing between categories is unknown. A **continuous score** may be summarised by a mean and standard deviation; reporting its mode alone would usually waste most of the information.

Level 2 Solutions

Solution 3.4

For the mindfulness variables, **zero** missing values in every complete pair; for practice minutes, some. The verified record shows exactly this: 72 cases for each mindfulness variable and 68 for practice minutes. If the mindfulness variables had missing values, the declared policy would have been violated, which is a finding in itself—it would mean either that the policy was unrealistic or that the collection did not follow it, and either way the report must say so rather than quietly analysing complete cases.

Solution 3.5

Software would permit every numeric operation on it: a mean, a standard deviation, a correlation with the outcome, even its use as a predictor. The specific nonsense is that **participant 72 would count as twice participant 36**, so any relationship involving the identifier would reflect the order of enrolment, which is not a property of anyone. Recording the level as an identifier is what makes the operation unavailable. **Transfer cue**: the measurement level is a guard rail, not a label.

Level 4 Solutions

Solution 3.6

N = 72 is consistent with the complete policy declared for both mindfulness variables. N = 68 is consistent with limited_synthetic_missingness for practice minutes. Alongside them the report must state the **declared**

policy, the **observed missing count** for each variable, and—critically—that any summary of practice minutes describes the 68 participants who reported rather than the study. **Transfer cue:** an N is only interpretable against a policy someone declared in advance.

Solution 3.7

The ranges can show **whether observed values fall inside the permitted range**: 5.000 to 31.700 minutes and 23.900 to 64.600 points are plausible for their instruments, and a value outside a permitted range would be a data-quality finding. They **cannot** show that the values are correct, that the instrument was administered properly, or that no plausible-looking value was mistyped for another plausible-looking value. **Claim ceiling:** ranges detect impossibility, not error.

Solution 3.8

The table cannot answer because **role is not a property of the data**. A Descriptives table reports whatever numeric columns exist and would look identical if every variable's role were different. The answer lives in the **data dictionary and the study protocol**, where roles are declared. Here neither of the two means shown is the covariate: 43.293 and 49.379 are the pre and post outcome scores, and practice minutes is the covariate.

Level 5 Solutions

Solution 3.9

The **identifier** may be counted, checked for uniqueness, and used to link the two occasions; nothing may be averaged over it. **Practice minutes** may be summarised by a mean and standard deviation as a continuous variable, with its sample size reported, and treated as a covariate rather than an outcome. The **mindfulness score** may be summarised the same way and, because the same instrument is used at both occasions, differenced within participants. Each follows from the measurement level and the declared role.

Solution 3.10

Reject it. Two things can no longer be checked. First, whether the **permitted range** was set from the instrument or from the observed data—a range inferred from what was collected can never flag an impossible value, because everything collected is inside it by construction. Second, whether the **missing-data policy** was a plan or a description; a policy written after seeing the missingness cannot constrain anything. **Rejected rule:** “the data will tell us what the variables are.”

Solution 3.11

It converts a **prespecified rule** into a **choice made with knowledge of the data**. Even if the handling chosen is defensible in isolation, it was selected from among alternatives whose consequences were visible, so the reported result is one of several that were implicitly available. The policy should have been fixed in the codebook; where it was not, the report should say so and show the result under more than one defensible handling.

Solution 3.12

Reject it. The instrument is a self-report mindfulness questionnaire; wellbeing is a broader construct that this instrument was not built to represent, and no property of the scores can bridge the gap. What the instrument supports is a claim about **self-reported mindfulness among the represented participants**. **Transfer cue:** name the construct the instrument names, not the one the programme is about.

Study in the Larger Evidence Program: Specifying the Mindfulness Measurement

- **What this study adds:** A versioned specification of what each variable means, what summaries it permits, and what missingness was expected — the record against which every later number can be audited.
- **What it cannot settle:** Whether the instrument represents the construct. Measurement validity is argued from evidence outside the file and is never demonstrated by the numbers in it.
- **Causal-support role:** None directly. Good measurement is a precondition for consistency evidence rather than a source of it; a well-specified variable in a one-group design still supports no causal claim.
- **Next evidence priority:** Describe what was actually collected against this specification, including the observed missing counts and the shape of each distribution.

Level 6 Solutions

Solution 3.13

Participant identifier — role: case identifier; measurement level: identifier; planned type: string; missing policy: complete, since a case without one cannot be linked; permitted values: one unique label per participant, no duplicates.

Practice minutes — role: covariate; measurement level: continuous; planned type: numeric; missing policy: limited missingness expected, because the log is self-reported and some participants will not return one; permitted range: non-negative and not exceeding the maximum daily minutes the protocol treats as plausible, with values outside it flagged for inspection rather than deleted.

Solution 3.14

The **recoding** is the first error: the codebook declares this variable an identifier of planned type string, and converting it to a number to make an operation available is defeating the guard rail rather than satisfying it. The **conclusion** is the second: 36.5 is arithmetically correct as the mean of 1 to 72, and it describes the numbering scheme rather than any participant. Labels carry order of enrolment at best and nothing at all at worst, so their average cannot “confirm a balanced sample”—balance is a property of the measured variables, not of the labels. Corrected: report the **number of participants** and, if balance matters, the distributions of the variables on which balance is being claimed. **Transfer cue:** when software refuses an operation, the question is why it refused, not how to get around it.

Solution 3.15

Missing: **who the figure describes** and **how many they are**. Corrected: “Among the 68 participants who returned a practice log, average daily practice was 20.185 minutes.” It should be reported against the declared `limited_synthetic_missingness` policy, stating that four logs were not returned and that nothing establishes those participants practised comparably.

Solution 3.16

Validity is not a property an instrument carries between settings. Evidence that a scale performs well in one population, administered one way, is evidence about that setting; it makes validity here **plausible** rather than established, and says nothing about whether administration conditions were comparable. Corrected: “The scale has established validity evidence in other populations. We report scores as self-reported mindfulness in this sample and did not collect validity evidence in this setting.” **Claim ceiling:** borrowed validity is a reason to expect, not a reason to assert.

Unit 4: Describing Distributions

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Unit Purpose

Describing a distribution is not a button. Every summary you report is a decision: which distribution answers the question, what its shape will tolerate, which observations are unusual, and what is missing. A mean and a standard deviation are the right answer to some of those questions and a misleading answer to others. This unit builds the habit of asking the four questions—center, spread, shape, unusual observations—separately, and of describing the distribution that matches the design rather than the one that matches the file.

Learning Objectives

After completing this unit, a learner should be able to:

- separate center, spread, shape, and unusual observations as four questions;
- identify which distribution answers a question about linked measurements;
- compute quartiles, an interquartile range, and outlier fences, and say what they mean;
- distinguish an unusual observation from a data error;
- choose numerical summaries that the observed shape supports;
- report missing values as part of the description rather than after it; and
- state the strongest claim that description alone permits.

Decision Map

Ask these questions in order:

1. What are the cases, and what is the quantitative outcome?
2. Does the question concern one measurement or the difference between linked measurements?
3. What is the shape of the distribution that answers the question?
4. Which center and which spread does that shape support?
5. Which observations sit far from the rest, and are they errors or evidence?
6. Which values are missing, from which variable, and for how many cases?
7. What can a description claim, and where does it stop?

Unit Study Files

Study	Data structure	Described here	Unit role
Study 02	Pre and post scores for 72 synthetic participants, plus a practice log	Mindfulness scores, their differences, and practice minutes	Shape, unusual observations, and missingness
Study 02C	Pre and post scores for 72 synthetic participants	Difference scores	Shape contrast against Study 02

Plausible Research Scenarios

The scenarios create decisions rather than documenting real research. At each **Pause and Decide**, stop reading and write an answer, reason, and confidence rating. The explanation that follows is feedback on that commitment.

Plausible Research Scenario: What the Intake Scores Look Like Before Anything Is Compared

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

A university counseling center has finished collecting questionnaires for its six-week mindfulness pilot. Nothing has been compared yet. Before any test is run, the center's fictional research team must write one page describing what it collected, because the director has asked a reasonable question: *what do these students actually look like?*

There are three quantitative variables. Each participant has a mindfulness score recorded before the interval and a second recorded after six weeks, and each was asked to keep a practice log reporting average daily practice minutes. One participant is the independent unit. The question at this stage is not whether anything changed. It is: **how are these scores distributed among the students who took part, and what should a reader be able to check?**

A description is not a warm-up for the analysis; it is the evidence a reader uses to decide whether to believe the analysis that follows.

Two descriptive questions hide in the same file. One is about the scores themselves: how high they run, how much they vary, whether they are symmetric. The other is about the change each participant showed, which lives in a different distribution. Two score columns each yield a mean and a standard deviation, but the average of the individual post-minus-pre differences is not something either column shows.

Study 02 supplies a distribution of differences that is mostly positive, with a tail running into negative values. Study 02C preserves the same intake scores and design while representing a smaller, more symmetric set of changes. Describing both before either is tested is what makes the later comparison honest: the shape of the evidence was fixed before anyone knew whether the result was welcome.

Pause and Decide

Before reading further, decide which distributions this report must describe. Then decide, for each, whether you would lead with a mean and standard deviation or with a median and interquartile range, and name the feature of the data that would change your answer. Finally, state what a reader would be entitled to conclude from your description alone.

Why This Model?

The descriptive model is a set of summaries chosen to match a shape, not a fixed template applied to every column. A mean and a standard deviation describe center and spread well when the distribution is roughly symmetric and no observation exerts extreme influence. A median and an interquartile range describe the same two features in a way a long tail cannot drag around.

Reporting both when they disagree is not indecision. The gap between a mean and a median is itself a description of shape, and a reader shown only one cannot recover it.

A paired-samples test is not the model here; no comparison has been claimed. Presenting the pre and post distributions side by side as two groups would also be wrong: the same students appear in both, so two marginal summaries describe the same people twice rather than how they changed.

What Would a Favorable Result Look Like?

There is no favorable result to hope for, and that is the point. A description succeeds when a reader can reconstruct the shape of the evidence, not when the numbers are encouraging. One that reveals a problem is worth more than one that does not: a skew, a low outlier, or a missing variable changes what the team does next, and finding it now is cheaper than finding it later.

Best Defensible Claim

Statistical evidence: The summaries describe the distributions of intake scores, post scores, and post-minus-pre differences among the represented participants, and how spread out and symmetric each one is.

Causal identification: None is attempted or available. Description compares nothing to a counterfactual, so nothing here speaks to why any value takes the level it does.

Claim ceiling: The team may describe the represented participants. It may not estimate a population value, claim the observed spread is what a future cohort would show, or treat a gap between two summaries as a finding.

What Must Be True?

Scores must be recorded on a comparable scale at both occasions, participant matching must be correct, and out-of-range values must have been investigated rather than silently retained. No small set of observations may dominate a mean or a standard deviation. The number of cases behind every variable must be reported, because a variable with missing entries describes fewer people than the file appears to contain.

Defensibility Procedures

Decide which distributions to describe before computing anything, plot each one rather than relying on numerical summaries, report the number of cases behind every summary, investigate values far from the rest without deleting them, and state the outcome definition and scoring procedure. Where the mean and median disagree, report both.

Next evidence priority: With the shapes on record, the next question is how much an estimate of this kind would move if the pilot were repeated—a question about sampling variability, not about these particular students.

Claim Audit

Repair this claim before continuing: “The average intake score was about 43, so the typical student entering this program has a mindfulness score of 43.” Distinguish a sample summary from a population claim, name what “typical” assumes about shape, and say what a reader would need in order to check the sentence.

Plausible Research Scenario: Missing Practice Logs and Three Low Difference Scores

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

The team now looks at two things its first draft glossed over.

The first is the practice log. Not every participant recorded average daily practice minutes, so that summary describes fewer people than the mindfulness summaries do—and a reader seeing both in one table without sample sizes will assume otherwise.

The second is three participants whose scores fell far enough to sit outside the usual fences. A junior colleague proposes removing them as “clearly outliers” dragging the mean down. The proposal is attractive and wrong, for a reason worth stating precisely: an outlier is a claim about a value's position; an error is a claim about how it came to exist. The first is visible in the data, the second is not.

Pause and Decide

Decide what the report may and may not say about the participants whose practice logs are absent. Then decide what happens to the three low difference scores, and state the evidence required before removing any observation.

Why This Model?

The descriptive model must carry its own sample sizes and missingness. A mean for practice minutes with no count beside it describes an unnamed subgroup as though it were the whole pilot.

Outlier fences are a screening rule, not a deletion rule. Deleting inconvenient observations changes the estimand from “average change among participants” to “average change among participants whose change was not extreme”—a quantity nobody asked about.

What Would a Favorable Result Look Like?

Complete logs and no extreme values would make the description simpler, but neither is a result to be achieved by editing. A description that reports both is stronger than a tidier one whose tidiness cannot be checked.

Best Defensible Claim

Statistical evidence: Practice minutes are described for the participants who reported them; differences for all complete pairs, including the unusual ones.

Causal identification: None. Nothing shows why some logs are absent, and the reasons could relate to how much those participants practiced.

Claim ceiling: The team may describe the reporting participants, and must say so. It may not present the practice summary as describing everyone.

What Must Be True?

Missing entries must be identified by variable and counted, and the reason treated as unknown rather than harmless. Values outside the fences must first be checked against the scale's permitted range and the data-entry record.

Defensibility Procedures

Report the number of cases behind every summary, count missing values by variable, inspect flagged observations individually, document any exclusion with its reason and effect, and repeat the description without it.

Next evidence priority: A collection procedure that does not depend on participants remembering to log, so practice can be described for everyone rather than for a self-selected subset.

Claim Audit

Repair this claim: “Participants practiced about twenty minutes a day, and after removing three outliers the average improvement was even larger than we thought.” Say who the figure describes, what the removal did to the estimand, and what would have justified it.

Problems

Do not consult the solutions until you have recorded an answer, reason, and confidence rating for every attempted item.

Level 1: Recall and Recognize

Problem 4.1 — Four questions, not one

Name the four separate questions a description of a quantitative distribution should answer. For each, name one summary that answers it.

Problem 4.2 — Choose the distribution before the summary

Study 02 contains a pre score and a post score for each participant. A colleague computes a mean and standard deviation for each column and says the description is finished. Which distribution has not been described, and why can it not be recovered from the two column summaries?

Problem 4.3 — Define the interquartile range

Define the first quartile, the third quartile, and the interquartile range in terms of the ordered observations. State what the interquartile range measures that the range does not.

Problem 4.4 — Outlier and error are different claims

A value sits outside the lower outlier fence. State what that establishes and what it does not. Name two pieces of evidence that would be needed before calling the value an error.

Level 2: Predict Before Clicking

Problem 4.5 — Predict the mean-median gap

Study 02's difference scores have a sample skewness of -0.606. Before looking at any summary, predict whether the mean of those differences will be above or below their median, and explain the reasoning that connects a negative skew to your prediction.

Problem 4.6 — Predict the spread of the differences

Study 02's pre scores have a standard deviation of 8.932 and its post scores 10.060. Predict whether the standard deviation of the post-minus-pre differences will be larger or smaller than both, and name the feature of the design that drives your prediction.

Problem 4.7 — Predict the effect of one bad value

Suppose one participant's post score were mistyped as 714 instead of 71.4. Predict the direction and rough severity of the effect on the mean, the median, the standard deviation, and the interquartile range of the post scores. Rank the four summaries from most to least affected.

Problem 4.8 — Predict what the practice summary describes

Every participant has a mindfulness score, but not every participant reported practice minutes. Predict what happens to the number of cases behind the practice summary, and state what a reader would wrongly conclude if both summaries were printed in one table without their sample sizes.

Level 3: Calculate and Explain

Problem 4.9 — Compute the interquartile range

Study 02's difference scores have a first quartile of 3.825 and a third quartile of 9.175. Compute the interquartile range using $IQR = Q_3 - Q_1$ and state, in one sentence, what the value describes.

Problem 4.10 — Compute the outlier fences

Using the same quartiles, compute the lower and upper fences from $Q_1 - 1.5 \times IQR$ and $Q_3 + 1.5 \times IQR$. State what a value beyond either fence has and has not been shown to be.

Problem 4.11 — Place the extreme value

Study 02's smallest difference is -8.700, its mean difference is 6.086, and the standard deviation of the differences is 4.993. Compute how many standard deviations below the mean that observation lies. Explain why this quantity is a description of position and not a test.

Problem 4.12 — The mean of differences and the difference of means

Study 02's pre mean is 43.293 and its post mean is 49.379; the mean of the individual post-minus-pre differences is 6.086. Verify the arithmetic relationship, then explain why the same relationship does *not* hold for the standard deviations.

Level 4: Read jamovi Output

Problem 4.13 — Read a descriptives table

A jamovi Descriptives table for Study 02 reports a mean of 43.293 and a median of 43.850 for the pre scores, and a mean of 49.379 and a median of 48.350 for the post scores. Describe the shape each pair implies, and say which distribution shows the larger departure from symmetry.

Problem 4.14 — Read shape statistics

For Study 02's difference scores, jamovi reports a skewness of -0.606 and an excess kurtosis of 0.755. Interpret each in words a reader without a statistics background could follow, and state what neither of them settles.

Problem 4.15 — Read a row with missing values

A Descriptives table shows $N = 68$ for practice minutes while every mindfulness row shows 72, with a mean of 20.185, a median of 20.400, and a standard deviation of 5.808. Write the sentence describing practice minutes that this row does and does not support.

Problem 4.16 — Compare two difference distributions

Study 02's differences have a mean of 6.086 and a median of 6.300. Study 02C's have a mean of 0.400 and a median of 0.400, with a standard deviation of 5.198. Compare the two distributions on center, spread, and symmetry, and state which comparison the descriptions alone do not license.

Level 5: Choose the Model

Problem 4.17 — Mean and standard deviation, or median and interquartile range?

For Study 02's difference scores, decide which pair you would lead with in a one-page report and defend the choice using the shape evidence. Then state the circumstance under which you would report both.

Problem 4.18 — Reject the deletion

A colleague proposes deleting the three difference scores below the lower fence because “they are outliers.” Reject the proposal, and describe the specific evidence that *would* justify removing an observation.

Problem 4.19 — Reject two marginal summaries

A draft report describes the pre and post distributions separately and concludes that the two summaries answer the question about change. Explain what the draft has described instead, and name the summary it is missing.

Problem 4.20 — Reject the two-bar figure

A draft figure shows two bars, one for the pre mean and one for the post mean, with no other information. Explain what a reader cannot see, and name a figure that would answer the same question honestly.

Level 6: Report and Critique**Problem 4.21 — Write the description**

Write two sentences describing Study 02's difference scores for a non-specialist reader. Include center, spread, and shape, and make it possible for the reader to tell how many participants the description covers.

Problem 4.22 — Repair a description that became a conclusion

Repair this sentence: “Scores rose by an average of about six points, so the mindfulness program is working.” Identify every step where the sentence exceeds what a description can support.

Problem 4.23 — Repair a silent exclusion

Repair this sentence: “After removing the three outliers, the average change was 6.674 points.” The arithmetic is correct. State what the reader has not been told and what the sentence now describes.

Problem 4.24 — Repair a summary of an unnamed subgroup

Repair this sentence: “Participants practiced for an average of 20.185 minutes a day.” Say who the figure actually describes and what would have to be true for the original sentence to be defensible.

Worked Solutions

Each solution identifies the decisive cue, reasoning path, relevant evidence, rejected alternative, claim boundary, and a transfer cue.

Level 1 Solutions

Solution 4.1

The four questions are **center** (where the distribution sits), **spread** (how much values differ from one another), **shape** (whether the distribution is symmetric, skewed, or heavy-tailed), and **unusual observations** (whether any values sit far from the rest). A mean or median answers the first; a standard deviation or interquartile range the second; skewness, kurtosis, and a plot the third; and quartile-based fences with a plot the fourth. **Transfer cue:** if a report answers only the first two, the reader cannot tell whether to trust them.

Solution 4.2

The distribution of **post-minus-pre differences** has not been described. It cannot be recovered from the two column summaries because the mean of the differences equals the difference of the means, but nothing else does: the spread of the differences depends on how strongly the two scores move together within participants, and a correlation is not visible in either marginal summary. Two students could show identical column summaries and completely different patterns of individual change.

Solution 4.3

Order the observations. The **first quartile** is the value below which about a quarter of them fall, the **third quartile** the value below which about three-quarters fall, and the **interquartile range** is $Q_3 - Q_1$, the width of the middle half. The range depends on the two most extreme observations and can be changed entirely by one value; the interquartile range describes the bulk of the distribution and is unaffected by how far the extremes reach.

Solution 4.4

Sitting beyond a fence establishes only that the value is **far from the middle half relative to the spread of the middle half**. It does not establish that the value is wrong, impossible, or uninformative. Before calling it an error you would need evidence about how the value came to exist: that it falls outside the instrument's permitted range, or that the data-entry or scoring record shows a specific fault. **Claim ceiling:** position is visible in the data; provenance is not.

Level 2 Solutions

Solution 4.5

The mean should sit **below** the median. A negative skew means the longer tail runs toward low values; the mean is computed from the values themselves and is pulled toward that tail, while the median depends only on rank and stays with the bulk. The verified summaries show a mean of 6.086 against a median of 6.300, which is the predicted direction. **Transfer cue:** the sign of the mean-median gap is a shape statement.

Solution 4.6

Smaller than both. The two scores are measured on the same participants, so a student who starts high tends to finish high; that shared person-level variation is subtracted out when the difference is formed. The verified value is 4.993, below both 8.932 and 10.060, and the recorded pre-post correlation of 0.868 is what drives it. If

the two columns had come from unrelated people, the spread of the differences would have been *larger* than either. **Decisive cue:** pairing removes between-person variation from the quantity being described.

Solution 4.7

The **mean** moves most: one value inflated by roughly 640 points raises a 72-case mean by about nine points. The **standard deviation** moves even more in relative terms, because the deviation is squared. The **median** barely moves—it shifts by at most one rank position. The **interquartile range** is essentially unaffected, since the corrupted value simply becomes the largest observation and the quartiles are unchanged. Ranked from most to least affected: standard deviation, mean, median, interquartile range. **Transfer cue:** the summaries that are easiest to interpret are the ones that a single bad value can destroy.

Solution 4.8

The practice summary is computed from 68 cases rather than 72, because four participants did not report practice minutes. Printed in one table without sample sizes, a reader would conclude that the practice figure describes the same students as the mindfulness figures. It does not: it describes the subgroup who reported, and nothing in the description rules out that those participants practiced differently from those who did not.

Study in the Larger Evidence Program: Describing the Mindfulness Intake

- **What this study adds:** A complete, checkable account of what was measured, on how many participants, with what shape, and with which observations unusual—the evidence a reader needs before any comparison is credible.
- **What it cannot settle:** Anything comparative or causal. Description does not estimate a population value, and it does not establish that any distribution would look like this again.
- **Causal-support role:** None directly; it is measurement evidence, and it supports later responsiveness claims only by showing that the outcome was recorded in a way worth comparing.
- **Next evidence priority:** Quantify how much a summary of this kind would move under repetition, which is the sampling-variability question Unit 5 takes up.

Level 3 Solutions

Solution 4.9

$IQR = 9.175 - 3.825 = 5.350$ points. The middle half of the participants changed by amounts spanning 5.350 points. It is a description of the bulk of the distribution and says nothing about how far the extremes reach.

Solution 4.10

Lower fence: $3.825 - 1.5 \times 5.350 = -4.200$. Upper fence: $9.175 + 1.5 \times 5.350 = 17.200$. A value beyond either has been shown to be **unusually far from the middle half**, and nothing more. It has not been shown to be an error, an invalid measurement, or a case that should be excluded.

Solution 4.11

$(-8.700 - 6.086) / 4.993 = -2.96$ standard deviations below the mean, approximately. This locates the observation within the distribution it belongs to. It is not a test: no null hypothesis has been stated, no sampling distribution invoked, and no probability computed. A value about three standard deviations from the mean is unremarkable in a sample of 72 with a long tail. **Transfer cue:** a standardized position is a description, not a verdict.

Solution 4.12

$49.379 - 43.293 = 6.086$, matching the mean of the individual differences: the mean is a linear operation, so the mean of the differences equals the difference of the means. The relationship fails for standard deviations because variance is not linear—the variance of a difference depends on the two variances *and* on the covariance between them. That covariance term is exactly the information the paired design supplies and the two marginal summaries omit.

Level 4 Solutions

Solution 4.13

For the pre scores the mean of 43.293 sits slightly below the median of 43.850, implying a mild negative skew. For the post scores the mean of 49.379 sits above the median of 48.350, implying a mild positive skew. The **post** distribution shows the larger gap, about 1.0 point against about 0.6, so it departs further from symmetry. Both gaps are small relative to the spread, so neither by itself argues against a mean.

Solution 4.14

A skewness of -0.606 says the distribution of changes leans left: most participants cluster on the higher side, with a thinner tail of larger decreases pulling the average down. An excess kurtosis of 0.755 says the

distribution has somewhat heavier tails than a normal curve—extreme values are a little more common than a bell shape would predict. Neither settles whether the mean is an appropriate summary, whether any observation is an error, or whether the change matters; those are judgments about purpose and provenance, not about shape.

Solution 4.15

The row supports: “Among the 68 participants who reported a practice log, average daily practice was 20.185 minutes, with a median of 20.400 and a standard deviation of 5.808.” It does not support any sentence beginning “Participants practiced...”, because the four non-reporting participants are not described and there is no evidence that they resemble the ones who are. **Transfer cue:** the sample size belongs in the sentence, not in a footnote.

Solution 4.16

Center: Study 02's changes are centered near 6.086 points; Study 02C's near 0.400. **Spread:** the two are similar, 4.993 against 5.198, so the distributions are about equally wide. **Symmetry:** Study 02's mean sits below its median (6.086 against 6.300), indicating a left lean; Study 02C's mean and median coincide at 0.400, indicating a distribution balanced around its center. What the descriptions do not license is any statement that one program worked better than the other: these are two synthetic representations of the same design, and description supplies no basis for comparing them as evidence.

Level 5 Solutions

Solution 4.17

Lead with the **mean and standard deviation**. The shape evidence supports it: a skewness of -0.606 is a mild lean rather than a severe one, the mean of 6.086 and median of 6.300 are close on a scale whose middle half spans 5.350 points, and the quantity a reader wants—average change—is the mean. Report **both** pairs whenever the mean and median disagree enough that a reader's impression would change, or when a small number of observations is visibly driving the mean. **Rejected rule:** “skewed data always require a median.”

Solution 4.18

Reject it. The three values are unusual in position, which is all a fence establishes, and they are the participants whose experience most differs from the rest—the least disposable cases in the file. Removing them silently changes the estimand from average change among participants to average change among participants who did not decline much. Removal would be justified only by evidence about provenance: a score outside the instrument's permitted range, a documented data-entry or scoring fault, or a recorded protocol violation for that case. Any exclusion is then reported with its reason and its effect.

Solution 4.19

The draft has described **where the group stood at two moments**, not how individuals moved between them. Two very different patterns of individual change—everyone improving slightly, or most improving substantially while some decline—produce the same pair of marginal summaries. The missing summary is the distribution of post-minus-pre differences: its center, its spread, and its shape. **Decisive cue:** the design links measurements within a person, so the description must too.

Solution 4.20

The reader cannot see spread, shape, sample size, uncertainty, or how many participants moved in each direction; two bars encode two numbers and imply a comparison the figure does not support. An honest alternative is a plot of the individual difference scores—a histogram or dot plot with the mean and median marked—which shows the whole distribution the summary came from.

Level 6 Solutions

Solution 4.21

“Among the 72 participants with complete pairs, mindfulness scores changed by an average of 6.086 points from before to after the six-week interval, with a standard deviation of 4.993 points. The middle half of participants changed by between 3.825 and 9.175 points, and the distribution leans slightly toward larger decreases, with three participants declining by more than the lower fence of -4.200 points.” The sample size, center, spread, shape, and unusual observations are all recoverable.

Solution 4.22

“Scores rose by an average of about six points” is defensible as a description of these participants. “So the mindfulness program is working” is not: it converts a described change into a causal effect with no comparison condition, treats the represented participants as a population, and takes a favorable direction as proof. A defensible version: “Among the represented participants, scores rose by an average of 6.086 points over the interval. Because there is no comparison group, this describes change and does not show what caused it.”

Solution 4.23

The arithmetic is right and the sentence is still indefensible, which is the point. Dropping the three differences below the -4.200 fence moves the average from 6.086 to 6.674, an inflation of about six-tenths of a point. The reader has not been told which observations were removed, why, on what evidence, or what the average was before removal—so the inflation is invisible. The sentence now describes average change among participants whose change was not extreme: a quantity nobody set out to estimate, biased upward by construction whenever the removed values are the low ones. A defensible version reports the full-sample figure first, then the sensitivity check, with the reason for each exclusion. **Transfer cue:** correct arithmetic on the wrong estimand is still the wrong answer.

Solution 4.24

The figure describes the 68 participants who reported a practice log, not the pilot. A defensible version: “Among the 68 participants who reported practice, average daily practice was 20.185 minutes.” The original would be defensible only if every participant had reported, or if there were positive evidence that the non-reporting participants practiced comparably—and no such evidence exists here. Since reporting a log may itself be related to practicing, the gap is not safely ignorable. **Transfer cue:** name the group a number describes, every time.

Study in the Larger Evidence Program: Reporting What Is Missing and What Is Unusual

- **What this study adds:** A record of how complete the evidence is and which observations are unusual, so that a later comparison can be audited rather than taken on trust.
- **What it cannot settle:** Why the four practice logs are absent, or whether the participants who reported resemble those who did not.
- **Causal-support role:** Consistency evidence only, and weakly: it establishes that the measurement record is intact enough for repeated study, not that any relationship holds.
- **Next evidence priority:** A collection procedure that does not rely on self-reported logs, so that practice can be described for every participant rather than for a self-selected subset.

PART II

Uncertainty and evidence

Unit 5: Sampling Variability and Standard Error

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Unit Purpose

An estimate computed from a sample would have come out differently if a different sample had been drawn. That is not a defect and not a mistake; it is the reason a standard error exists. This unit builds the habit of asking, before interpreting any difference between two numbers, how much the numbers would move on their own—and of recognising that a standard error describes the behaviour of an estimate, not the spread of the people it was computed from.

Learning Objectives

After completing this unit, a learner should be able to:

- distinguish the distribution of observations from the distribution of an estimate;
- explain what a standard error measures and what it does not;
- connect sample size to the precision of an estimate through $SE = s / \sqrt{n}$;
- predict how much a summary would move under repetition before comparing two summaries;
- read a simulated sampling distribution and relate it to the formula-based standard error;
- reject the inference that two different sample means indicate a real change; and
- state what sampling variability can and cannot explain.

Decision Map

Ask these questions in order:

1. What quantity is being estimated, and from how many independent cases?
2. Is the number in front of me an observation, a summary, or an estimate?
3. How much would this estimate move if the study were repeated?
4. Does the standard error describe the people or the summary?
5. Is the gap between two numbers larger than the movement I should expect anyway?
6. What would change that expected movement—sample size, spread, or design?
7. What does sampling variability explain, and what remains to be explained?

Unit Study Files

Study	Data structure	Used here for	Unit role
Study 02	Post-practice scores for 72 synthetic participants	Observed spread, standard error, and a simulated sampling distribution	Estimate behaviour under repetition
Study 02	The same scores, resampled at two sample sizes	Comparing 72-case and 36-case sampling distributions	The role of sample size

Plausible Research Scenarios

The scenarios create decisions rather than documenting real research. At each **Pause and Decide**, stop reading and write an answer, reason, and confidence rating. The explanation that follows is feedback on that commitment.

Plausible Research Scenario: The Second Cohort Came Out Different

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

The counseling center has run its six-week mindfulness pilot a second time with a new group of students. Nothing about the program was changed: same materials, same facilitator, same questionnaire, same six-week interval.

The second cohort's average post-program score came out about a point and a half below the first cohort's. The director, reading the two numbers side by side, asks whether something went wrong—a distracted cohort, a drift in how the questionnaire was administered. Each is a real possibility. But the fictional research team must answer a prior question first, and it is the question this unit exists for: **how far apart would two cohort averages be expected to fall even if nothing had changed?**

Answering it requires a shift in what is being described. The previous unit described participants: how their scores were distributed. The question now is about a *summary*. If the same program were run again and again with different groups of 72 students drawn the same way, the average would land somewhere slightly different each time, and those averages would form a distribution of their own. Its spread is what tells the director whether a point and a half is remarkable.

There are two routes to that spread, and both are worth seeing. One is a formula: divide the spread of the individual scores by the square root of the number of participants. The other is to simulate—draw new samples of 72 from the observed scores, compute the average each time, and look at how those ten thousand averages scatter. When the two agree, the formula stops being an incantation and becomes a description of something the learner has watched happen.

Neither route says whether the procedure was any good, whether the students represent anyone, or whether a real change occurred between cohorts. A standard error is a statement about arithmetic stability, not about validity.

Pause and Decide

Decide whether the standard error you would compute here describes the participants or the average, and how you would know from the number itself. Then predict whether a gap of about a point and a half between two

cohort averages should be treated as surprising. Finally, name one explanation for the gap that sampling variability could never rule out.

Why This Model?

The model is the sampling distribution of the mean: the distribution an estimate would follow if the study were repeated. The standard error is its spread, and $SE = s / \sqrt{n}$ estimates that spread from a single sample. Dividing by the square root of the sample size is what makes a larger study more precise, and also what makes precision expensive: halving a standard error takes four times the participants.

The simulated distribution is a teaching device, not a replacement. It shows the same quantity from a direction learners can watch.

Two alternatives are wrong here. Testing the two cohorts against each other answers a different question and would still require knowing how much an average moves. And reporting the *spread of scores* where the spread of the *average* belongs mistakes variation among students for uncertainty in a summary of them.

What Would a Favorable Result Look Like?

There is nothing to hope for in a standard error. A small one means the estimate is stable under repetition; a large one means the study cannot resolve much, and knowing that early is better than discovering it after a conclusion has been announced.

The team would like the simulated and formula-based values to agree, because a large disagreement would signal that something about the distribution makes the formula a poor description here.

Best Defensible Claim

Statistical evidence: The standard error quantifies how much the average post-program score would vary across repeated samples of the same size drawn the same way.

Causal identification: None. Sampling variability describes arithmetic under repetition and cannot distinguish a real change between cohorts from an incidental one.

Claim ceiling: The team may say how precisely the average is estimated and whether an observed gap is within the range of ordinary variation. It may not conclude that nothing changed, and it may not treat the represented students as a random sample of any wider population.

What Must Be True?

Participants must contribute independent observations; the formula assumes it, and clustered or repeated observations would make the computed value too small. The sample size must be the number of independent cases, not of measurements. For a single spread to summarize the estimate's distribution usefully, no small set of observations may dominate the average.

Defensibility Procedures

State the estimand and the sample size behind it, report a standard error beside every estimate, say which quantity each reported spread describes, keep the simulation's seed and repetition count on record, and compare the simulated and formula-based values rather than reporting whichever is smaller.

Next evidence priority: Turn the standard error into an interval, so the report states a range of values compatible with the evidence rather than a single number with an unquantified margin.

Claim Audit

Repair this claim: "The second cohort scored lower, so the program is losing effectiveness." Distinguish a difference between two estimates from a difference between two underlying quantities, name what would have to be shown instead, and say what sampling variability does not rule out.

Plausible Research Scenario: Running the Next Pilot With Half the Students

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

Facilitator time is the center's binding constraint, and the director proposes running the next round with 36 students instead of 72, framing the cost as “a bit of precision.” The team is asked whether that is fair.

It is not, though the correction is not the obvious one either. Halving the sample does not double the standard error; because sample size enters through its square root, the error grows by a factor of about 1.4. Whether that is “a bit” depends on the study's purpose. For describing a cohort it may be tolerable; for detecting a change the center cares about, it may decide whether the study can answer its question at all.

The team can show this rather than assert it: simulating repeated samples at both sizes yields two sampling distributions from the same scores, and the wider one is visibly wider.

Pause and Decide

Predict how much larger the standard error becomes when the sample is halved, and state the reasoning behind your factor. Then decide what you would tell the director: what is lost, and what is not.

Why This Model?

The comparison is between two sampling distributions of the same estimate, differing only in sample size. Holding the scores fixed isolates the effect of n : any difference in spread comes from sample size alone.

The formula alone would satisfy a statistician and not a director; the simulation alone would leave a learner unable to generalize past the two sizes simulated.

What Would a Favorable Result Look Like?

A favorable answer is one the center can act on, not one that endorses the proposal. If the smaller study can still resolve differences of the size that matter, the saving is real. If not, the cheaper study buys less than it costs.

Best Defensible Claim

Statistical evidence: The two simulated distributions quantify how much the estimate would move at each sample size, and the ratio of their spreads follows the square-root relationship.

Causal identification: None is at issue; this concerns the precision of a procedure, not any effect.

Claim ceiling: The team may state how much precision is lost, not that the smaller study would fail—which depends on what difference must be detected.

What Must Be True?

The two designs must be identical apart from sample size, observations must stay independent at both sizes, and the smaller sample must be drawn by the same procedure rather than by taking whoever is convenient.

Defensibility Procedures

Fix the estimand before comparing designs, hold everything but sample size constant, record the seed and repetition count, report both distributions rather than the preferred one, and state the difference the study must resolve.

Next evidence priority: Decide what magnitude of change would matter here, since precision is only interpretable against a difference worth detecting.

Claim Audit

Repair this claim: “Using 36 students instead of 72 will only cost a bit of precision.” Quantify the change, name what determines whether it matters, and identify the question the sentence assumes answered.

Problems

Do not consult the solutions until you have recorded an answer, reason, and confidence rating for every attempted item.

Level 1: Recall and Recognize

Problem 5.1 — Two distributions, not one

Distinguish the distribution of the observed scores from the sampling distribution of their mean. State what each one's spread describes.

Problem 5.2 — What a standard error is the spread of

A report gives a mean of 49.379 with a standard error of 1.186 and a standard deviation of 10.060. Say precisely what each of the last two numbers is the spread of, and which one shrinks as the study grows.

Problem 5.3 — Identify the independent units in the formula

The formula $SE = s / \sqrt{n}$ contains a sample size. For a paired study with 72 participants each measured twice, state what n must be and what would go wrong if the number of measurements were used instead.

Problem 5.4 — Name what a standard error cannot fix

A study has a very small standard error and a convenience sample from one setting. Name two things the small standard error does not establish.

Level 2: Predict Before Clicking

Problem 5.5 — Predict the effect of a larger sample

Two studies measure the same outcome with the same spread of scores, one with 72 participants and one with 288. Predict the ratio of their standard errors before computing anything, and state the rule you used.

Problem 5.6 — Predict which spread the simulation reproduces

Ten thousand samples of 72 are drawn from Study 02's post scores and a mean is computed for each. Predict whether the spread of those ten thousand means will be closer to 10.060 or to 1.186, and explain why.

Problem 5.7 — Predict the centre of the sampling distribution

Predict roughly where the average of those ten thousand resampled means will fall, relative to the original sample mean of 49.379. Explain what property of the mean produces your prediction.

Problem 5.8 — Predict the cost of halving the sample

Predict the factor by which the standard error grows when the sample size falls from 72 to 36. State the factor and the reasoning, then say what would have to happen to the sample size to halve the standard error instead.

Level 3: Calculate and Explain

Problem 5.9 — Compute a standard error from the spread

Study 02's post scores have a standard deviation of 10.060 across 72 participants. Use $SE = s / \sqrt{n}$ to compute the standard error, and state which quantity's variability the answer describes.

Problem 5.10 — Recompute at the smaller sample size

Using the same standard deviation of 10.060, compute the standard error that would apply to a sample of 36. Compare it with your answer to the previous problem and express the change as a ratio.

Problem 5.11 — Compute a paired standard error

Study 02's post-minus-pre differences have a standard deviation of 4.993 across 72 pairs. Compute the standard error of the mean difference and explain why the separate score standard deviations are not used.

Problem 5.12 — Compare the simulated and formula-based values

At a sample size of 72 the simulation gives an empirical standard error of 1.156 while the formula gives 1.186. Compute the ratio of the two, and explain what their closeness demonstrates and what a large gap would have indicated.

Level 4: Read jamovi Output**Problem 5.13 — Read a descriptives row as an estimate**

A Descriptives table reports a mean of 49.379, a standard deviation of 10.060, and a standard error of 1.186 for 72 participants. Write the sentence about precision that this row supports, and the sentence it does not.

Problem 5.14 — Read a simulated sampling distribution

The 72-case simulation places 5860 of its ten thousand resampled means between 48 and 50, and only 19 below 46. Describe the shape and concentration of the distribution, and say what the counts tell a reader that the standard error alone does not.

Problem 5.15 — Read two distributions at different sample sizes

The 36-case simulation reports an empirical standard error of 1.646 and a middle 95% range from 46.233 to 52.653. The 72-case simulation reports 1.156 and a range from 47.164 to 51.710. Compare the two and state what the comparison isolates.

Problem 5.16 — Two standard errors from one file

For Study 02, jamovi reports a standard error of 1.186 for the post-score mean and 0.588 for the mean post-minus-pre difference. Both come from 72 participants. Explain what each describes and why the second is so much smaller.

Level 5: Choose the Model**Problem 5.17 — Standard deviation or standard error?**

A results section needs one number beside the mean. Decide which to report when the audience wants to know how variable the students were, and which when they want to know how well the average is pinned down. Justify each.

Problem 5.18 — Reject the two-number comparison

Two cohort averages differ by about a point and a half. Explain why comparing them directly is the wrong first move, and name the quantity that has to be computed before the gap can be interpreted.

Problem 5.19 — Reject the count of measurements

A colleague computes a standard error using 144 as the sample size for Study 02, reasoning that there are 144 scores. Reject the choice and state the consequence for the reported precision.

Problem 5.20 — Reject precision as a substitute for validity

A study reports a very small standard error from a large convenience sample and concludes that its estimate is trustworthy. Identify the substitution being made and name what would actually be required.

Level 6: Report and Critique

Problem 5.21 — Report an estimate with its precision

Write one sentence reporting Study 02's mean post-program score with its precision, in a form a non-specialist can read, that makes clear how many participants it rests on.

Problem 5.22 — Repair a comparison of two cohorts

Repair this sentence: “The second cohort averaged a point and a half lower, so the program's effect is fading.” Name each step that exceeds the evidence.

Problem 5.23 — Repair a mislabelled spread

Repair this sentence: “Scores averaged 49.379 with a standard error of 10.060, so most students scored within about ten points of the average.” Identify the confusion and correct both halves.

Problem 5.24 — Repair a claim about a smaller study

Repair this sentence: “Cutting the sample from 72 to 36 will barely change anything.” Quantify what changes, and state the question that determines whether the change matters.

Worked Solutions

Each solution identifies the decisive cue, reasoning path, relevant evidence, rejected alternative, claim boundary, and a transfer cue.

Level 1 Solutions

Solution 5.1

The **distribution of observed scores** describes how the 72 participants differ from one another; its spread is a standard deviation. The **sampling distribution of the mean** describes how the average itself would vary across repeated samples of the same size; its spread is a standard error. One is about people, the other about a summary of people. **Transfer cue:** ask what a spread is the spread *of* before interpreting it.

Solution 5.2

The **standard deviation of 10.060** is the spread of the individual scores: it says students differ from one another by roughly ten points. The **standard error of 1.186** is the spread of the average: it says that a repeat of the same study would produce an average about a point away, give or take. Only the standard error shrinks as the study grows; the standard deviation estimates a feature of the students and does not depend on how many were measured.

Solution 5.3

n must be **72**, the number of independent participants. Using 144 would treat each participant's two measurements as two independent sources of information, when the two are linked. The computed standard error would be too small by a factor of about the square root of two, and every interval built from it would be too narrow. **Decisive cue:** the formula counts independent cases, not cells.

Solution 5.4

It does not establish that the estimate is **unbiased**, and it does not establish that it **generalizes** to anyone outside the setting sampled. A standard error describes movement under repetition of the same flawed procedure; repeating a biased procedure precisely gives a precisely wrong answer. **Claim ceiling:** precision is not accuracy, and neither is representativeness.

Level 2 Solutions

Solution 5.5

The larger study's standard error is **half** the smaller one's. Sample size enters through its square root, and 288 is four times 72, so the standard error falls by a factor of $\sqrt{4} = 2$. **Transfer cue:** precision improves with the square root of effort, which is why the second half of a sample buys much less than the first.

Solution 5.6

Much closer to **1.186**. Each of the ten thousand values is itself an average of 72 scores, and averaging is what removes variation: individual scores scatter by about ten points, but averages of 72 of them scatter by about one. The simulation is estimating the same quantity the formula estimates, from the other direction.

Solution 5.7

Very close to 49.379. The mean of the sampling distribution of the mean is the quantity being resampled from, so repeated averaging recovers the original average rather than drifting away from it. The simulation's recorded

average of resampled means is 49.399, about two-hundredths of a point away. **Transfer cue:** sampling variability spreads an estimate out; it does not systematically move it.

Solution 5.8

It grows by a factor of about **1.4**, not 2. Halving n multiplies the standard error by $\sqrt{2} \approx 1.414$. To halve the standard error instead, the sample size would have to be **quadrupled**. The simulated values show the same thing: 1.646 at 36 against 1.156 at 72, a ratio of about 1.42.

Study in the Larger Evidence Program: How Stable the Mindfulness Estimate Is

- **What this study adds:** A quantified account of how much the pilot's central estimate would move under repetition, which is what makes any later comparison between cohorts interpretable.
- **What it cannot settle:** Whether the estimate is right. Sampling variability describes repetition of the same procedure and is silent about bias, sampling frame, and measurement quality.
- **Causal-support role:** Consistency evidence in a limited sense: it establishes how much apparent inconsistency between repetitions should be expected before any is treated as meaningful.
- **Next evidence priority:** Express the precision as an interval, so that a range of compatible values is reported rather than a point estimate with an unstated margin.

Level 3 Solutions

Solution 5.9

$SE = 10.060 / \sqrt{72} = 1.186$ points, approximately. It describes the variability of the **average**, not of the students: repeated samples of 72 drawn the same way would produce averages differing from one another by roughly this much.

Solution 5.10

$SE = 10.060 / \sqrt{36} = 1.677$ points, approximately. The ratio to the 72-case value is $1.677 / 1.186 = 1.41$, which is $\sqrt{2}$ as expected. Halving the sample costs about 41% more uncertainty in the estimate, not 100%.

Solution 5.11

$SE = 4.993 / \sqrt{72} = 0.588$ points, approximately. The paired analysis forms one difference per participant, so the relevant variability is variation among those differences. The separate score standard deviations describe how students differ from one another, which includes the between-person variation that pairing removes.

Decisive cue: the standard error must be the spread of the quantity being estimated.

Solution 5.12

$1.156 / 1.186 = 0.975$: the two agree to within about two and a half percent. The closeness demonstrates that the formula is describing something real—the scatter of averages a learner can watch accumulate—rather than asserting a convention. A large gap would have indicated that the distribution's shape makes the formula a poor approximation here, for instance because a few extreme observations dominate the averages.

Level 4 Solutions

Solution 5.13

Supported: “The average post-program score was 49.379 among the 72 participants, estimated with a standard error of 1.186, so a repeat of the same study would be expected to produce an average within roughly a point of that.” Not supported: any sentence about how much individual students varied—that is the standard deviation of 10.060—and any sentence about a population, which the design does not license.

Solution 5.14

The distribution is concentrated and roughly symmetric: 5860 of ten thousand resampled means, well over half, fall in a two-point window from 48 to 50, and only 19 fall below 46. The counts show the **shape** of the distribution, not just its spread. A standard error is a single number and would look identical for a distribution with much heavier tails; the counts let a reader see that extreme averages are genuinely rare rather than merely assumed to be.

Solution 5.15

The 36-case distribution is wider on both measures: a standard error of 1.646 against 1.156, and a middle 95% range spanning about 6.4 points against about 4.5. Because the two simulations resample from the same observed scores, the comparison isolates the effect of **sample size alone**. Nothing about the students, the program, or the measurement differs between them.

Solution 5.16

The first is the precision of the **post-score average**; the second is the precision of the **average change**. They differ because they estimate different quantities from differently variable raw material: post scores vary by 10.060 across students, while individual changes vary by only 4.993, because the change subtracts out each student's own starting level. Same participants, same sample size, half the uncertainty—produced by the design, not by the arithmetic. **Transfer cue**: a standard error is only interpretable once you know which estimate it belongs to.

Level 5 Solutions**Solution 5.17**

Report the **standard deviation** when the audience wants to know how much students differed—for judging whether the group was homogeneous, or whether an individual score is unusual. Report the **standard error** when the audience wants to know how well the average is pinned down—for judging whether a difference between two averages is interpretable. Reporting the wrong one is not a stylistic slip: it changes the claim by roughly a factor of the square root of the sample size.

Solution 5.18

Because two estimates always differ. The question is not whether they differ but whether they differ by more than the movement expected under repetition, and that requires the **standard error of each estimate**—or, better, of their difference. Comparing two numbers without it is comparing two draws from distributions whose widths are unknown. **Rejected rule**: “different numbers mean a different result.”

Solution 5.19

Reject it. The 144 scores come from 72 participants, and the two scores from one participant are linked rather than independent. Using 144 would divide by $\sqrt{144}$ instead of $\sqrt{72}$, understating the standard error by a factor of about 1.41. Every interval built on it would be about 30% too narrow, and every claim of precision correspondingly overstated.

Solution 5.20

The substitution is **precision for validity**. A standard error describes how much an estimate moves when the same procedure is repeated; if the procedure systematically misses the target, repeating it more times moves the estimate less without moving it closer. What would actually be required is evidence about the sampling mechanism, the measurement, and the design—none of which a standard error can supply. **Transfer cue**: a narrow interval around the wrong quantity is still the wrong quantity.

Level 6 Solutions**Solution 5.21**

“Among the 72 participants, the average post-program mindfulness score was 49.379, with a standard error of 1.186—so an estimate from a repeat of the same study would be expected to land within about a point of this one.” The sample size, the estimate, and what its precision means are all recoverable.

Solution 5.22

“The second cohort averaged a point and a half lower” is a fair description of two numbers. “So the program's effect is fading” fails three times over: it treats a gap between two estimates as a gap between two underlying quantities without asking how much estimates move on their own; it asserts a trend from two points; and it attributes the gap to the program when nothing in the design rules out anything else. A defensible version reports both averages with their standard errors and observes whether the gap is within ordinary variation.

Solution 5.23

Two things are wrong. The number 10.060 is the standard **deviation**, not the standard error, and the standard error is 1.186. The conclusion drawn from it— that most students scored within about ten points of the average —happens to be a reasonable reading of the standard deviation, so the sentence reaches a defensible statement about students by way of the wrong label. Corrected: “Scores averaged 49.379 with a standard deviation of 10.060, so most students scored within about ten points of the average. The average itself is estimated with a standard error of 1.186.”

Solution 5.24

The standard error rises from about 1.186 to about 1.677, a factor of $\sqrt{2}$, and the middle 95% range of the estimate widens from roughly 4.5 points to roughly 6.4. Whether that matters depends on a question the sentence assumes has been answered: **what size of difference does the center need to be able to detect?** Precision has no interpretation on its own, only against a difference worth resolving—which is where study planning begins.

Study in the Larger Evidence Program: What a Smaller Pilot Would Cost

- **What this study adds:** A quantified comparison of estimate precision at two sample sizes, holding everything else fixed, so a design decision can be argued from evidence rather than intuition.
- **What it cannot settle:** Whether the smaller design is adequate, which depends on the magnitude the center needs to detect and on the decision the result will inform.
- **Causal-support role:** None directly; it is planning evidence that determines whether a future responsiveness test could resolve anything.
- **Next evidence priority:** Establish the smallest change that would matter in this program context, since precision is only interpretable against it.

Unit 6: Confidence Intervals and Estimation

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Unit Purpose

An interval answers a better question than a point estimate does, and it is misread more often than almost anything else in an applied report. This unit builds two habits: reading an interval as a range of values the data are compatible with, and refusing the four sentences that turn it into something else—a probability statement about a fixed quantity, a claim that the true value is definitely inside, a proof of no effect when it contains zero, and a significance test wearing different clothes.

Learning Objectives

After completing this unit, a learner should be able to:

- construct an interval from an estimate, a critical value, and a standard error;
- state what the confidence level describes and what it does not;
- read interval width as precision and interval position as magnitude;
- explain why an interval containing zero is not evidence of no effect;
- distinguish an interval for a mean difference from an interval for an effect size;
- choose estimation language over threshold language in a written result; and
- state the claim ceiling an interval cannot raise.

Decision Map

Ask these questions in order:

1. What quantity is the interval an interval for?
2. What estimate, critical value, and standard error produced it?
3. How wide is it, and is that width useful for the decision at hand?
4. Where does it sit—are all the values inside it practically similar, or not?
5. Does it contain a value of special interest, and what does that mean?
6. What does the confidence level describe: this interval, or the procedure?
7. What does the design permit me to say about the interval's contents?

Unit Study Files

Study	Data structure	Used here for	Unit role
Study 02	Pre and post scores for 72 synthetic participants	An interval for the mean change, and an interval for a standardized effect	A precise, clearly positive estimate
Study 02C	Pre and post scores for 72 synthetic participants	An interval containing zero	An imprecise, inconclusive estimate

Plausible Research Scenarios

The scenarios create decisions rather than documenting real research. At each **Pause and Decide**, stop reading and write an answer, reason, and confidence rating. The explanation that follows is feedback on that commitment.

Plausible Research Scenario: One Page for the Director

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

The counseling center's director has asked for one page on the mindfulness pilot. She is not a statistician, she is deciding whether to fund a larger evaluation, and she has said she wants “the number, and how sure we are.”

The fictional research team has the estimate: an average post-minus-pre change of 6.086 points among 72 participants, with a standard error of 0.588. Turning that into an interval is arithmetic. Turning it into a sentence the director can act on, without licensing a conclusion the design cannot support, is the actual work.

Four candidate sentences circulate in the team's drafts. The first says the true average change is between 4.913 and 7.259 points. The second says there is a 95% probability that the true average change lies in that range. The third says the program produces an improvement of about six points. The fourth says the interval excludes zero, so the effect is statistically significant.

Every one is wrong, in four different ways worth separating. One overstates certainty about a fixed quantity. One misplaces the probability: the interval varies from study to study, not the underlying value. One promotes a described change to a causal effect the design cannot identify. And one converts an estimate back into a threshold decision, trading magnitude and precision for a yes-or-no.

The interval's real content is more useful and more modest. It says the data, under this model, are compatible with average changes from roughly five to seven points, and not with a change near zero or near fifteen. For a director deciding whether to fund a larger evaluation that is genuinely informative—provided she is also told that a one-group pre-post design cannot say what caused it.

The team also has a second interval, for the standardized effect: 0.911 to 1.522. It answers a different question in different units.

Pause and Decide

Write the sentence you would put in the director's brief. Check it against four failure modes: does it claim certainty about a fixed value, place a probability on that value, attribute the change to the program, or reduce the interval to a significance verdict? Then state what the interval would have to look like before you would advise against a larger evaluation.

Why This Model?

The interval is built from the paired estimate, its standard error, and the critical value from the t distribution with 71 degrees of freedom: 6.086 plus or minus 1.994 times 0.588. Each piece has a job: the estimate sets the position, the standard error the scale, the critical value how much of the sampling distribution the procedure covers.

The confidence level describes the **procedure**, not this interval. Across repeated applications to samples from the same process, 95% of the intervals produced would contain the underlying value. This one either contains it or does not, and nothing in the data says which. That is what makes the second draft sentence wrong.

A point estimate alone reports magnitude with no precision. A significance test alone reports a threshold decision with neither—it answers whether zero is plausible, and the director asked how large the change is.

What Would a Favorable Result Look Like?

An interval narrow enough to distinguish possibilities the director cares about, positioned where the change would be worth the program's cost. These are two separate properties: an interval can be narrow around a trivial value or wide around a large one, and the two call for different decisions.

A wide interval is not a failed study. It is a study reporting honestly that it cannot resolve the question—information the director needs before spending more.

Best Defensible Claim

Statistical evidence: The interval reports the range of average post-minus-pre changes compatible with the observed data under the paired model, among the represented participants.

Causal identification: None. The one-group design supplies no counterfactual, so the interval bounds a described change, not an effect.

Claim ceiling: The team may state the estimated change and its precision. It may not say the program produces that change, may not attach a probability to the underlying value, and may not present the interval as covering a population it did not sample.

What Must Be True?

Participants must contribute independent difference scores, the mean difference must be a meaningful summary, and the difference distribution must not be so distorted by a few observations that the t -based interval misstates the uncertainty. The interval also assumes the estimand was fixed before the data were seen: one computed for whichever quantity looked best does not have the coverage its label claims.

Defensibility Procedures

Fix the estimand and confidence level before analysis, report the interval beside the estimate rather than instead of it, name the quantity and units it bounds, avoid probability language about the underlying value, state the sample size and design limits, and report the interval whether or not it excludes zero.

Next evidence priority: Establish what magnitude of change would justify the program's cost, since an interval is only interpretable against a difference worth having.

Claim Audit

Repair this claim before continuing: “We are 95% confident that the program improves mindfulness by between 4.913 and 7.259 points.” Your repair should fix the causal attribution, correct what the confidence level describes, and keep the sentence usable by a non-specialist.

Plausible Research Scenario: The Interval That Contains Zero

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

A second pilot, represented by Study 02C, uses the same design, questionnaire and number of participants, and produces an average change of 0.400 points with a 95% interval from -0.821 to 1.621.

A draft memo summarises this as “There was no effect.” The sentence does something the interval does not support, and the failure is instructive: it is the most common misreading in applied work.

An interval containing zero says zero is among the values compatible with the data. It also says a decrease of eight-tenths of a point is compatible, and an increase of a point and a half, and everything between. Reporting only the value that happens to sit inside, as though the data had selected it, discards the rest.

The real claim is worth stating: this study found no evidence of an important average change and was not precise enough to rule one out. Establishing that a change is *negligible* is a different question, needing a stated margin and an analysis built to test against it. Nobody set one here.

Pause and Decide

Decide what this interval licenses and what it does not. Write the sentence you would put in the memo instead of “there was no effect,” and name the extra ingredient needed to claim the change is negligible.

Why This Model?

The model is the same paired interval: 0.400 plus or minus 1.994 times the standard error of 0.613. Nothing about the procedure changed—model choice was fixed by the design, not by whether the result was favorable.

An equivalence test would answer “is the change negligible,” and is rejected here not because it is wrong but because it needs an input this study lacks: a prespecified margin.

What Would a Favorable Result Look Like?

The team would prefer a narrow interval, wherever it sat. A narrow one around zero would be informative: it would say the change is small, precisely. This is not that. It spans nearly two and a half points, so it mostly reports that the study could not resolve its question.

Best Defensible Claim

Statistical evidence: The data are compatible with average changes from a small decrease to a modest increase; the estimate is small and imprecise.

Causal identification: None, for the same design reason as before.

Claim ceiling: The team may report an inconclusive estimate with its interval—not that the program has no effect, that the true change is zero, or that the two pilots differ.

What Must Be True?

The same conditions as the first pilot, plus one: the decision to report must not depend on how the interval came out. Reporting intervals only when they exclude zero produces a published record that is systematically wrong.

Defensibility Procedures

Report estimate and interval whatever their position, state the width in the audience's units, avoid “no effect” unless an equivalence margin was prespecified and met, record the sample size, and say plainly when the study was not precise enough to settle the question.

Next evidence priority: A prespecified margin for a negligible change, and a design precise enough to distinguish it from a change that matters.

Claim Audit

Repair this claim: “The interval included zero, so the program had no effect.” Distinguish absence of evidence from evidence of absence, say what the interval did report, and name what a negligible-change claim would require.

Problems

Do not consult the solutions until you have recorded an answer, reason, and confidence rating for every attempted item.

Level 1: Recall and Recognize

Problem 6.1 — Name the quantity

An interval runs from 4.913 to 7.259. Before interpreting it, state what has to be known about it, and why an interval with no named quantity and no units is uninterpretable.

Problem 6.2 — What the confidence level describes

State what “95%” refers to in a 95% confidence interval. Then state what it does not refer to, in terms a non-specialist would follow.

Problem 6.3 — Width and position are different readings

Distinguish what the *width* of an interval tells you from what its *position* tells you. Give one decision that depends on each.

Problem 6.4 — Two intervals, two quantities

Study 02 reports an interval of 4.913 to 7.259 for one quantity and 0.911 to 1.522 for another. Identify each quantity and its units, and explain why the two cannot be compared directly.

Level 2: Predict Before Clicking

Problem 6.5 — Predict the effect of a larger sample

A paired study is repeated with four times as many participants and produces the same estimate and the same spread of differences. Predict what happens to the interval's width, and state the factor.

Problem 6.6 — Predict the effect of a higher confidence level

Predict what happens to an interval's width when the confidence level rises from 95% to 99%, holding the data fixed. Explain what is being traded for what.

Problem 6.7 — Predict whether the interval contains zero

Study 02C's mean change is 0.400 with a standard error of 0.613. Before computing, predict whether a 95% interval will contain zero, and state the comparison you used.

Problem 6.8 — Predict the interval's relationship to a test

A 95% interval for a mean difference excludes zero. Predict what a two-sided test of the same difference at the 5% level will conclude, and state why the two agree.

Level 3: Calculate and Explain

Problem 6.9 — Build the interval

Study 02's mean difference is 6.086 with a standard error of 0.588, and the critical value for 71 degrees of freedom is 1.994. Use $estimate \pm t^* \times SE$ to compute the 95% interval, and state what each of the three inputs contributes.

Problem 6.10 — Build the inconclusive interval

Study 02C's mean difference is 0.400 with a standard error of 0.613 and the same critical value of 1.994. Compute its 95% interval and compare its width with the one from the previous problem.

Problem 6.11 — Recover the standard error from an interval

An interval runs from 4.913 to 7.259 around an estimate of 6.086, at 95% confidence with 71 degrees of freedom. Work backwards to recover the standard error, and explain why this is possible.

Problem 6.12 — Compute the half-width and interpret it

Compute the half-width of Study 02's interval and express it as a percentage of the estimate. Explain what that ratio does and does not tell a reader about whether the change matters.

Level 4: Read jamovi Output**Problem 6.13 — Read a paired interval**

jamovi reports a mean difference of 6.086 with a 95% interval from 4.913 to 7.259 for 72 participants. Write the sentence this output supports and one it does not.

Problem 6.14 — Read an interval containing zero

jamovi reports a mean difference of 0.400 with a 95% interval from -0.821 to 1.621. Write the sentence this output supports, and identify the sentence a careless reader would write instead.

Problem 6.15 — Read an effect-size interval

jamovi reports Cohen's d_z of 1.219 with an interval from 0.911 to 1.522. Interpret it in words, and say what it adds to the interval for the mean difference and what it cannot replace.

Problem 6.16 — Read two intervals side by side

Study 02's interval spans 4.913 to 7.259; Study 02C's spans -0.821 to 1.621. Compare their widths and positions, and state which comparison between the two studies the output does not license.

Level 5: Choose the Model**Problem 6.17 — Interval or test?**

A results section must report one of an interval or a p value for a mean difference. Decide which serves a reader deciding whether to fund a program, and justify the choice in terms of what each reports.

Problem 6.18 — Reject the probability statement

A draft says there is a 95% probability the true mean change lies between 4.913 and 7.259. Reject it, and state precisely which quantity the 95% attaches to.

Problem 6.19 — Reject the equivalence claim

A draft treats Study 02C's interval as establishing that the program's effect is zero. Reject it, and name the two ingredients an equivalence claim would require.

Problem 6.20 — Reject the interval as a threshold

A draft reports only whether each interval excludes zero and reports nothing else. Explain what has been discarded, and name the reader whose question is no longer answered.

Level 6: Report and Critique

Problem 6.21 — Write the director's sentence

Write the one-sentence result for the director's brief, reporting Study 02's estimate and interval in a form that survives all four failure modes identified in the scenario.

Problem 6.22 — Repair the causal interval

Repair this sentence: “We are 95% confident the program improves mindfulness by between 4.913 and 7.259 points.” Fix both the causal attribution and the misplaced probability.

Problem 6.23 — Repair the null report

Repair this sentence: “The interval contained zero, so the program had no effect.” State what the data do support and what an equivalence claim would need.

Problem 6.24 — Repair the mixed-up interval

Repair this sentence: “The average change was 6.086 points, 95% CI [0.911, 1.522].” Identify the error, and write the two correct sentences that report both quantities.

Worked Solutions

Each solution identifies the decisive cue, reasoning path, relevant evidence, rejected alternative, claim boundary, and a transfer cue.

Level 1 Solutions

Solution 6.1

You must know **what quantity it bounds, in what units, from what design**. The same two numbers could bound a mean difference in points, a difference in percentages, or a ratio. Without the quantity, no reader can judge whether the width is useful or the position important. **Transfer cue**: an interval is a statement about a named quantity; the numbers alone are not the statement.

Solution 6.2

The 95% describes the **procedure**: if the same interval-building procedure were applied repeatedly to samples from the same process, about 95% of the intervals produced would contain the underlying value. It does **not** describe this interval, which either contains the value or does not, and it does not assign a probability to the underlying value, which is treated as fixed rather than random. A non-specialist phrasing: “the method gets it right about 19 times in 20; we cannot tell which kind of time this is.”

Solution 6.3

Width reports precision—how much the data narrow the field of compatible values. **Position** reports magnitude—where those compatible values sit relative to a value that matters. A decision about whether to run a larger study depends chiefly on width; a decision about whether the change is worth its cost depends chiefly on position. **Transfer cue**: report both, because either alone can mislead.

Solution 6.4

The first bounds the **mean post-minus-pre change, in points on the mindfulness scale**. The second bounds **Cohen's d^* , a standardized effect with no units**, expressing the change relative to the standard deviation of individual changes. They cannot be compared directly because they are measured on different scales; the second is a rescaling of the first, not a second opinion about it.

Level 2 Solutions

Solution 6.5

The width **halves**. The interval's half-width is a critical value times $s/\sqrt{n}$, so quadrupling n halves the standard error and therefore the width (the critical value also changes very slightly with degrees of freedom, but the effect is negligible at these sample sizes). **Transfer cue**: the square-root relationship governs interval width just as it governs the standard error.

Solution 6.6

The interval gets **wider**. A higher confidence level demands a procedure that captures the underlying value more often, and the only way to do that with the same data is to name a larger set of compatible values. The trade is **coverage for precision**: more confidence, less information. Neither level is intrinsically correct; 95% is a convention, not a discovery.

Solution 6.7

Yes, it will contain zero. The estimate of 0.400 is less than one standard error from zero—about two-thirds of one—and a 95% interval reaches roughly two standard errors either side. The comparison is estimate against standard error, made before any interval is computed. **Transfer cue:** an estimate smaller than its own standard error will always produce an interval containing zero.

Solution 6.8

The test will reject the null hypothesis of zero difference at the 5% level. The two agree because they are the same calculation read in two directions: the interval contains exactly those null values that would not be rejected at that level. This equivalence is also why reporting only the test discards information—the interval reports every value compatible with the data, not just whether one particular value survived.

Study in the Larger Evidence Program: Estimating the Mindfulness Change

- **What this study adds:** A bounded estimate of average six-week change with its precision stated, which is what a funding decision needs and what a point estimate alone cannot supply.
- **What it cannot settle:** What caused the change. The interval bounds a described quantity; the one-group design supplies no counterfactual for it.
- **Causal-support role:** Weak responsiveness evidence at best. It shows a change coincided with the program, with quantified uncertainty, and no comparison condition.
- **Next evidence priority:** A randomized comparison condition, so that an interval of the same width bounds an effect rather than a change.

Level 3 Solutions

Solution 6.9

Half-width = $1.994 \times 0.588 = 1.173$ points, giving $6.086 \pm 1.173 = [4.913, 7.259]$. The **estimate** sets the interval's position, the **standard error** sets the scale of plausible movement, and the **critical value** sets how much of the sampling distribution the procedure covers—raising it widens the interval without adding any information.

Solution 6.10

Half-width = $1.994 \times 0.613 = 1.221$ points, giving $0.400 \pm 1.221 = [-0.821, 1.621]$. It is slightly *wider* than Study 02's interval, because the individual changes were slightly more variable. The two studies are therefore about equally precise; what differs is where the intervals sit, not how much either study could resolve. **Decisive cue:** precision and magnitude are independent.

Solution 6.11

The half-width is $(7.259 - 4.913) / 2 = 1.173$, and dividing by the critical value gives $1.173 / 1.994 = 0.588$, recovering the standard error. This works because the interval is a deterministic function of three quantities: given any three of estimate, half-width, critical value and standard error, the fourth follows. It is a useful check when a paper reports an interval but not a standard error.

Solution 6.12

The half-width is 1.173 points, about 19% of the 6.086-point estimate. That ratio describes **relative precision**: the estimate is pinned down to roughly a fifth of its own size. It does not tell a reader whether the change matters, because that depends entirely on what a point on this scale means in the program's context—a question the arithmetic cannot answer.

Level 4 Solutions

Solution 6.13

Supported: “Among the 72 participants, mindfulness scores changed by an average of 6.086 points over the six-week interval, 95% CI [4.913, 7.259]; the data are compatible with average changes of roughly five to seven points.” Not supported: any sentence attributing the change to the program, which the one-group design cannot identify, and any sentence placing a probability on the underlying value.

Solution 6.14

Supported: “The average change was 0.400 points, 95% CI [-0.821, 1.621]. The data are compatible with a small decrease, no change, or a small increase; this study was not precise enough to distinguish among them.”

The careless sentence is “there was no effect,” which promotes one value inside the interval to a conclusion and discards the rest.

Solution 6.15

The estimated change is about 1.2 standard deviations of the individual change scores, and the data are compatible with values from about 0.9 to about 1.5 on that scale. It adds **comparability**: a standardized effect can be set beside results measured on other instruments. It cannot replace the interval in points, because only the raw scale can be judged against what a change is worth in this program. **Transfer cue**: standardization buys comparison and costs meaning.

Solution 6.16

Study 02's interval spans 2.347 points and lies entirely above zero; Study 02C's spans 2.443 points and straddles zero. They are of nearly equal width, so neither study is meaningfully more precise; they differ in **position**. The output does not license any statement that the two studies differ from one another—comparing two intervals by whether they overlap is not a test of their difference, and no such comparison was specified.

Level 5 Solutions

Solution 6.17

Report the **interval**. The director's question is how large the change is and how well it is pinned down; the interval answers both, and the p value answers neither—it reports only whether one particular value, zero, is compatible with the data. A funding decision turns on whether the plausible values are worth the cost, which a threshold verdict cannot express. **Rejected rule**: “report the significance and let the reader infer the size.”

Solution 6.18

Reject it. The 95% attaches to the **procedure's long-run coverage**, not to the underlying mean change, which is treated as a fixed unknown rather than a random quantity. Under this framework the statement “there is a 95% probability the value lies in [4.913, 7.259]” is not merely imprecise but incoherent: the value is either in that range or it is not, and the data cannot say which.

Solution 6.19

Reject it. An equivalence claim requires two things this analysis has neither of: a **prespecified margin** stating what magnitude would count as negligible in this context, and an **analysis designed to test against that margin** rather than against zero. Study 02C's interval reaches 1.621 points in one direction, which may well exceed any margin the center would set. **Claim ceiling**: inconclusive, not null.

Solution 6.20

What has been discarded is **magnitude and precision**—everything the interval was carrying. Two studies can both exclude zero while one estimates a trivial change precisely and the other a large change vaguely; reduced to threshold verdicts they become indistinguishable. The reader no longer served is anyone making a decision that depends on how big the change is, which is every reader outside a hypothesis-testing exercise.

Level 6 Solutions

Solution 6.21

“Among the 72 students in the pilot, mindfulness scores rose by an average of 6.086 points over six weeks, and the data are compatible with average rises of roughly 4.913 to 7.259 points. Because everyone in the pilot

received the program, this describes the change we observed rather than showing the program caused it.” No certainty about a fixed value, no probability attached to it, no causal attribution, and no reduction to a verdict.

Solution 6.22

Two errors. The causal one: “the program improves mindfulness” asserts an effect a one-group design cannot identify. The probabilistic one: 95% confidence describes the procedure, not the underlying value. Corrected: “Among the represented participants, average mindfulness scores rose by 6.086 points, 95% CI [4.913, 7.259]. The design does not establish that the program caused the rise.”

Solution 6.23

The data support: “This study did not find evidence of an important average change, and it was not precise enough to rule one out.” An equivalence claim would need a prespecified margin for what counts as negligible and an analysis testing against that margin. **Transfer cue:** absence of evidence is not evidence of absence, and the interval's width is what tells you which one you have.

Solution 6.24

The interval quoted belongs to the standardized effect, not to the mean change in points, so the sentence attaches the wrong interval to its estimate. Correct: “The average change was 6.086 points, 95% CI [4.913, 7.259]. The corresponding standardized effect was $dz = 1.219$, 95% CI [0.911, 1.522].” Each estimate carries the interval computed for it, in its own units.

Study in the Larger Evidence Program: Reporting an Inconclusive Result

- **What this study adds:** A precise account of what the second pilot could and could not resolve, which keeps an uninformative result from entering the record as a null finding.
- **What it cannot settle:** Whether the average change is negligible. That requires a prespecified margin and an analysis built to test against it.
- **Causal-support role:** None on its own; its contribution to consistency evidence depends entirely on being reported whatever its direction.
- **Next evidence priority:** Agree a margin for a negligible change, then design a study precise enough to distinguish it from a change that matters.

Unit 7: Tests, Effect Sizes, and Evidence Strength

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Unit Purpose

A result usually arrives as three numbers—an estimate, a p value, and an effect size—and each answers a different question. Confusing them is the most consequential reading error in applied statistics, because it lets a small p stand in for importance and a large effect stand in for evidence. This unit builds the habit of assigning each number its own job, and of noticing what none of them supplies.

Learning Objectives

After completing this unit, a learner should be able to:

- state precisely what a p value is the probability of;
- distinguish statistical significance from practical importance and from scientific success;
- interpret a standardized effect size and name its denominator;
- explain why the same effect size can produce very different p values;
- read an inconclusive result without converting it into evidence of no effect;
- integrate estimate, interval, test, and effect size into one defensible sentence; and
- state what no test result can supply about design.

Decision Map

Ask these questions in order:

1. What quantity is estimated, and how large is it in the units of the problem?
2. How precisely is it estimated?
3. What null value is being tested, and was it chosen before the data?
4. What exactly is the p value the probability of?
5. What effect size is reported, and what is its denominator?
6. Do the estimate, interval, test, and effect size tell a consistent story?
7. What does the design permit, regardless of how any of them came out?

Unit Study Files

Study	Data structure	Used here for	Unit role
Study 02	Pre and post scores for 72 synthetic participants	A clear result: large effect, small p , precise interval	Integrating four numbers that agree
Study 02C	Pre and post scores for 72 synthetic participants	An inconclusive result: tiny effect, large p	Reading a non-significant result honestly

Plausible Research Scenarios

The scenarios create decisions rather than documenting real research. At each **Pause and Decide**, stop reading and write an answer, reason, and confidence rating. The explanation that follows is feedback on that commitment.

Plausible Research Scenario: Three Numbers That Are Not Interchangeable

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

The counseling center's pilot has produced a result, and the fictional research team is writing it up. The analysis reports an average post-minus-pre change of 6.086 points, a t statistic of 10.343 on 71 degrees of freedom with a p value below .001, and a standardized effect of 1.219 with an interval from 0.911 to 1.522.

A colleague reads the output and says: “Highly significant, and a huge effect size—this is strong evidence the program works.” Every word of that sentence is doing work that one of the numbers cannot do, and disentangling it is the whole exercise.

Take them one at a time. The **6.086 points** is the estimate: how much scores changed on average, in units the center thinks in. It is the only number that speaks to whether the change is worth the program's cost, and using it requires knowing what a point on this scale means—a judgment the statistics cannot make.

The *** p * value** is the probability of obtaining a t statistic at least this extreme *if the average change in the underlying process were exactly zero*. It is small here, which means data like these would be very unusual under that assumption. It is not the probability that the null is true, not the probability the result was luck, and not a measure of how large anything is. A large enough study finds a small p for a trivial change.

The **standardized effect of 1.219** expresses the change relative to how much individual changes varied. It travels between instruments, which is its value, and loses the scale, which is its cost. Its denominator must be named: a paired dz divides by the spread of the differences, and quoting it beside an effect computed with a different denominator invites an invalid comparison.

And the word doing the most damage is **“works.”** Nothing in the output bears on it. A one-group pre-post design has no comparison condition, so the estimate bounds a change that occurred, and neither a small p nor a large dz supplies the counterfactual the claim requires. Statistical strength cannot be traded for design strength; the exchange rate is zero.

Pause and Decide

Before reading further, write down what each of the three numbers answers, in one line each. Then decide which of them speaks to whether the program is worth funding, which to whether the data are surprising under a null, and which to neither. Finally, name what the colleague's sentence assumes that no number in the output supports.

Why This Model?

The paired t test is the model the design fixed in earlier units, and nothing here changes it. What changes is what is read off it. The estimate and interval report magnitude and precision; the test reports compatibility with one null value; the effect size reports magnitude scale-free. Each answers a question the others cannot.

Reporting the test alone discards magnitude. Reporting the effect size alone carries no precision and no units, and hides its denominator.

What Would a Favorable Result Look Like?

An estimate large enough to matter in context, determined precisely enough to distinguish it from values that would not, with a test and an effect size telling the same story. The instructive cases are when they disagree: a large effect with a wide interval, or a small p on a change too small to care about.

A favorable pattern is not a small p —worth stating flatly, because almost every incentive in applied research pushes the other way.

Best Defensible Claim

Statistical evidence: Scores changed by an estimated 6.086 points on average, precisely enough determined to exclude values near zero, and this change is large relative to the variation among individual changes.

Causal identification: None. The design has no comparison condition, and no test statistic or effect size can supply one.

Claim ceiling: The team may report a large, precisely estimated change among the represented participants. It may not say the program works, that the effect is important without saying what a point is worth, or that a small p makes any of it more credible.

What Must Be True?

Participants must contribute independent difference scores, the null value must have been fixed before the data were seen, and the effect size's denominator must be stated. A p value computed after inspecting several possible outcomes and choosing the most striking does not mean what its label says, however correctly the arithmetic was performed.

Defensibility Procedures

Prespecify the estimand, the null value and the effect-size definition; report the estimate with its units, the interval, the test statistic with its degrees of freedom, and the effect size with its denominator; never report a p value alone; state what change would be practically meaningful before seeing the result; and describe the design's limits in the same paragraph as the numbers.

Next evidence priority: A randomized comparison condition, so that the same four numbers describe an effect rather than a change.

Claim Audit

Repair this claim before continuing: “The result was highly significant with a huge effect size, so the program clearly works.” Your repair should separate significance, magnitude and causal identification, and say what each number in the output actually established.

Plausible Research Scenario: What $p = .516$ Does Not Say

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

The second pilot, Study 02C, produces an average change of 0.400 points, a t statistic of 0.653 on 71 degrees of freedom, a p value of .516, and a standardized effect of 0.077.

The draft says: “The test was not significant, so the program had no effect.” A second draft says: “The effect size was 0.077, essentially zero, confirming there is nothing there.” Both are wrong, and differently, which is why both are worth working through.

The first converts a failure to reject into a positive claim. A p value of .516 says data like these are unsurprising if the underlying average change were zero. It says nothing about whether that change is zero, because these data would also be unsurprising under many other values—the interval from -0.821 to 1.621 lists them.

The second treats a point estimate as though it had no uncertainty. An effect of 0.077 is small, but it is an estimate with its own imprecision, and this study cannot distinguish it from values several times larger.

The honest reading is less satisfying and more useful: this study did not detect an important change and was not sensitive enough to rule one out. That is a statement about the study, not about the program.

Pause and Decide

Decide what claim the three numbers jointly support. Then identify which piece of information is missing that would be required before the word “negligible” could be used at all.

Why This Model?

The model is the same paired t test, applied because the design says so rather than because of how it came out.

An equivalence test would answer the question the drafts reach for, and is rejected for lack of a prespecified margin. Reporting the effect size alone hides the imprecision that makes both drafts wrong.

What Would a Favorable Result Look Like?

A precise estimate, wherever it fell. One tightly bounded near zero would let the center redirect its effort with confidence. This study offers neither a detected change nor a confident absence, so its favorable outcome is an honest account of its own limits.

Best Defensible Claim

Statistical evidence: The estimated change is small and imprecisely determined; the data are compatible with values from a small decrease to a modest increase.

Causal identification: None, for the same design reason as the first pilot.

Claim ceiling: The team may report an inconclusive result. It may not say the program has no effect, that the change is negligible, or that this pilot contradicts the first.

What Must Be True?

The same conditions as before, plus one that only bites here: the analysis and the decision to report must not depend on the result. A literature in which inconclusive studies go unreported is systematically misleading even when every individual analysis is correct.

Defensibility Procedures

Report the estimate, interval, test and effect size together whatever their values; state the sample size; avoid “no effect” and “negligible” unless a margin was prespecified and met; describe what the study could and could not detect; and publish the result regardless of direction.

Next evidence priority: A prespecified margin for a negligible change, and enough precision to distinguish it from a change that matters.

Claim Audit

Repair this claim: " $p = .516$ and $d_z = 0.077$, so we can conclude the program does nothing." Distinguish failing to detect from establishing absence, and name what would be needed for the stronger claim.

Problems

Do not consult the solutions until you have recorded an answer, reason, and confidence rating for every attempted item.

Level 1: Recall and Recognize

Problem 7.1 — State what a p value is the probability of

Complete the sentence precisely: “A p value is the probability of...”. Then name two quantities it is commonly mistaken for.

Problem 7.2 — Three questions, three numbers

A result reports an estimate in points, a p value, and a standardized effect size. State the question each one answers, in one line each.

Problem 7.3 — Name the denominator

Cohen's d_z for a paired design is reported as 1.219. Name the quantity it divides by, and explain why an effect size without its denominator named cannot be compared with another.

Problem 7.4 — Separate three meanings

Distinguish statistical significance, practical importance, and scientific success. Give one example of a result that has the first without the second.

Level 2: Predict Before Clicking

Problem 7.5 — Predict the p value from the estimate and its precision

Study 02's mean change is 6.086 with a standard error of 0.588. Before looking, predict roughly how many standard errors from zero the estimate lies and whether the p value will be large or small.

Problem 7.6 — Predict the inconclusive case

Study 02C's mean change is 0.400 with a standard error of 0.613. Predict the approximate t statistic and whether the p value will be near or far from .05.

Problem 7.7 — Predict the effect of a larger sample on each number

A study is repeated with four times the participants, and the estimate and the spread of differences come out the same. Predict what happens to the estimate, the interval width, the t statistic, and the standardized effect size.

Problem 7.8 — Predict which number moves with the units

Suppose the mindfulness scale were rescaled so every score doubled. Predict what happens to the mean change, the standard deviation of the differences, the standardized effect size, and the p value.

Level 3: Calculate and Explain

Problem 7.9 — Compute the test statistic

Divide Study 02's mean difference of 6.086 by its standard error of 0.588 to recover the t statistic. State what the resulting quantity measures and in what units.

Problem 7.10 — Compute the standardized effect

Use Study 02's mean difference of 6.086 and difference-score standard deviation of 4.993 to compute $d_z = M_{\text{difference}} / SD_{\text{difference}}$. Explain why this denominator differs from the one used for the t statistic.

Problem 7.11 — Compute the inconclusive effect

Use Study 02C's mean difference of 0.400 and difference-score standard deviation of 5.198 to compute its standardized effect. Compare the two denominators used in this problem and the previous one.

Problem 7.12 — Show that t and d are not the same information

Using $t = d_z \times \sqrt{n}$, verify the relationship for Study 02 with 72 pairs, and explain what this identity reveals about why a p value cannot report magnitude.

Level 4: Read jamovi Output**Problem 7.13 — Read a clear result**

jamovi reports $t(71) = 10.343$, $p < .001$, mean difference 6.086, 95% CI [4.913, 7.259], Cohen's $d_z = 1.219$. Write the one-sentence result and state what it does not establish.

Problem 7.14 — Read an inconclusive result

jamovi reports $t(71) = 0.653$, $p = .516$, mean difference 0.400, 95% CI [-0.821, 1.621], Cohen's $d_z = 0.077$. Write the one-sentence result, and the sentence a careless reader would write instead.

Problem 7.15 — Read an effect size with its interval

Study 02's standardized effect is 1.219 with a 95% interval from 0.911 to 1.522. State what the interval adds, and what a reported effect size with no interval leaves a reader unable to judge.

Problem 7.16 — Read two results together

Study 02 gives $d_z = 1.219$ and Study 02C gives 0.077, from the same design and the same number of participants. State what this comparison shows and the comparison it does not license.

Level 5: Choose the Model**Problem 7.17 — Which number answers the funding question?**

A director asks whether the change is worth the program's cost. Choose which of the four reported numbers answers her, justify it, and name what else she needs that no analysis supplies.

Problem 7.18 — Reject the significance-only report

A draft reports only $p < .001$. Name the three things a reader can no longer determine, and state the minimum set of numbers that should replace it.

Problem 7.19 — Reject the null claim

A draft concludes from $p = .516$ that the program has no effect. Reject it and name the analysis that would answer the question being asked.

Problem 7.20 — Reject the post-hoc null value

A colleague, seeing that the change was about six points, tests whether it differs from five and reports the result as though five had been the hypothesis all along. Explain why the reported p value no longer means what it says.

Level 6: Report and Critique

Problem 7.21 — Write the integrated result

Write a complete APA-style result sentence for Study 02 that reports estimate, interval, test, and effect size, and that states the design limit in the same breath.

Problem 7.22 — Repair the significance-for-importance substitution

Repair this sentence: “The result was highly significant, so the program has a large effect on mindfulness.” Identify both substitutions being made.

Problem 7.23 — Repair the null conclusion

Repair this sentence: “ $p = .516$, so the program does nothing.” State the defensible claim and what a stronger one would require.

Problem 7.24 — Repair the borrowed effect size

Repair this sentence: “Our d of 1.219 is much larger than the d of 0.4 reported in the published trial, so our program is better.” Name every reason the comparison fails.

Worked Solutions

Each solution identifies the decisive cue, reasoning path, relevant evidence, rejected alternative, claim boundary, and a transfer cue.

Level 1 Solutions

Solution 7.1

“A p value is the probability of obtaining a test statistic at least as extreme as the one observed, **if the null hypothesis were true.**” It is commonly mistaken for **the probability that the null hypothesis is true** and for **a measure of effect magnitude**. It is neither: it is a statement about data under an assumption, not about the assumption, and not about size. **Transfer cue:** the conditional runs from hypothesis to data, never the other way.

Solution 7.2

The **estimate in points** answers *how much changed, in units we understand*. The *** p * value** answers *how surprising these data would be if the underlying change were zero*. The **standardized effect** answers *how large the change is relative to how much individuals varied*. No two of these are substitutes.

Solution 7.3

It divides by the **standard deviation of the difference scores**—how much individual participants' changes varied from one another. Without the denominator named, an effect size is uninterpretable and dangerous to compare, because the same estimate divided by a different spread (pooled group SD, control-group SD, raw score SD) gives a different number. Two studies reporting “ $d = 0.5$ ” may not be reporting the same quantity.

Solution 7.4

Statistical significance is a threshold decision about compatibility with a null value. **Practical importance** is a judgment about whether the magnitude matters in context, which requires knowing what the units mean.

Scientific success is whether the study produced a credible answer to its question with an honest account of its uncertainty and limits—which an inconclusive study can achieve and a significant one can fail. An example of the first without the second: a study of 40,000 people finding a reliably nonzero change of a tenth of a point on a 100-point scale.

Level 2 Solutions

Solution 7.5

About $6.086 / 0.588 \approx 10.3$ standard errors from zero, which is very far. The p value will be extremely small. **Transfer cue:** estimate divided by standard error is the whole of what a t test computes, so the p value is predictable from those two numbers before any software is opened.

Solution 7.6

$0.400 / 0.613 \approx 0.65$, well under one standard error from zero, so the p value will be far from .05—somewhere near a half. The verified value is $t(71) = 0.653$ with $p = .516$. An estimate smaller than its own standard error can never approach significance.

Solution 7.7

The **estimate** is unchanged, since it is the same average. The **interval width** halves, because the standard error falls by a factor of two. The **** statistic** doubles, because the same estimate is divided by half the standard error, so the p value shrinks dramatically. The **standardized effect size** is unchanged, because it divides by the spread of the differences, which did not change. **Decisive cue:** sample size drives evidence, not magnitude.

Solution 7.8

The **mean change** doubles and the **standard deviation of the differences** doubles, since both are in the original units. The **standardized effect size** is unchanged, because the doubling cancels in the ratio. The **** value** is unchanged, for the same reason. This is what “scale-free” means, and it is why an effect size travels between instruments while a raw estimate does not.

Study in the Larger Evidence Program: Reading the Mindfulness Result

- **What this study adds:** A precisely estimated, contextually large change with its evidence strength quantified, reported in a form that separates magnitude from surprise.
- **What it cannot settle:** Whether the program caused it. No test statistic or effect size compensates for a missing comparison condition.
- **Causal-support role:** Responsiveness evidence only in the weakest sense—a change coincided with an intervention. Without a counterfactual, it does not establish that the outcome responded to the program.
- **Next evidence priority:** A randomized comparison condition, so that the same four numbers describe an effect instead of a change.

Level 3 Solutions

Solution 7.9

$6.086 / 0.588 = 10.35$, approximately, recovering $t = 10.343$ before rounding. It measures the estimate's distance from the null value of zero **in standard errors**—a unit of evidence, not a unit of mindfulness. Doubling the sample would change it without changing anything about how much scores moved.

Solution 7.10

$d_z = 6.086 / 4.993 = 1.219$, approximately. The denominators differ because the questions differ: the t statistic divides by the **standard error**, which describes how much the *estimate* would move under repetition, while the effect size divides by the **standard deviation**, which describes how much *individual participants* differed. One scales by precision, the other by variability among people. **Transfer cue:** naming the denominator names the question.

Solution 7.11

$d_z = 0.400 / 5.198 = 0.077$, approximately. Both problems divide by the standard deviation of the difference scores—4.993 in one study and 5.198 in the other—so the two effect sizes are computed the same way and are comparable with each other. That the denominators are nearly equal is what makes the contrast between 1.219 and 0.077 a contrast in magnitude rather than in scaling.

Solution 7.12

$d_z \times \sqrt{n} = 1.219 \times \sqrt{72} = 10.34$, recovering the t statistic. The identity shows that t carries exactly the effect size **plus the sample size**, and nothing else. A p value derived from t therefore cannot report magnitude: any effect size can be made to produce any p value by choosing n . **Decisive cue:** significance is magnitude times evidence, and only one factor is about the world.

Level 4 Solutions

Solution 7.13

“Among the 72 participants, mindfulness scores rose by an average of 6.086 points, 95% CI [4.913, 7.259]; $t(71) = 10.343$, $p < .001$, Cohen's $d_z = 1.219$ using the standard deviation of the difference scores.” It does not establish that the program caused the change, that the change matters in the program's context, or that the result generalizes beyond the represented participants.

Solution 7.14

“Among the 72 participants, mindfulness scores changed by an average of 0.400 points, 95% CI [-0.821, 1.621]; $t(71) = 0.653$, $p = .516$, Cohen's $d_z = 0.077$. The study did not detect an important average change and

was not precise enough to rule one out.” The careless sentence is “there was no effect,” which converts a failure to reject into a positive claim.

Solution 7.15

The interval adds **precision**: it says the data are compatible with standardized effects from about 0.9 to about 1.5, all of them substantial. Without it, a reader cannot tell whether a reported effect size of 1.219 came from a study that pinned it down or from one that barely constrained it—and those two studies contribute very differently to a body of evidence.

Solution 7.16

It shows that **two studies of identical design and size can produce very different magnitudes**, and that the difference is in the data rather than in the analysis, because both effect sizes use the same denominator definition and both studies have 72 pairs. The comparison does **not** license any claim that the two programs differ: no test of the difference between the two studies was specified or performed, and comparing two effect sizes by eye is not such a test.

Level 5 Solutions

Solution 7.17

The **estimate with its interval**, in points: roughly 4.9 to 7.3 points of average change. Only that is in units she can weigh against cost. The p value reports surprise under a null she has no interest in, and the standardized effect reports magnitude in units of between-person variability rather than in anything budgetable. What no analysis supplies is **what a point on this scale is worth** in her setting—a judgment that has to come from the program's purpose. And the design limit stands regardless: this is a change, not a demonstrated effect.

Solution 7.18

A reader can no longer determine **how large the change was, how precisely it was estimated, and whether the null value tested was the one that mattered**. The minimum replacement is the estimate with its units, a confidence interval, the test statistic with degrees of freedom, and an effect size with its denominator named. **Rejected rule**: “significance implies the rest.”

Solution 7.19

Reject it: a large p value means the data are unsurprising under the null, not that the null is true. The question being asked—whether the change is small enough not to matter—is answered by an **equivalence test against a prespecified margin**, which requires deciding in advance what magnitude would count as negligible. Nothing in this analysis supplies that margin, so the question cannot be answered from these data.

Solution 7.20

Because the null value was chosen **after seeing the estimate**. A p value's long-run properties depend on the hypothesis being fixed in advance; a hypothesis selected to sit near the observed value produces a p value whose stated meaning no longer holds, however correct the arithmetic. Reporting it as though five had been prespecified misrepresents the procedure that generated it. **Transfer cue**: a p value describes a procedure, and changing the procedure changes what it describes.

Level 6 Solutions

Solution 7.21

“Among the 72 participants who completed both questionnaires, mindfulness scores increased by an average of 6.086 points over the six-week interval, 95% CI [4.913, 7.259], $t(71) = 10.343$, $p < .001$, Cohen's $d_z = 1.219$, 95% CI [0.911, 1.522], computed using the standard deviation of the difference scores. Because all participants received the program and no comparison condition was observed, this estimates change over the interval rather than the program's effect.”

Solution 7.22

Two substitutions. **Significance for magnitude:** a small p reports that the data are surprising under a null, not that anything is large; magnitude is reported by the estimate and the effect size. **Association for causation:** “effect on mindfulness” asserts a causal relationship the one-group design cannot identify. Corrected: “Scores rose by an estimated 6.086 points, a large change relative to individual variation. The design does not establish that the program produced it.”

Solution 7.23

Defensible: “The study did not detect an important average change; the data are compatible with values from a small decrease to a modest increase, so the question remains open.” A stronger claim—that the program's effect is negligible—would require a prespecified equivalence margin and an analysis testing against it, plus enough precision to exclude values beyond that margin. **Claim ceiling:** inconclusive, not null.

Solution 7.24

Every element of the comparison fails. The two effect sizes may use **different denominators**—a paired d_z divides by the spread of within-person differences, a trial's d typically by a pooled between-group standard deviation—so they are not the same quantity. Neither figure is reported with an interval, so their apparent gap may be within joint uncertainty. The two studies measure different populations with possibly different instruments. And most decisively, this is a **one-group pre-post estimate** while the trial's is a controlled effect: even if the numbers were comparable, one describes change and the other an effect. **Transfer cue:** before comparing two effect sizes, check that they are the same quantity.

Study in the Larger Evidence Program: Reporting a Result That Did Not Replicate

- **What this study adds:** An inconclusive result reported with enough detail that it can enter a cumulative record rather than disappearing, which is what keeps the record from being biased toward striking findings.
- **What it cannot settle:** Whether the change is negligible, and whether the two pilots genuinely differ—neither question was posed in a form these data can answer.
- **Causal-support role:** Consistency evidence, and only if it is reported. An unreported inconclusive study makes the surrounding literature look more consistent than it is.
- **Next evidence priority:** A prespecified margin for a negligible change, and a design precise enough to distinguish it from one that matters.

Unit 8: Power, Precision, and Study Planning

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Unit Purpose

“How many participants do we need?” has no answer until someone says how large a difference would matter. Power calculations look like arithmetic and are mostly a disguised judgement call: the number that comes out is a consequence of the smallest effect you decide is worth detecting. This unit builds the habit of locating that judgement, stating it, and refusing the two calculations that hide it—one performed after the fact, and one that quietly substitutes the effect you hope for.

Learning Objectives

After completing this unit, a learner should be able to:

- state the four quantities linking sample size, effect, alpha, and power;
- identify the smallest effect worth detecting as a judgement, not an estimate;
- read a sample-size table and say what assumption each row encodes;
- explain why power computed from an observed effect adds nothing;
- distinguish an underpowered study from one that found nothing; and
- report a planned sample size with the reasoning that produced it.

Decision Map

Ask these questions in order:

1. What quantity is being estimated, and in what units?
2. What is the smallest difference that would change a decision?
3. Where did that number come from, and who agreed to it?
4. What error rate and what power are being targeted?
5. What sample size follows, and how sensitive is it to the assumed effect?
6. Is the resulting study affordable, and if not, what changes?
7. If the study runs and finds nothing, what will that mean?

Unit Study Files

Study	Data structure	Used here for	Unit role
Study 02	Pre and post scores for 72 synthetic participants	A prospective power record: detectable effect at the achieved size, and pairs required at three assumed effects	Planning the next study

Plausible Research Scenario

The scenario creates decisions rather than documenting real research. At the **Pause and Decide**, stop reading and write an answer, reason, and confidence rating. The explanation that follows is feedback on that commitment.

Plausible Research Scenario: How Many Students Does the Next Study Need?

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

The counseling centre has decided to run a proper comparison study, and the director asks the planning question: *how many students do we need?*

The team cannot answer it, and saying so is the first useful move. A sample size follows from four quantities: three conventional — the error rate, the target power, the form of the test — and one nobody has supplied. The missing one is **the smallest change that would matter**, not a statistical quantity at all but a decision about the programme: how much improvement would justify the facilitator time and the students' six weeks.

Two shortcuts beckon: planning on **the effect the pilot observed**, and computing power **after** a study from the effect it found. Both are wrong, and both are worked through below.

What the team can do is show the director the trade. At the pilot's 72 pairs, a standardised change of about 0.335 is detectable with 80% power; smaller changes need more pairs, and the requirement grows steeply as the target shrinks.

That turns an unanswerable question into an answerable one: not *how many do we need*, but *how small a change must we see, and can we afford to?*

Pause and Decide

Decide which of the four quantities the director must supply and which the team can set by convention. Then state what you would do if the implied sample size is larger than the centre can recruit.

Why This Model?

The calculation is prospective: assume an effect, fix alpha and power, solve for the number of pairs, using the exact noncentral-*t* distribution. Its inputs are assumptions, so its output is conditional—*if the underlying change is this large, a study of this size would detect it this often*—and not a prediction.

A retrospective calculation from the observed effect is rejected: it restates the *p* value in a form inviting misinterpretation. Planning from the pilot's point estimate is rejected because it treats an imprecise estimate as a target.

What Would a Favorable Result Look Like?

A favourable outcome is a defensible plan — possibly the finding that the study the centre wants cannot be afforded. Discovering that before recruiting is the point: a study too small to resolve the difference that matters consumes the same participants and answers nothing.

Best Defensible Claim

Statistical evidence: For a paired design at the stated error rate and power, the required number of pairs follows from the assumed standardised change.

Causal identification: Unaffected by sample size. A larger one-group study estimates change more precisely and identifies nothing.

Claim ceiling: The team may say what a design could detect under stated assumptions — not that the assumed effect is true, or that a well-powered study finding nothing has shown there is nothing.

What Must Be True?

The smallest change worth detecting must be set before the calculation and recorded with its justification; the assumed variability must be defensible for the population studied; and the planned analysis and design must be those the calculation assumed.

Defensibility Procedures

Record who set the smallest effect worth detecting and why; report the alpha, power, test and distributional assumption; show a range of assumed effects rather than one; state what the achievable sample size can detect; and never report power computed from an observed effect.

Next evidence priority: A comparison condition — power governs what a design can resolve, never what it can identify.

Claim Audit

Repair this claim: “Our study had 80% power, so the null result is trustworthy.” Say what would have to be true for a power statement to support a null result, and what the sentence probably means.

Problems

Do not consult the solutions until you have recorded an answer, reason, and confidence rating for every attempted item.

Level 1: Recall and Recognize

Problem 8.1 — Name the four quantities

Name the four quantities that determine one another in a power calculation. State which one is a judgement about the research question rather than a statistical convention.

Problem 8.2 — Power and precision

Distinguish statistical power from the precision of an estimate. State which one a confidence interval reports.

Problem 8.3 — Underpowered versus null

Distinguish a study that was too small to detect an important effect from a study that provides evidence against one. Name the piece of output that separates them.

Level 2: Predict Before Clicking

Problem 8.4 — Predict how required size moves with the target

Predict what happens to the number of participants required when the smallest effect worth detecting is halved. State whether the requirement doubles, and explain.

Problem 8.5 — Predict the effect of a stricter alpha

Predict what happens to the required sample size if the error rate is tightened from .05 to .01, holding the assumed effect and target power fixed.

Level 4: Read jamovi Output

Problem 8.6 — Read a detectable-effect statement

A planning record states that at 72 complete pairs, with two-sided alpha of .05, a standardised change of 0.335 is detectable with 80% power. Write what this sentence establishes and what a reader might wrongly take from it.

Problem 8.7 — Read a sample-size table

A planning table gives the complete pairs required for 80% power at three assumed standardised changes: 90 pairs at 0.30, 52 at 0.40, and 34 at 0.50. Describe the shape of the relationship and what each row assumes.

Problem 8.8 — Read a power claim in a results section

A results section reports a non-significant result and adds that the study “had adequate power.” State what you would need to see before accepting that, and what you would suspect if it were absent.

Level 5: Choose the Model

Problem 8.9 — Choose the input the calculation needs

Choose between the pilot's observed effect and the smallest effect worth detecting as the input to a prospective calculation. Justify the choice.

Problem 8.10 — Reject the retrospective calculation

A colleague computes power from the effect the study observed and reports it as evidence the design was adequate. Reject the calculation and say what it is a restatement of.

Problem 8.11 — Reject the affordable-effect substitution

A team cannot recruit the number of pairs its target requires, so it raises the assumed effect until the required number matches what it can recruit. Name what has been decided and by what.

Problem 8.12 — Reject power as a substitute for design

A draft argues that a large enough one-group study would establish that the programme works. Reject the argument.

Level 6: Report and Critique

Problem 8.13 — Write the planning paragraph

Write the sample-size paragraph for the next study: the smallest change worth detecting, its justification, the error rate and power, the resulting number of pairs, and what the study will be able to say if it finds nothing.

Problem 8.14 — Repair a post-hoc power claim

Repair this sentence: "Post-hoc power was 34%, so the study was underpowered and the null result should be discounted." Name the circularity and write what the report should say.

Problem 8.15 — Repair a plan with no stated target

Repair this sentence: "We recruited 90 participants, which gave us adequate power." Supply what is missing.

Problem 8.16 — Repair a conclusion from an underpowered study

Repair this sentence: "With only 34 pairs we found no significant difference, so the two approaches are equivalent." Distinguish the three claims being run together.

Worked Solutions

Each solution identifies the decisive cue, reasoning path, relevant evidence, rejected alternative, claim boundary, and a transfer cue.

Level 1 Solutions

Solution 8.1

Sample size, the **assumed effect size**, the **error rate** (alpha), and **power**. Fix any three and the fourth follows. The **assumed effect size** is the judgement: alpha and power are conventions, and sample size is usually the thing being solved for, but how large a change would matter is a question about the programme and its costs that no statistical procedure can answer. **Transfer cue**: find the judgement hiding inside the arithmetic.

Solution 8.2

Power is the probability that a study will produce a significant result *if* an effect of an assumed size exists; it is a property of a design, computed before data. **Precision** is how narrowly the study pins down the estimate; it is a property of a result. A confidence interval reports **precision**. Power concerns detection, precision concerns estimation, and a study can be adequately powered for a large effect while estimating it very imprecisely.

Solution 8.3

A study **too small** to detect an important effect produces a wide interval that includes both zero and effects that would matter. A study providing **evidence against** an important effect produces a narrow interval that excludes them. The piece of output that separates them is the **confidence interval**—specifically its width, not whether it contains zero. **Decisive cue**: both studies report a non-significant result, and only the interval tells you which one you are reading.

Level 2 Solutions

Solution 8.4

The requirement rises by a factor of about **four**, not two. Detectable effect scales roughly with one over the square root of the sample size, so halving the target effect requires quadrupling the sample. The planning table shows the same pattern: 34 pairs at an assumed change of 0.50 rises to 90 at 0.30 — the target falls by a factor of about 1.7 and the requirement rises by about 2.6. **Transfer cue**: small effects are disproportionately expensive.

Solution 8.5

The required sample size **increases**. A stricter error rate demands a more extreme test statistic before rejecting, so a larger sample is needed to reach it as often. This is the same trade seen with confidence levels in Unit 6: buying a lower false-positive rate costs either sample size or power, and the calculation makes the price explicit.

Level 4 Solutions

Solution 8.6

It establishes a **conditional statement about a design**: if the underlying standardised change were 0.335, a paired study with 72 complete pairs and two-sided alpha of .05 would produce a significant result about 80%

of the time. A reader might wrongly take it as a claim that the true change is 0.335, or that a change smaller than 0.335 does not exist — the statement is about what the design can detect, not about what is there.

Solution 8.7

The relationship is **steeply decreasing and non-linear**: 90 pairs at 0.30, 52 at 0.40, and 34 at 0.50. Raising the assumed effect from 0.30 to 0.40 saves 38 pairs; raising it again by the same amount, to 0.50, saves only 18. Each row assumes that the underlying standardised change is exactly that value, that the test is a two-sided paired t test at the stated alpha, and that 80% power is the target. **Transfer cue**: every row of a sample-size table is a different assumed world, not a range of uncertainty.

Solution 8.8

You would need the **assumed effect the power statement refers to**, its justification, and confirmation that the calculation was **prospective**. If those are absent, suspect a retrospective calculation from the observed effect — which is what “adequate power” usually means in a results section, and which carries no information. What should appear instead is the confidence interval, which shows directly what the study could and could not rule out.

Level 5 Solutions

Solution 8.9

Choose the **smallest effect worth detecting**. The calculation's purpose is to size a study against a decision, and the decision is defined by what magnitude would change what the centre does. The pilot's observed effect is an imprecise estimate of what happened once; using it as a target assumes the future study should be sized to detect exactly what a small study happened to see, which is neither a decision nor a reliable number. **Rejected rule**: “power the study for the effect we found.”

Solution 8.10

Reject it. Power computed from the observed effect is a **deterministic function of the p value**: a significant result always yields high computed power and a non-significant one always yields low, regardless of the design. It therefore cannot distinguish an adequate design from an inadequate one, which is the question being asked. It is a restatement of the result in different units. **Claim ceiling**: it supports nothing the p value did not already say.

Solution 8.11

What has been decided is the **smallest effect the study will be able to detect**, and it has been decided by the **recruitment budget** rather than by anything about the programme. That may be an acceptable constraint, but it must be reported as one: the honest version says the study can detect changes of at least the given size and would miss smaller ones, so the centre can decide whether such a study is worth running at all.

Solution 8.12

Reject it. Sample size governs **precision and power**, which determine what a study can resolve; it has no bearing on **identification**, which determines what a study can attribute. A very large one-group study estimates change with great precision and still observes no comparison condition, so every alternative explanation available to the small study remains available to the large one. **Transfer cue**: more data narrows an interval and never supplies a counterfactual.

Study in the Larger Evidence Program: Sizing the Next Mindfulness Study

- **What this study adds:** A stated smallest-change-worth-detecting and the sample size it implies, which converts the pilot's imprecision into a concrete design decision.
- **What it cannot settle:** Whether the assumed effect is the real one, and whether the programme works. Planning constrains what a future study could resolve, not what is true.
- **Causal-support role:** None on its own. It determines whether a future responsiveness test could resolve a difference worth resolving, which is a precondition for that evidence rather than the evidence itself.
- **Next evidence priority:** A comparison condition with a recorded assignment mechanism, sized by this calculation.

Level 6 Solutions

Solution 8.13

“The centre judged that an average improvement smaller than a standardised change of 0.40 would not justify the programme's cost; this threshold was set by the counseling team before any planning calculation. For a two-sided paired t test with alpha of .05 and 80% power, detecting a change of that size requires 52 complete pairs, and we will recruit to allow for attrition. If the study finds no significant difference, that will indicate the data are compatible with changes smaller than the threshold we set, not that the programme has no effect; the confidence interval will be reported and interpreted directly.”

Solution 8.14

The circularity is that post-hoc power is computed **from the observed effect**, so a non-significant result mechanically produces low computed power. The sentence therefore reduces to “the result was not significant, so the result should be discounted,” which is not an argument. Corrected: “The estimated change was [value], 95% CI [lower, upper]. The interval includes effects large enough to matter, so this study does not rule them out; it was not precise enough to settle the question.”

Solution 8.15

Missing: **what effect** the power was calculated for, **how much power**, at **what alpha**, and **where the assumed effect came from**. As written, “adequate power” is unfalsifiable — every sample size is adequate for a large enough assumed effect. Corrected: “We recruited 90 participants, which provides 80% power to detect a standardised change of 0.30 in a two-sided paired t test at alpha .05. The threshold of 0.30 was set by [whom] on the grounds that [reason].”

Solution 8.16

Three claims are being run together. “**We found no significant difference**” is a description of the test result and is fine. “**So the two approaches are equivalent**” is a positive claim of no difference, which requires a prespecified equivalence margin and an analysis testing against it. And “**with only 34 pairs**” concedes that the study was sized for a large effect, which makes the first observation unsurprising and the second less supportable, not more. Corrected: “With 34 pairs the study was sized to detect standardised changes of about 0.50 or larger. It found no significant difference, and its interval remains compatible with changes that would matter, so it does not establish equivalence.”

PART III

Choosing and using models

Unit 9: Relationships Between Quantitative Variables

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Unit Purpose

A relationship between two quantitative variables is not a fact waiting to be found. It is an estimate with a direction, a magnitude, a unit of measurement, an uncertainty, and a set of cases it actually describes. This unit develops the habit of reading a scatterplot, a correlation, and a fitted line as three views of the same estimate, and of stopping before the sentence that turns an association into a cause.

Two of the traps in this unit are quieter than causal overreach. The first is the sample the analysis silently describes: four practice logs are absent, so the relationship is estimated on sixty-eight participants while the change score itself is available for seventy-two. The second is a weak relationship read as though it were an absence of one.

Learning Objectives

After completing this unit, a learner should be able to:

- state which cases a complete-case relationship analysis actually describes;
- read direction, form, spread, and unusual points from a scatterplot before computing anything;
- interpret a correlation as a standardized, unitless summary of linear association;
- interpret an intercept and a slope in the measured units of both variables;
- read R^2 as proportion of variance accounted for rather than as importance or accuracy;
- distinguish leverage, a large residual, and influence, and respond to each without deleting data;
- separate a statistically undetectable relationship from a demonstrated absence of one; and
- reject causal, predictive, and generalizing language that the design does not support.

Decision Map

Ask these questions in order:

1. What is the scientific question, and which variable is being explained?
2. What are the cases, and which cases does the available data actually cover?
3. Is the relationship plausibly linear, and does the scatterplot show anything the summaries would hide?
4. What does one unit of the predictor mean, and what would a slope in those units mean?
5. What is the estimate, its interval, and its uncertainty?
6. Do any observations sit far out on the predictor, far from the line, or both?
7. What does the design identify, what is the claim ceiling, and what evidence should come next?

Study Files

Study	Cases used	Predictor and outcome	Unit role
Study 02 practice log	68 of 72 synthetic participants	Practice minutes predicting post-minus-pre change	Weak, imprecisely estimated relationship
Study 02 baseline record	All 72 synthetic participants	Pre score predicting post score	Strong relationship with a design-created explanation

Both analyses use the pinned synthetic file `study_02_mindfulness_paired.csv`. Its data dictionary and supporting public files are available at <https://github.com/nicholaskarlson/data>. Reported quantities were independently recomputed rather than copied from printed output.

Plausible Research Scenarios

The scenarios create decisions rather than documenting real research. At each **Pause and Decide**, stop reading and write an answer, reason, and confidence rating. The explanation that follows is feedback on that commitment.

Plausible Research Scenario: Does More Practice Go With More Change?

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

The fictional counseling centre has already estimated how much mindfulness scores moved across the six-week interval. A staff member now asks the question that always follows: did the participants who practised more change more? The centre's log records an average number of practice minutes per day for each participant who returned it, and the team treats logged practice as the predictor and post-minus-pre change as the outcome.

The question sounds like a small addition. It is not: it changes which participants are described. The change score exists for all seventy-two, but four participants did not return a usable practice log, and a relationship analysis needs both variables on the same person, so it runs on the 68 participants with complete records. That reduction must be settled before any coefficient is read. The record does not document why four logs are absent, so if those participants differed systematically the estimate is not merely less precise but slightly off-question.

The second decision is what a relationship would mean if one were found. Logged practice was chosen, not assigned, and whatever led a participant to practise more — motivation, symptom severity, free time, belief in the programme — may also be related to how much that participant changed. Practice minutes are self-reported, so the predictor carries measurement error that pulls an estimated relationship toward zero rather than away from it.

The team therefore prespecifies the analysis: a scatterplot, Pearson's correlation with a confidence interval, and a simple linear regression of change on practice minutes reported with the intercept, the slope, its standard error and interval, and R^2 . It also prespecifies what happens if one observation dominates the fit — that case is investigated and reported, and the model refitted without it as a stated sensitivity check alongside the primary fit rather than in place of it. The slope is in change-score points per logged minute per day, so a slope of one tenth would mean that ten extra minutes go with a one-point difference in average change. Whether that is large is a judgement about the mindfulness scale, not one the software can make.

Pause and Decide

Before reading further, write down which participants the analysis will describe, what the slope's units are, what a positive slope would and would not license, and what you would conclude if the slope's interval contained zero.

Why This Model?

Simple linear regression of change on practice minutes matches the question: one quantitative outcome, one quantitative predictor, and an interest in direction and magnitude in the outcome's own units. Pearson's correlation answers the standardized version of the same question; for one predictor the two are the same fit on two scales. Splitting practice into "high" and "low" halves would discard information and move the cut point into the analyst's hands. Regressing practice on change would answer a question nobody asked.

What Would a Favorable Result Look Like?

The staff member hopes for a clearly positive slope, large enough that a realistic difference in practice corresponds to a difference in change a student would notice, and estimated precisely enough to separate that from a negligible relationship. Precision, not the size of a p value, makes such a result usable. A near-zero estimate is also a scientifically successful result: it constrains an expectation the centre might otherwise build a programme around and redirects effort toward a design that could settle the question.

Best Defensible Claim

Statistical evidence: The regression estimates how much post-minus-pre change differs, on average, between represented participants whose logged practice differs by one minute per day, with its uncertainty. **Causal identification:** Practice was chosen rather than assigned, so any relationship is confounded with whatever else distinguishes participants who practise more, and the predictor's measurement error further distorts it. **Claim ceiling:** Describe an estimated association among the sixty-eight participants who returned a usable log; do not claim that practising more produces more change, that the relationship holds among the missing, or that a small estimate proves practice does not matter.

What Must Be True?

Each participant must contribute one independent pair of values, the pairing must be correct, the relationship must be close enough to linear over the observed range that a straight line is fair, residual spread should be roughly similar across that range, and the fit must not be driven by one or two observations. These model conditions are separate from what a causal claim would need.

Defensibility Procedures

Define the predictor, outcome, direction, and sensitivity rule before inspecting results. Plot before reading any coefficient, report the number of complete cases beside every estimate, and say that the analysis describes the reporting subgroup. Examine leverage, residuals, and influence together, investigate an influential case rather than deleting it, and report both fits. Report the correlation, the slope with its standard error and interval, the intercept, R^2 , the residual standard deviation, and the limits of a self-reported, unassigned predictor.

Next evidence priority: If the effect of practice dose is the target, assign practice amount rather than observe it and measure adherence directly. Merely enlarging this sample would estimate the same confounded quantity more precisely.

Claim Audit

Repair this claim: "Practice minutes did not predict change, so practice does not matter." Distinguish an imprecise estimate from a demonstrated absence, name the participants actually described, and say what the stronger claim would need.

Plausible Research Scenario: A Strong Relationship That Explains Nothing New

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

The same team asks a second relationship question, this time using all 72 participants: how well does a participant's baseline mindfulness score predict that participant's follow-up score? The predictor is the pre score and the outcome is the post score, and no participant is dropped because both values were recorded for everyone.

This scenario exists to make a point that a strong result can hide. The temptation is to treat strength as discovery. But the two scores measure the same construct on the same person six weeks apart, and a strong relationship is what the design guarantees when a measure is reliable and people differ from one another. It says a great deal about measurement stability and nothing about whether the programme worked.

The slope carries the second lesson. A slope of one would mean a one-point difference at baseline goes with a one-point difference at follow-up — a parallel shift preserving the ranking of participants — while a slope clearly below one would indicate regression toward the mean. The team prespecifies that it will compare the slope's interval with one, and will not read a slope near one as evidence of a uniform programme effect.

Pause and Decide

Identify what the pre-post relationship is a property of, predict whether it will be stronger or weaker than the practice relationship and say why, and name the claim this analysis cannot make no matter how strong it is.

Why This Model?

Regression of the post score on the pre score gives the slope in the score's own units and an interval comparable with one, which a correlation alone cannot do; the correlation is reported with it as the standardized summary. A paired comparison of the two means answers a different question — average change, not the relationship between the scores — and belongs to Unit 10.

What Would a Favorable Result Look Like?

There is no favorable direction here: the team is characterizing measurement behaviour, not testing a hypothesis. A useful result is a well-estimated slope and an honest account of how much follow-up variation baseline standing leaves unexplained.

Best Defensible Claim

Statistical evidence: The fit estimates how follow-up scores track baseline scores among the represented participants, and how much variation in follow-up scores baseline standing accounts for. **Causal identification:** None is attempted or possible; baseline standing is not manipulable and the design has no comparison condition. **Claim ceiling:** Describe the represented relationship and its implications for measurement and precision; do not treat a high R^2 as evidence about the programme.

What Must Be True?

The two scores must be scored comparably at both occasions, participants must be independent, and the relationship must be adequately linear across the observed baseline range. A ceiling or floor on the scale would bend it, and that must be visible in the plot before the line is trusted.

Defensibility Procedures

Plot before fitting. Report the slope with its interval and compare it with one explicitly. Report R^2 and the residual standard deviation together, so a reader sees what is explained and what is not. Record the relationship as a precision input for planning, not as a finding about the programme.

Next evidence priority: Use the estimated pre-post relationship to plan a comparison study — it determines how much a baseline adjustment would sharpen a future treatment-versus-control estimate — rather than reporting it as a result in its own right.

Claim Audit

Explain what is wrong with: “Baseline scores explained 75 percent of follow-up scores, so the programme's effect is highly predictable.” Name at least two distinct errors in that sentence.

Problems

Do not consult the solutions until you have recorded an answer, reason, and confidence rating for every attempted item.

Level 1: Recall and Recognize

Problem 9.1 — Name the analyzed cases

The Study 02 file contains 72 rows, and every row has a pre score and a post score. The practice-and-change analysis reports 68 cases. Explain where the other four went and state, in one sentence, who the reported relationship describes.

Problem 9.2 — Separate the three quantities

A report gives a correlation, a slope, and an R^2 for the same fit. Say what each one is measured in, and identify which of the three changes if the outcome is rescaled from points to tenths of a point.

Problem 9.3 — Define the slope in words

For the practice-and-change fit, the slope is expressed in mindfulness change-score points per logged minute of daily practice. Write the sentence that begins “Two represented participants whose logged practice differs by one minute per day differ, on average, by ...” and finish it without using the word *causes*.

Problem 9.4 — Distinguish leverage, residual, and influence

Define leverage, residual, and Cook's distance in one sentence each, then state which of the three can be large for an observation that sits exactly on the fitted line.

Level 2: Predict Before Clicking

Problem 9.5 — Predict from the plot description

A scatterplot of change against practice minutes shows points filling an approximately oval cloud with no visible tilt, spanning roughly five to thirty minutes on the horizontal axis. Predict the approximate size of the correlation and whether its confidence interval will contain zero.

Problem 9.6 — Predict the two relationships' relative strength

One analysis relates practice minutes to change; the other relates the pre score to the post score. Before seeing either result, predict which correlation will be larger in absolute value and give the design reason, not a guess about the programme.

Problem 9.7 — Predict the effect of measurement error

Logged practice is a self-reported daily average. Predict the direction in which random measurement error in the predictor moves an estimated correlation, and say whether that makes a near-zero estimate easier or harder to interpret.

Problem 9.8 — Predict what dropping a point will do

One participant has a change score far below the fitted line at a moderate practice value. Predict whether removing that participant would move the slope toward zero or away from it, and predict whether the conclusion about the relationship would change.

Level 3: Calculate and Explain

Problem 9.9 — Recover the slope from the correlation

For the practice-and-change fit, the correlation is -0.098, the standard deviation of change is 5.092 and the standard deviation of practice minutes is 5.808. Use $b = r \times (SD_y / SD_x)$ to approximate the slope, and explain why the correlation and the slope have the same sign but different sizes.

Problem 9.10 — Recover the intercept

The complete-case means are 20.185 logged minutes and 6.043 change-score points, and the slope is -0.086. Use $a = M_y - b \times M_x$ to approximate the intercept. Then say why the resulting value should not be read as “the change expected for a participant who practised zero minutes.”

Problem 9.11 — Turn a slope into a test statistic

The slope is -0.086 with a standard error of 0.107. Compute the ratio and state what quantity you have recovered. Then compute the approximate 95% interval using a critical value of 1.997 and check whether it contains zero.

Problem 9.12 — Compute and interpret the explained proportion

The practice-and-change fit has a total sum of squares of 1737.026 and a residual sum of squares of 1720.368. Compute R^2 . Then explain why the adjusted value for this fit is negative, and what a negative adjusted value indicates.

Level 4: Read jamovi Output

Problem 9.13 — Read a correlation matrix

The reconstructed jamovi Correlation Matrix for the complete-case analysis is below.

Correlation Matrix: practice_minutes with change_post_minus_pre	Value
Pearson's r	-0.098
p-value	.427
95% CI lower	-0.329
95% CI upper	0.144
N	68

Write two sentences: what the estimate and interval say, and what the p value does and does not add.

Problem 9.14 — Read a regression panel

The reconstructed jamovi Linear Regression output for the same fit is below.

Model Fit Measures and Overall Model Test	Value
R	0.098
R^2	0.010
Adjusted R^2	-0.005
Residual SD	5.106

Model Fit Measures and Overall Model Test	Value
F (df1 = 1, df2 = 66)	0.639
p	.427

Predictor	Estimate	SE	Lower	Upper	t	p
Intercept	7.776	2.254	3.274	12.277	3.449	< .001
practice_minutes	-0.086	0.107	-0.300	0.129	-0.799	.427

State what the slope row licenses, and explain why the intercept row's small p value is of no scientific interest here.

Problem 9.15 — Read the strong relationship

The baseline-and-follow-up fit reports the panel below.

Predictor	Estimate	SE	Lower	Upper	t	p
Intercept	7.037	2.950	1.153	12.921	2.385	.020
mindfulness_pre	0.978	0.067	0.845	1.111	14.650	< .001

Its correlation is 0.868, its R^2 is 0.754 and its residual standard deviation is 5.024. Summarize the relationship, compare the slope's interval with one, and state the one thing this strong result does not tell the centre.

Problem 9.16 — Read the influence and sensitivity record

For the practice-and-change fit, the largest Cook's distance is 0.255 for participant P069, whose standardized residual is -2.802, and the largest leverage is 0.117 for a different participant. The prespecified refit without P069 gives a slope of -0.019, $p = .855$. Explain what changed, what did not, and why the refit is reported rather than substituted.

Level 5: Choose the Model

Problem 9.17 — Relationship or comparison?

A researcher has one quantitative outcome and asks (a) whether it differs on average between two measurement occasions for the same people, and (b) whether it tracks a quantitative background variable across people. Choose a model for each and name the feature of the question that decides it.

Problem 9.18 — Reject the median split

A colleague proposes splitting practice minutes at the median and comparing mean change in the two halves, arguing that it is easier to explain. Give three reasons to keep the quantitative predictor, and name the one situation in which a grouped comparison would genuinely be the right analysis.

Problem 9.19 — Choose the direction of the fit

For the baseline-and-follow-up analysis, explain why regressing post on pre and regressing pre on post give different slopes but the same correlation. State which fit answers the team's question and why the choice cannot be made from the output.

Problem 9.20 — Mixed relationship clinic

Choose a starting analysis for each case and name the decisive cue: (a) daily step counts and daily mood ratings recorded for thirty days on each of forty people; (b) one anxiety score and one sleep-hours value per person for ninety unrelated adults; (c) a curved relationship between arousal and performance that rises then falls; and (d) an ordinal five-point satisfaction rating related to income category. For the first case, name the information that must be preserved.

Level 6: Report and Critique**Problem 9.21 — Repair a causal reading**

Critique: “More practice caused greater improvement, though the effect did not reach significance.” Identify at least four problems and write a defensible replacement using the practice-and-change fit.

Problem 9.22 — Repair a no-effect reading

Critique: “There was no relationship between practice and change.” Rewrite it so that it reports the estimate, the interval, the cases described, and what remains compatible with the data.

Problem 9.23 — Report the complete-case boundary

Write a two-sentence methods note for the practice-and-change analysis that states how many participants were analyzed, why the others were excluded, and what a reader should not assume about the excluded participants.

Problem 9.24 — Critique an R^2 claim

A draft states: “The model explained 75.4 percent of the variance, so follow-up scores can be predicted accurately from baseline scores.” Explain what R^2 does and does not license here, use the residual standard deviation to describe the actual prediction error, and write a corrected sentence.

Worked Solutions

Each solution identifies the decisive cue, reasoning path, relevant evidence, rejected alternative, claim boundary, and a transfer cue.

Level 1 Solutions

Solution 9.1

Four participants did not return a usable practice log, so their predictor value is absent. A relationship analysis needs both variables on the same case, and those four rows cannot contribute a point to the scatterplot; the analysis therefore runs on 68 complete cases. The reported relationship describes the sixty-eight participants who reported their practice, not the seventy-two participants in the file. **Transfer cue:** the analyzed sample is defined by the variables the analysis uses, not by the size of the file.

Solution 9.2

The correlation is unitless and bounded between -1 and 1. The slope is in outcome units per one unit of the predictor — here mindfulness change-score points per logged minute per day. R^2 is a proportion and is also unitless. Rescaling the outcome to tenths of a point multiplies the slope by ten and leaves the correlation and R^2 unchanged. **Rejected rule:** “a bigger slope means a stronger relationship.”

Solution 9.3

“Two represented participants whose logged practice differs by one minute per day differ, on average, by -0.086 mindfulness change-score points — that is, the participant with more logged practice shows the slightly smaller estimated change.” The sentence compares participants who already differ; it does not describe what would happen if one participant practised more. **Claim ceiling:** association among the represented reporting participants.

Solution 9.4

Leverage measures how far a case's predictor value sits from the mean of the predictor; it is a property of the predictor alone. A residual is the vertical distance between the observed outcome and the fitted line. Cook's distance combines the two and measures how much the fitted line moves when the case is removed. Leverage can be large for an observation lying exactly on the line: an extreme predictor value with a perfectly fitting outcome has high leverage, a zero residual, and no influence.

Study in the Larger Evidence Program: Practice Dose and Change

- **What this study adds:** A first, imprecise estimate of whether logged practice tracks change among the participants who reported it, plus an honest record of how many did not.
- **What it cannot settle:** Whether practice amount affects change, what the missing logs would have shown, or whether a dose-response relationship exists at all.
- **Causal-support role:** Weak responsiveness evidence at best; an observed dose measure with no assignment and self-report error cannot establish that outcomes respond to practice.
- **Next evidence priority:** Assign practice amount rather than observe it, and measure adherence directly, before treating any dose-response question as answerable.

Level 2 Solutions

Solution 9.5

An untitled oval cloud implies a correlation near zero — plausibly within about 0.2 of it in either direction — and a confidence interval that comfortably contains zero. The verified estimate is -0.098 with a 95% interval of [-0.329, 0.144]. Predicting from the picture first is the point: a plot that shows no tilt should not be followed by surprise at a small coefficient.

Solution 9.6

The pre-post relationship should be much stronger, and the reason is structural rather than substantive. The pre and post scores are two administrations of the same measure to the same people, so they inherit the stable differences between participants. Practice minutes and change scores are different constructs on different scales, and the change score has already removed each participant's baseline standing — the very thing that makes the pre-post relationship strong. **Transfer cue:** ask what a strong relationship would be a property of before predicting its size.

Solution 9.7

Independent random error in a predictor attenuates the estimated relationship — it pulls the correlation and slope toward zero. That makes a near-zero estimate harder, not easier, to interpret: the estimate is compatible with a genuinely absent relationship and with a real but modest one that the measure was too noisy to resolve. Reporting the interval rather than a verdict is what keeps both possibilities visible.

Solution 9.8

Removing a point that lies far below the line at a moderate predictor value pulls the fitted line upward at that end and moves the slope toward zero, because that case was the main source of the apparent downward tilt. The conclusion does not change: both the primary fit and the refit produce intervals that comfortably contain zero. A conclusion that survived only one version of the analysis would be a conclusion about that one observation.

Level 3 Solutions

Solution 9.9

$b = -0.098 \times (5.092 / 5.808) = -0.086$ change-score points per logged minute, approximately. The correlation and the slope always share a sign because both are driven by the same cross-product term. They differ in size because the slope carries the ratio of the two standard deviations, which converts a standardized association into the outcome's own units. **Decisive cue:** ask what units the answer must be in before choosing which quantity to report.

Solution 9.10

$a = 6.043 - (-0.086 \times 20.185) \approx 7.78$ change-score points, which matches the verified intercept of 7.776 once the rounded slope is allowed for. It should not be read as the expected change at zero practice because no represented participant logged zero minutes; the observed range starts at five minutes per day. The intercept here is the point where the fitted line meets the vertical axis, an arithmetic consequence of the fit rather than an estimate for a case type the data contain.

Solution 9.11

$-0.086 / 0.107 = -0.80$, approximately, recovering the t statistic for the slope; the verified value is -0.799. The approximate interval is $-0.086 \pm 1.997 \times 0.107$, which runs from about -0.300 to 0.129 and therefore contains

zero. The estimate is less than one standard error from zero, so the data do not distinguish a small negative relationship from a small positive one.

Solution 9.12

$R^2 = 1 - (1720.368 / 1737.026) = 0.010$. About one percent of the variation in change scores is accounted for by logged practice. The adjusted value is

-0.005 because the adjustment charges the model for the parameter it spent: when a predictor accounts for less variation than one parameter would be expected to absorb by chance, the adjusted value goes below zero. A negative adjusted value is a signal that the predictor has bought nothing, not a computational error.

Level 4 Solutions

Solution 9.13

Among the 68 participants with complete records, logged practice and post-minus-pre change were essentially unrelated: $r = -0.098$, 95% CI [-0.329, 0.144]. The interval is wide and spans both directions, so the data are compatible with a modest negative relationship, no relationship, and a modest positive one. The p value of .427 adds only that the estimate is not distinguishable from zero at a conventional threshold; it does not measure the size of the relationship and does not establish that the relationship is absent.

Solution 9.14

The slope row licenses this: among represented participants, a one-minute difference in logged daily practice corresponds to an estimated -0.086-point difference in mindfulness change, 95% CI [-0.300, 0.129], $t(66) = -0.799$, $p = .427$. Nothing in the row supports a causal or predictive claim. The intercept's small p value merely says the fitted line does not pass through the origin, which is unsurprising — participants changed by about six points on average regardless of practice — and tests a value of zero change at zero practice that no one hypothesized and no observation approaches. $R^2 = 0.010$ and an adjusted value of -0.005 confirm that practice accounts for essentially none of the variation in change.

Solution 9.15

Follow-up scores tracked baseline scores closely among the represented participants: $r = 0.868$, slope 0.978 points per baseline point, 95% CI [0.845, 1.111], $t(70) = 14.650$, $p < .001$, $R^2 = 0.754$. The slope's interval contains one, so the represented follow-up scores are consistent with a roughly parallel upward shift that preserves the ordering of participants rather than with strong regression toward the mean. What this does not tell the centre is anything about the programme: a reliable measure administered twice to people who genuinely differ will produce this relationship whether or not any intervention occurred.

Solution 9.16

What changed is the point estimate. Excluding P069 — the case with the largest Cook's distance, 0.255, and a standardized residual of -2.802 — moves the slope from -0.086 to -0.019, roughly four fifths of the way to zero. What did not change is the conclusion: both fits produce intervals containing zero and neither supports a relationship claim. The refit is reported alongside the primary fit because deleting a case that is not a demonstrated error and then reporting only the survivor would be a data-dependent choice made after seeing its effect. Note also that the highest leverage, 0.117, belongs to a different participant: being unusual on the predictor and being poorly fitted are different things.

Study in the Larger Evidence Program: Baseline Standing and Follow-Up

- **What this study adds:** Evidence that the mindfulness measure behaves stably across six weeks and that participants differ substantially from one another, which is a precision input for planning the next study.

- **What it cannot settle:** Anything about the programme; the relationship would look much the same with no programme at all.
- **Causal-support role:** No responsiveness or mechanism evidence; it contributes measurement-consistency evidence only.
- **Next evidence priority:** Use the estimated relationship to size and design a randomized comparison in which baseline standing is used for adjustment rather than reported as a finding.

Level 5 Solutions

Solution 9.17

Question (a) asks about an average difference between two linked measurements and calls for a paired comparison of means. Question (b) asks how one quantitative variable tracks another across cases and calls for correlation and simple regression. The decisive feature is whether the second quantitative variable plays the role of an outcome measured again or the role of a predictor measured once. **Rejected rule:** “two numeric columns means a relationship analysis.”

Solution 9.18

Keep the quantitative predictor because a median split discards the ordering and spacing within each half, reduces precision by replacing a graded predictor with two levels, and lets an analyst move the cut point until a result appears. It also makes the estimate depend on the sample's own median, so the same substantive relationship yields different groupings in different samples. A grouped comparison is genuinely right when the grouping is a real design feature — assigned conditions, or a category that exists independently of the observed distribution — not when it is manufactured from a continuous measurement.

Solution 9.19

The slope of post on pre divides the cross-product by the variation in pre; the slope of pre on post divides the same cross-product by the variation in post. The two differ whenever the two standard deviations differ, while the correlation, being standardized by both, is identical either way. The team's question is how follow-up standing relates to baseline standing, so post on pre is the fit that answers it. The output cannot make that choice: both fits are arithmetically valid, and only the question distinguishes them.

Solution 9.20

(a) A model that respects repeated days within person — the thirty observations from one person are not thirty independent cases; preserve the person identifier and the day ordering, and do not pool all 1,200 rows into a single correlation. (b) Correlation and simple linear regression on ninety independent cases, after plotting. (c) A relationship model that permits curvature, chosen because the scatterplot shows a rise and then a fall; a straight line through a curve can return a near-zero slope while a strong relationship is present. (d) A method suited to ordinal data, such as a rank-based association, because the five satisfaction points are ordered but not equally spaced. **Transfer cue:** the decisive cue is the structure of the cases and the measurement level, not the number of columns.

Level 6 Solutions

Solution 9.21

The statement asserts causation from an unassigned, self-reported predictor; treats a p value above a threshold as a weakened version of a demonstrated effect rather than as an absence of evidence; reports no estimate, interval, or sample; and ignores that the sign of the estimate is negative, so even the claimed direction is not what the data show. A defensible replacement is: “Among the 68 represented participants who returned a usable practice log, logged practice showed essentially no association with post-minus-pre mindfulness

change: $r = -0.098$, slope -0.086 points per logged minute, 95% CI $[-0.300, 0.129]$, $t(66) = -0.799$, $p = .427$, $R^2 = 0.010$. Because practice was chosen rather than assigned and was self-reported, no causal interpretation is warranted.”

Solution 9.22

“There was no relationship” asserts an exact zero that the data cannot support. A defensible replacement is: “Among the sixty-eight participants with complete practice logs, the estimated association between logged practice and change was

-0.098 , 95% CI $[-0.329, 0.144]$. The data are compatible with a modest negative association, no association, and a modest positive one; they do not provide evidence of a meaningful relationship in either direction, and they do not establish its absence. Establishing absence would require a prespecified equivalence margin and a design able to resolve it.”

Solution 9.23

“Relationship analyses used the 68 participants with both a practice log and a complete pair of mindfulness scores; four participants were excluded because no usable practice log was recorded. The reason for the missing logs is not documented, so readers should not assume that the excluded participants resemble the analyzed ones in either practice or change, and the estimates describe the reporting subgroup rather than the full sample.”

Solution 9.24

$R^2 = 0.754$ says that baseline standing accounts for about three quarters of the variance in follow-up scores across the represented participants. It does not say that individual follow-up scores can be predicted accurately, and it says nothing about the programme's effect. The residual standard deviation of 5.024 points is the honest accuracy statement: typical prediction errors are around five points on the mindfulness scale, so a prediction interval for one participant spans roughly twenty points. A corrected sentence is: “Baseline scores accounted for 75.4 percent of the variance in follow-up scores among the represented participants, with a residual standard deviation of about five points; individual follow-up scores therefore remain substantially uncertain, and this relationship carries no information about the programme's effect.”

Unit Reflection Log

After attempting the unit, record:

- the problem number;
- your original answer and confidence;
- the cue you used and the cue you missed;
- the corrected decision rule;
- the claim ceiling you should have noticed;
- what the study adds to a larger evidence program; and
- the date on which you will attempt a new version.

Unit 10: Comparing Two Conditions

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Unit Purpose

Two score columns do not automatically require a paired test, and two group labels do not automatically require an independent test. The decisive question is how observations are related. This unit develops the habit of identifying the cases, outcome, condition structure, comparison direction, and source of dependence before selecting a model.

Learning Objectives

After completing this unit, a learner should be able to:

- distinguish linked measurements from measurements on independent groups;
- define the estimand and comparison direction before interpreting a sign;
- explain why pairing changes the estimated standard error;
- interpret an estimate, interval, test statistic, p value, and effect size together;
- prespecify Welch's independent-samples procedure when unequal spreads and group sizes make pooling unattractive;
- separate statistical evidence from causal identification; and
- state what one study adds to a larger evidence program.

Decision Map

Ask these questions in order:

1. What is the scientific question and quantitative outcome?
2. What are the cases—the independent units that contribute information?
3. Are two measurements linked within the same case or a defensible matched pair?
4. If not linked, do observations come from independent groups or from a clustered structure?
5. What direction defines the reported difference?
6. What estimate, interval, effect definition, and diagnostic evidence answer the question?
7. What does the design identify, what is the claim ceiling, and what evidence should come next?

Study Files

The primary paired and independent-groups datasets are `study_02_mindfulness_paired.csv` and `study_03_feedback_two_groups.csv`. Scenario variants are fully specified in the problems and worked solutions. Public CSV files and dictionaries are available at <https://github.com/nicholaskarlson/data>.

Study	Data structure	Primary outcome	Unit role
Study 02	Pre and post scores for 72 synthetic participants	Mindfulness score	Clear paired comparison
Study 02C	Pre and post scores for 72 synthetic participants	Mindfulness score	Near-zero paired contrast
Study 03	48 process-feedback and 48 outcome-feedback cases	Motivation score	Balanced independent comparison
Study 03A	28 process-feedback and 68 outcome-feedback cases	Motivation score	Unequal-size, unequal-variance contrast

Plausible Research Scenarios

The scenarios create decisions rather than documenting real research. At each **Pause and Decide**, stop reading and write an answer, reason, and confidence rating. The explanation that follows is feedback on that commitment.

Plausible Research Scenario: A Mindfulness Pilot Before and After Six Weeks

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

A university counseling center is considering a six-week mindfulness program for students who voluntarily seek help managing everyday stress. Before investing in a larger evaluation, the center's fictional research team wants to learn whether the program and measurement procedures are feasible and how mindfulness scores change across the six-week interval among participants who complete both questionnaires.

The team recruits volunteers from one program setting. Each participant receives an identifier, completes the same mindfulness questionnaire immediately before the interval, and completes it again after six weeks. The research question is: **How much did mindfulness scores change, on average, among the represented participants from before to after the interval?** The target quantity is the mean post-minus-pre score difference among those participants. One participant—not one score—is the independent unit. The two scores in a row remain linked through the participant identifier.

Study 02 represents a clear positive average change. Study 02C preserves the same design and starting scores but represents a near-zero, imprecisely estimated average change. The contrast prevents model choice from depending on whether the eventual result is favorable.

Pause and Decide

Before reading further, identify the independent unit, explain why the two measurements are paired, define the subtraction direction, choose the model, and state the strongest claim this one-group design could support. Then name one result that would be scientifically useful even if it were not favorable.

Why This Model?

A paired-samples model analyzes one post-minus-pre difference for each participant. That matches the within-person question and preserves the linkage created by repeated measurement. An independent-samples test would pretend that the pre and post scores came from unrelated people. A between-groups one-way ANOVA would represent the same wrong independence structure.

The model estimates change; the design does not uniquely identify why change occurred. There is no comparison condition and no random assignment in the represented evidence. History, maturation, testing effects, changing measurement conditions, attrition, and regression toward the mean remain possible explanations.

What Would a Favorable Result Look Like?

The researchers would hope to see a positive mean post-minus-pre change large enough to matter in the program context, accompanied by an interval narrow enough to distinguish that change from negligible or harmful possibilities. The direction and magnitude matter more than whether a threshold is crossed. A small p value would not show that the program caused the change.

A near-zero estimate can also be scientifically successful. If the procedures were implemented faithfully and the interval is informative, it can challenge an exaggerated expectation, guide planning, or reveal that a more sensitive measure or stronger design is needed. An interval containing zero is not proof that the true effect is exactly zero.

Best Defensible Claim

Statistical evidence: The paired analysis estimates the average post-minus-pre difference among the represented participants and quantifies its uncertainty. Study 02 and Study 02C answer the same estimand under the same paired logic even though their numerical patterns differ.

Causal identification: The one-group pre-post comparison does not identify the program's causal effect because the counterfactual change without the program is not observed or represented by a defensible comparison group.

Claim ceiling: The researchers may describe estimated change over the represented interval among these participants. They may not claim that the program uniquely caused the change or that the result automatically generalizes to other students or settings.

What Must Be True?

Participant matching must be correct, each participant's difference must be independent of other participants' differences, and the same outcome must be scored comparably at both occasions. The mean difference must be a meaningful summary, and data errors or extreme influence must not make the t -based summary misleading. These model conditions are separate from the design requirements needed for a causal claim.

Defensibility Procedures

The team should define post minus pre and the primary outcome before inspecting results. It should document recruitment, timing, program exposure, attrition, concurrent events, questionnaire administration, reliability, and scoring. It should inspect missing pairs and the distribution of difference scores, investigate influential observations without deleting them mechanically, and report the mean change, interval, test, effect-size definition, number of complete pairs, and limitations.

Next evidence priority: If the causal effect of the program is the target, the next study should introduce a credible comparison condition—ideally with random assignment—while preserving measurement quality and monitoring implementation and attrition. Later studies could test a proposed mechanism and repeat the comparison in settings that challenge its generality.

Claim Audit

Repair this claim before continuing: “The paired t test will prove that the program works if p is below .05.” Your repair should distinguish an estimated change from a causal effect, statistical significance from scientific success, and the current study from the larger evidence program.

Plausible Research Scenario: Motivation in Two Existing Feedback Contexts

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

A learning-support office observes students in two existing feedback contexts. One context usually emphasizes process feedback; the other emphasizes outcome feedback. Each student participates in only one context and contributes one motivation score. The question is how different the two represented context means are, defined as process feedback minus outcome feedback. Recruitment is by convenience, and no random assignment is represented.

Study 03 has equal group sizes and similar spreads. Study 03A has 28 process cases with a standard deviation of about 12 and 68 outcome cases with a standard deviation of about 5. That contrast makes the variance model consequential.

Pause and Decide

Identify why the groups are independent, choose the comparison and its variance model before viewing any p values, define the direction, and state the observational claim ceiling.

Why This Model?

Welch's independent-samples t procedure compares two independent means without imposing a pooled equal-variance model. It is a defensible default here and is especially well matched to Study 03A's unequal sizes and spreads. A paired model would invent links that do not exist. Selecting Student's pooled line because it produces a smaller p value would be result shopping.

What Would a Favorable Result Look Like?

A favorable pattern would be a practically meaningful process-minus-outcome mean difference estimated precisely enough to rule out important alternatives. Scientific success remains an honest estimate with uncertainty even if its interval contains zero.

Best Defensible Claim

Statistical evidence: Welch's procedure estimates the difference between the two represented context means.

Causal identification: Pre-existing groups do not identify the effect of feedback framing because selection and other context differences remain. **Claim ceiling:** The result describes an observational difference in the represented contexts, not a causal effect or a population-wide law.

What Must Be True?

Participants must contribute independent scores unless clustering is modeled; the motivation scores must be comparable across contexts; and the means must be useful summaries not dominated by errors or a few observations.

Defensibility Procedures

Prespecify Welch, the process-minus-outcome direction, and the primary outcome. Record recruitment, clustering, prior differences, implementation, missingness, and measurement conditions. Report both group summaries, the raw difference, Welch interval and test, the effect-size denominator, sample analyzed, and the observational limitation.

Next evidence priority: Randomly assign a manipulable feedback strategy if the causal effect of feedback is the intended estimand; separately replicate across instructors and settings to study consistency and generalization.

Claim Audit

Explain why “use whichever t test gives $p < .05$ ” is scientifically indefensible even when the two software lines are both available.

Problems

Do not consult the solutions until you have recorded an answer, reason, and confidence rating for every attempted item.

Level 1: Recall and Recognize

Problem 10.1 — Find the independent units

Study 02 contains a pre score and a post score in every row. Are there 72 or 144 independent units for the condition comparison? Name the feature that supports your answer.

Problem 10.2 — Name the model-changing fact

Study A measures every person in both conditions. Study B measures different people in the two conditions. What fact changes the appropriate comparison even when both studies use exactly two condition labels?

Problem 10.3 — State the paired estimand

Study 02 defines its difference as mindfulness post minus mindfulness pre. State the estimand in words. Why is “the difference between two columns” inadequate?

Problem 10.4 — Separate analysis from causal identification

The paired analysis is correctly specified for Study 02. Does that fact identify the causal effect of the fictional program? Explain using the missing counterfactual comparison.

Level 2: Predict Before Clicking

Problem 10.5 — Commit to a direction

Study 02's post mean is 49.379 and pre mean is 43.293. Predict the sign and rough size of the post-minus-pre mean difference. What would a negative sign mean?

Problem 10.6 — Predict uncertainty

A new paired dataset has the same mean change and number of pairs as Study 02 but a much larger standard deviation of the difference scores. Predict what happens to the standard error, confidence-interval width, and absolute t statistic.

Problem 10.7 — Predict the near-zero case

Study 02C has a mean post-minus-pre change of 0.400 and a standard error of 0.613. Before reading its test output, predict the approximate t statistic and whether a two-sided 95% interval is likely to contain zero.

Problem 10.8 — Predict the Welch consequence

In Study 03A, the smaller process group has the larger spread. Explain why the Welch standard error should be larger than a pooled equal-variance standard error that lets the larger, less-variable group dominate the pooled variance. Predict which interval will be wider.

Level 3: Calculate and Explain

Problem 10.9 — Reconstruct a paired standard error

For Study 02, the standard deviation of 72 post-minus-pre differences is 4.993. Use $SE = SD_{\text{difference}} / \sqrt{n}$ to approximate the standard error. Explain why the separate pre and post standard deviations are not substituted.

Problem 10.10 — Connect estimate and test statistic

Divide Study 02's mean difference of 6.086 by its standard error of 0.588. What quantity do you recover, and what does it represent?

Problem 10.11 — Calculate a paired standardized effect

Use Study 02C's mean difference of 0.400 and difference-score standard deviation of 5.198 to approximate Cohen's d_z . What does the small result say, and what does it not prove?

Problem 10.12 — Reconstruct the Welch standard error

Study 03A has process $n = 28$, $SD = 12.001$ and outcome $n = 68$, $SD = 4.992$. Use $\sqrt{(SD_1^2/n_1 + SD_2^2/n_2)}$ to approximate the Welch standard error. Divide the 4.000-point difference by that value.

Level 4: Read jamovi Output

Problem 10.13 — Integrate clear paired evidence

Study 02 reports a 6.086-point post-minus-pre difference, 95% CI [4.913, 7.259], $t(71) = 10.343$, $p < .001$, and Cohen's $d_z = 1.219$. Write three sentences: estimate and direction, uncertainty and standardized magnitude, then design boundary.

Problem 10.14 — Interpret an inconclusive paired result

Study 02C reports a 0.400-point difference, 95% CI [-0.821, 1.621], $t(71) = 0.653$, $p = .516$, and $d_z = 0.077$. What possibilities remain compatible with the interval? Why is “there was no effect” too strong?

Problem 10.15 — Read a balanced independent comparison

Study 03 reports process $M = 69.342$, $SD = 7.802$ and outcome $M = 63.952$, $SD = 7.837$. Welch's result is a 5.390-point process-minus-outcome difference, 95% CI [2.220, 8.559], $t(94.00) = 3.377$, $p = .001$, Hedges' $g = 0.684$. Summarize the evidence without making a causal claim.

Problem 10.16 — Resolve competing software lines

For Study 03A, Student's line gives $t(94) = 2.317$, $p = .023$, CI [0.572, 7.428]. Welch's line gives $t(30.92) = 1.704$, $p = .098$, CI [-0.788, 8.788]. Which line answers the prespecified analysis and why? What should be reported about the 4.000-point estimate?

Level 5: Choose the Model

Problem 10.17 — Same people or different people?

A researcher records anxiety before and after a workshop for the same 30 students. Another compares 30 workshop students with 30 different wait-list students. Choose the comparison for each and reject the tempting alternative.

Problem 10.18 — Ignore file shape

A repeated-measures dataset is stored in long format with one score column and two rows per participant. Which field reveals pairing, and which comparison is appropriate? Explain why counting columns fails.

Problem 10.19 — Choose the variance model prospectively

A planned comparison expects unequal group sizes and no scientific reason to assume equal variances. Choose Student's pooled test or Welch's test before data inspection. Explain the decision and why a preliminary variance-test threshold should not be used to switch between the two analyses.

Problem 10.20 — Mixed model-choice clinic

Choose a starting model for each case and state the decisive cue: (a) the same patients under two medication periods; (b) different volunteers in two therapy formats; (c) students in several classrooms assigned at the classroom level to two feedback conditions; and (d) twins deliberately matched across two conditions. For the clustered and matched cases, name the information that must be preserved.

Level 6: Report and Critique**Problem 10.21 — Repair causal overreach in a paired study**

Critique: “The paired t test proved the mindfulness program caused a large improvement because p was below .05.” Identify at least four problems and write a defensible replacement using Study 02.

Problem 10.22 — Repair significance-only reporting

Critique: “There was a significant difference between the feedback groups.” Rewrite it using Study 03's group summaries, mean difference, interval, test, effect size, and observational boundary.

Problem 10.23 — Report the near-zero case honestly

Write a concise result for Study 02C. It must avoid both “proved no effect” and “almost worked,” and it must identify the range of change compatible with the interval.

Problem 10.24 — Expose result shopping

A report presents only Student's $p = .023$ for Study 03A and omits the prespecified Welch result. Explain why this is misleading and write a transparent replacement.

Worked Solutions

Each solution identifies the decisive cue, reasoning path, relevant evidence, rejected alternative, claim boundary, and a transfer cue.

Level 1 Solutions

Solution 10.1

There are **72 independent units**: the synthetic participants. The 144 scores are not 144 unrelated observations because participant_id links each pre score to the corresponding post score. The analysis uses 72 within-participant differences. Counting measurements without identifying dependence would overstate the information. **Transfer cue**: count independent cases, not cells.

Solution 10.2

The model-changing fact is whether observations are **dependent or independent**. When the same person completes both conditions, the two scores form a pair and the analysis targets within-person differences. When each person contributes to only one group, group means are compared as independent. The number of labels alone cannot decide. **Rejected rule**: “two labels means an independent *t* test.”

Solution 10.3

The estimand is the **mean post-minus-pre mindfulness-score difference among the represented participants**. “Difference between two columns” omits the cases, linkage, direction, outcome, and target population. Columns are a storage choice; the estimand is a scientific quantity. The same paired estimand could be stored in long format with one score column and two rows per participant.

Solution 10.4

No. Correctly specifying the paired analysis estimates change but does not identify the program's causal effect. The unobserved counterfactual is how the same participants would have changed across the same interval without the program or under a relevant alternative. The design supplies no comparison condition for that change. History, maturation, testing effects, attrition, and regression toward the mean remain possible. **Claim ceiling**: describe within-sample change, not causal impact.

Study in the Larger Evidence Program: Mindfulness Pilot

- **What this study adds**: A reproducible estimate of six-week within-person change and evidence about feasibility, measurement, and individual variation.
- **What it cannot settle**: Whether the program caused change or whether the pattern generalizes beyond the represented participants.
- **Causal-support role**: Primarily preliminary consistency and measurement evidence; it does not provide strong responsiveness evidence without a defensible intervention comparison.
- **Next evidence priority**: A randomized comparison condition with implementation and attrition checks, followed by mechanism and generalization studies.

Level 2 Solutions

Solution 10.5

The difference should be positive and about $49.379 - 43.293 = 6.086$ points. Under post minus pre, a negative value would mean that the post score was lower than the pre score. A sign is not self-interpreting; its meaning comes from the prespecified subtraction direction.

Solution 10.6

With the number of pairs fixed, a larger standard deviation of difference scores produces a larger standard error because $SE = SD_{\text{difference}} / \sqrt{n}$. The confidence interval becomes wider. For the same nonzero mean change, the absolute t statistic becomes smaller because the estimate is divided by a larger standard error. **Transfer cue:** more variability weakens precision even when the estimated mean stays unchanged.

Solution 10.7

The approximate statistic is $0.400 / 0.613 = 0.65$. A two-sided 95% interval is likely to contain zero because the estimate is less than one standard error from zero. The verified result is $t(71) = 0.653$ with CI $[-0.821, 1.621]$. The prediction comes from estimate relative to uncertainty, not from hoping for a particular conclusion.

Solution 10.8

Welch's standard error allows each group variance to contribute as s^2/n . Study 03A's small process group has the larger variance, so its uncertain mean receives an appropriately large contribution. Pooling spreads lets the larger, less-variable outcome group heavily influence the common variance estimate. The Welch interval should therefore be wider. This choice should be made from the analysis problem, not after comparing p values.

Level 3 Solutions**Solution 10.9**

$SE = 4.993 / \sqrt{72} = 0.588$ points, approximately. A paired analysis first forms one difference per participant, so uncertainty is based on variation among those differences. Substituting the separate pre and post standard deviations would ignore their within-person correlation and answer an independent-groups question. **Decisive cue:** the same participants provide both measurements.

Solution 10.10

$6.086 / 0.588 = 10.35$, approximately, recovering the paired t statistic (10.343 before rounding). It represents the estimated mean difference measured in estimated standard-error units from the null value of zero. It does not measure practical importance or causal credibility.

Solution 10.11

$d_z = 0.400 / 5.198 = 0.077$, approximately. The estimated mean change is small relative to variation among individual changes. This does not prove an exact zero effect, universal absence of benefit, or equivalence within a prespecified margin. The interval and design must also be considered.

Solution 10.12

The Welch standard error is $\sqrt{(12.001^2/28 + 4.992^2/68)} = 2.347$ points, approximately. Dividing the 4.000-point difference by 2.347 gives $t = 1.704$. This matches the verified Welch line. The calculation shows why the less precise process-group mean matters strongly even though the total sample contains 96 cases.

Level 4 Solutions**Solution 10.13**

Among the represented participants, mindfulness scores increased by an estimated 6.086 points on average from pre to post. The 95% confidence interval of $[4.913, 7.259]$ indicates a positive, fairly precise average change under the model, and Cohen's $d_z = 1.219$ describes a large change relative to the standard deviation of individual differences; $t(71) = 10.343$, $p < .001$. Because this is a one-group pre-post design, the result

estimates change but does not establish that the fictional program caused it or that it generalizes to another population.

Solution 10.14

The point estimate is a small 0.400-point increase. The interval [-0.821, 1.621] remains compatible with a modest decrease, no change, or a modest increase on the score scale. The data do not provide precise evidence of an important positive average change, but “there was no effect” treats failure to exclude zero as proof of exact absence. Equivalence would require a justified equivalence margin and an analysis designed for that question.

Solution 10.15

The represented process-feedback context had a mean motivation score of 69.342 (SD = 7.802) compared with 63.952 (SD = 7.837) in the outcome-feedback context. The estimated process-minus-outcome difference was 5.390 points, 95% CI [2.220, 8.559], Welch's $t(94.00) = 3.377$, $p = .001$, with Hedges' $g = 0.684$. These statistics describe a moderate standardized observational difference in the represented contexts; they do not identify a causal effect of feedback framing because assignment was not randomized.

Solution 10.16

Report Welch because it was chosen from the unequal-size, unequal-variance analysis problem before inspecting results. The raw estimate remains a 4.000-point process-minus-outcome difference, but its uncertainty is substantial: Welch's 95% CI is [-0.788, 8.788], $t(30.92) = 1.704$, $p = .098$. Student's pooled line is narrower because it imposes a common variance poorly matched to the represented case. The disagreement illustrates a model consequence; it does not justify choosing the significant line.

Study in the Larger Evidence Program: Feedback Contexts

- **What this study adds:** Estimates of contextual mean differences and a demonstration of how variance modeling changes uncertainty.
- **What it cannot settle:** Whether feedback framing caused the difference, which pre-existing factors explain selection, or whether the result generalizes.
- **Causal-support role:** Possible consistency evidence if the pattern recurs across defensible settings; no strong responsiveness evidence is represented.
- **Next evidence priority:** Randomized manipulation of feedback with comparable delivery, followed by replication across instructors and tests of proposed motivational mechanisms.

Level 5 Solutions

Solution 10.17

The before-and-after workshop study requires a paired comparison because each student's two scores form a dependent pair. The workshop-versus-wait-list study requires an independent-groups comparison because each person contributes to one group. An independent test for the first discards pairing; a paired test for the second invents it. The group labels are less decisive than the link between observations.

Solution 10.18

The participant identifier reveals that two rows belong to the same person; the occasion or condition field identifies which measurement each row contains. The appropriate comparison remains paired. Wide versus long layout changes the shape of the file, not the dependence structure or estimand.

Solution 10.19

Choose Welch's independent-samples procedure. It does not require a pooled common variance and behaves well when group sizes or spreads differ. The choice comes from the planned estimand and variance model, not from a preliminary test. Student's pooled procedure adds an unnecessary restriction whose consequences can be serious when the smaller group has the larger variance. A nonsignificant variance test does not establish equality, and switching analyses at one noisy threshold creates a data-dependent procedure. Prespecifying Welch avoids that threshold-shopping rule.

Solution 10.20

(a) Paired comparison: the same patients contribute both medication-period scores. (b) Independent-samples comparison: different volunteers contribute to the two formats. (c) A clustered two-condition analysis that represents classroom-level assignment and within-classroom dependence; preserve classroom identifiers and the assignment record, and do not count students as fully independent. (d) Paired or matched-pairs comparison if each twin pair was defined before outcomes and one member received each condition. Preserve the twin-pair identifier and assignment within pairs. **Transfer cue:** locate every design-created link, not merely a similar group size.

Level 6 Solutions**Solution 10.21**

The statement confuses statistical significance with causality, uses “proved,” omits the estimate and interval, calls the effect large without naming the standardization, and ignores the one-group design. A defensible replacement is: “Among the 72 represented participants, mindfulness scores increased by an estimated 6.086 points from pre to post, 95% CI [4.913, 7.259], $t(71) = 10.343$, $p < .001$, Cohen's $d_z = 1.219$. The one-group pre-post comparison does not establish that the fictional program caused the change.”

Solution 10.22

“Mean motivation was 69.342 (SD = 7.802) in the represented process-feedback context and 63.952 (SD = 7.837) in the outcome-feedback context. The estimated process-minus-outcome difference was 5.390 points, 95% CI [2.220, 8.559], Welch's $t(94.00) = 3.377$, $p = .001$, Hedges' $g = 0.684$. Because the groups were pre-existing rather than randomly assigned, the result describes an observational difference and does not identify a causal effect of feedback.”

Solution 10.23

“The represented participants showed an estimated mean post-minus-pre change of 0.400 points, 95% CI [-0.821, 1.621], $t(71) = 0.653$, $p = .516$, Cohen's $d_z = 0.077$. The estimate was small and the interval remained compatible with a modest decrease through a modest increase; the analysis therefore did not provide precise evidence of a meaningful positive average change. It did not prove that change was exactly zero.”

Solution 10.24

Reporting only Student's significant line conceals the prespecified model and turns analysis selection into result shopping. A transparent replacement is: “The process context exceeded the outcome context by 4.000 points on average. The prespecified Welch analysis, appropriate to the unequal sizes and spreads, gave 95% CI [-0.788, 8.788], $t(30.92) = 1.704$, $p = .098$. Student's pooled line was narrower and crossed .05, but it relied on the common-variance model that the analysis did not adopt.”

Unit Reflection Log

After attempting the unit, record:

- the problem number;
- your original answer and confidence;
- the cue you used and the cue you missed;
- the corrected decision rule;
- the claim ceiling you should have noticed;
- what the study adds to a larger evidence program; and
- the date on which you will attempt a new version.

Unit 11: Comparing Three or More Conditions

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Unit Purpose

Unit 10 asked which observations are linked. That question does not go away when a third condition appears; it is joined by a second one. With three conditions there are three possible pairwise comparisons, and the analyst must decide, before seeing any of them, which comparisons the research question actually asks. An omnibus test answers whether the condition means differ somewhere. It does not say where, and it is not a licence to hunt.

This unit develops the habit of separating four things a three-group analysis routinely blurs: the omnibus question, the specific comparisons that answer the research question, the familywise cost of asking several questions at once, and the claim ceiling that an undocumented assignment mechanism imposes no matter how large the F statistic becomes.

Learning Objectives

After completing this unit, a learner should be able to:

- identify the independent units and condition structure before choosing a three-group model;
- state what an omnibus test does and does not establish;
- distinguish prespecified comparisons from comparisons selected after seeing the means;
- explain why a familywise procedure widens intervals and why that is a feature;
- read a one-way analysis-of-variance table, a post-hoc table, and a contrast table together;
- interpret η^2 and ω^2 with an interval rather than as a verdict;
- reject the tempting reduction of a three-group question to a series of two-group tests; and
- keep an undocumented assignment mechanism visible in every sentence of the conclusion.

Decision Map

Ask these questions in order:

1. What is the scientific question, and does it concern all three conditions or particular ones?
2. What are the cases, and does each contribute to exactly one condition?
3. Which comparisons were prespecified, and how many questions are being asked?
4. What does the omnibus test contribute, and what would it leave unanswered?
5. What estimate, interval, and effect size answer the question in the outcome's units?
6. Do the variances and residuals support the pooled error term the procedure uses?
7. What does the design identify, what is the claim ceiling, and what evidence should come next?

Study Files

Study	Data structure	Primary outcome	Unit role
Study 04	One retention score from each of 120 synthetic participants	Delayed retention score	Three independent conditions
Study 04 comparisons	40 massed, 40 spaced, 40 interleaved	Delayed retention score	Omnibus, familywise, and contrast evidence

Both scenarios use the pinned synthetic file `study_04_practice_spacing_anova.csv`. Its data dictionary and supporting public files are available at <https://github.com/nicholaskarlson/data>. The reported omnibus, interval, familywise, and contrast quantities were independently recomputed.

Plausible Research Scenarios

The scenarios create decisions rather than documenting real research. At each **Pause and Decide**, stop reading and write an answer, reason, and confidence rating. The explanation that follows is feedback on that commitment.

Plausible Research Scenario: Three Practice Schedules and Delayed Retention

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

A fictional teaching-and-learning centre compares three ways of scheduling practice for the same course material: massed practice in one concentrated block, spaced practice distributed across several days, and interleaved practice that is distributed and also mixes problem types. Each of the 120 represented participants follows one schedule and contributes one delayed retention score, so the independent unit is a participant and the observations are independent between conditions.

The research question has two halves: do the three schedules differ in average delayed retention, and if so, how? The first is an omnibus question about the set of three means. The second is answered only by particular comparisons, and which comparisons those are is a decision the team must make before seeing the data.

The literature the centre draws on suggests two: whether distributing practice at all helps — the average of the two distributed schedules against massed — and whether mixing problem types adds anything beyond distributing them — interleaved against spaced. The team prespecifies exactly these two contrasts, notes that they are orthogonal in a balanced design, and records that all three pairwise differences will also be reported with a familywise procedure.

The evidence record is where the scenario turns. Study 04 does not document how participants came to be in one schedule rather than another: no randomization procedure, no record of who decided, no record of whether those who chose a demanding schedule differed beforehand in motivation, prior knowledge, or available time. The study-design matrix therefore treats the assignment mechanism as undocumented, and that is a gap rather than a permission.

That matters more here than in a two-group comparison, because an ordered three-condition story is inviting. Massed, spaced, and interleaved line up as a ladder, and a result that climbs it reads like a demonstration that the schedule produced the climb. It is not: a large F , three separated intervals, and a substantial proportion of variance accounted for are all compatible with participants having sorted themselves into the schedules that suited them.

The team therefore prespecifies its reporting too: the three group means with intervals in retention points, the omnibus test with its degrees of freedom, an effect size with a confidence interval rather than a bare number, the two planned contrasts, and the full familywise-adjusted pairwise set. It will describe the pattern among the represented participants and will not write that any schedule improved or produced retention.

Pause and Decide

Before reading further, decide which comparisons the research question actually asks, whether the omnibus test alone could answer its second half, and what the undocumented assignment mechanism does to the strongest available conclusion.

Why This Model?

A one-way analysis of variance matches the design: one quantitative outcome, one condition with three levels, independent participants within each. It supplies a single pooled error term estimated from all 120 cases, which every later comparison uses, and answers the omnibus half of the question in one test rather than three. Three separate two-group tests would instead estimate the error term three times from smaller pieces of the data and ask three questions while reporting each as though it were the only one. Dropping a schedule discards a third of the evidence, and a repeated-measures model would invent links this design does not create.

What Would a Favorable Result Look Like?

The centre hopes to see the distributed schedules separated from massed practice by enough retention points to matter for course design, with intervals narrow enough to rule out a trivial difference, and a clear answer on whether interleaving adds more. Precision in retention points, not the size of F , makes such a result usable. Overlapping conditions would also be a scientific success, constraining a widely repeated claim in the centre's own setting.

Best Defensible Claim

Statistical evidence: The analysis estimates the differences among the three represented condition means and quantifies their uncertainty, for both the prespecified contrasts and the full pairwise set. **Causal identification:** The record does not document how participants came to be in a schedule, so the comparison does not identify an effect of scheduling. **Claim ceiling:** Describe the pattern of differences among the represented participants; do not claim that any schedule caused or improved retention, and do not extend the pattern to other material or delays.

What Must Be True?

Each participant must contribute one score to one condition, the measure must be scored comparably across conditions, and the group means must be meaningful summaries. The pooled procedures assume similar variance across conditions and residuals without unaddressed errors or extreme influence. These model conditions are separate from what a causal claim would need.

Defensibility Procedures

Prespecify the primary outcome, the contrasts, the familywise procedure, and the reporting set before inspecting results. Record recruitment, how condition membership arose, exposure, timing, and the retention delay. Inspect group counts, distributions, variance diagnostics, and residuals. Report group means with intervals, the omnibus test, an effect size with its interval, both contrasts, and all three familywise-adjusted comparisons — including those that do not support the story.

Next evidence priority: Randomly assign the schedule and document the procedure. Until then, a larger sample only estimates the same self-selected contrast more precisely.

Claim Audit

Repair this claim: “Interleaved practice produced the best retention, so courses should interleave.” Separate an estimated difference from a produced effect, name what the design does not document, and say what the second half would require.

Plausible Research Scenario: Which Comparison Was the Question?

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

A colleague reviews the same three-schedule evidence and proposes a shorter report: look at the three means, find the largest gap, test that one, present it. The proposal sounds efficient and describes how a good deal of reporting is actually done.

It is also a different procedure from the one the centre prespecified, and the difference is not cosmetic. Choosing which comparison to test after seeing which gap is largest means the comparison is no longer one question among three; it is the maximum of three, and the maximum of several noisy quantities is larger than any one of them on average. Selecting on the result and then reporting it as though it had been chosen in advance overstates the evidence, and no correct arithmetic afterwards repairs it.

The scenario sets two reporting rules against each other on identical data. One reports the prespecified contrasts and the full familywise set; the other reports the single biggest gap. The learner must say what changes, what does not, and why the wider familywise interval is the honest one rather than the timid one. The undocumented assignment mechanism applies to both: neither rule turns an observed difference into a caused one.

Pause and Decide

Decide what the difference between “prespecified” and “selected after seeing the means” does to the meaning of an interval, and what a reader would need to be told to evaluate the shorter report.

Why This Model?

The prespecified contrasts answer the questions the centre asked, in the outcome's units, with intervals that mean what they say. The familywise-adjusted pairwise set answers the different question of where the three conditions separate once the analyst is allowed to look at all three gaps. Reporting both is what lets a reader check the story rather than take it. A single unadjusted test of the largest observed gap answers neither question honestly, because its sampling behaviour is that of a maximum and not that of a planned comparison.

What Would a Favorable Result Look Like?

There is no favorable direction here; the scenario is about reporting integrity. A good result is a report in which a reader can tell which comparisons were planned, which were exploratory, and how many were run.

Best Defensible Claim

Statistical evidence: The two reporting rules produce different intervals on the same data, and the difference is a property of the selection procedure rather than of the participants. **Causal identification:** Neither rule identifies an effect of scheduling; the assignment mechanism is undocumented either way. **Claim ceiling:** Describe the represented differences and say plainly which comparisons were prespecified and how many were examined.

What Must Be True?

The prespecification must be genuine and recorded before the data are seen, the reported set must include comparisons that did not support the story, and the number examined must be stated. A reader must be able to reconstruct the analysis path from what is printed.

Defensibility Procedures

Record the analysis plan before inspecting results. Report every comparison in the prespecified set, label exploratory comparisons as exploratory, name the familywise procedure, and give intervals rather than decisions. Never present a comparison chosen for its size as though it had been planned.

Next evidence priority: Preregister the comparison set for the next study, and treat any comparison discovered in this one as a hypothesis for that study rather than as a finding of this one.

Claim Audit

Explain what is wrong with: “We compared the best and worst conditions and found a highly significant difference, so the effect is clearly real.” Name at least three distinct problems.

Problems

Do not consult the solutions until you have recorded an answer, reason, and confidence rating for every attempted item.

Level 1: Recall and Recognize

Problem 11.1 — Name what the omnibus test answers

Study 04 compares three practice schedules. State in one sentence what a significant omnibus result establishes and what it leaves entirely open.

Problem 11.2 — Count the questions

Three conditions permit three pairwise comparisons. Explain why the number of comparisons a report *runs* matters even when each individual comparison is computed correctly.

Problem 11.3 — Locate the pooled error term

The one-way analysis reports a mean square within of 63.418 on 117 degrees of freedom. Say what that quantity estimates, which cases contribute to it, and why every later comparison in this unit uses it.

Problem 11.4 — Separate the design from the analysis

The three-condition analysis is correctly specified for Study 04. Does that fact license the sentence “spacing improved retention”? Answer using the assignment record rather than the size of any statistic.

Level 2: Predict Before Clicking

Problem 11.5 — Predict from the group means

The three condition means are 63.388, 72.378 and 78.320 retention points, and each group has 40 participants with standard deviations near eight. Predict whether the omnibus test will be clearly significant, and say which quantity in the prediction did most of the work.

Problem 11.6 — Predict the familywise consequence

A familywise procedure is applied to all three pairwise comparisons. Predict whether each interval becomes wider or narrower than an unadjusted interval for the same difference, and explain what is being bought with that change.

Problem 11.7 — Predict the contrast standard error

The prespecified contrast compares the average of two distributed schedules with the massed schedule. Predict whether its standard error will be larger or smaller than the standard error of a single pairwise difference, and why.

Problem 11.8 — Predict the effect of unequal variances

The three condition standard deviations are 8.101, 7.051 and 8.656 with equal group sizes. Predict whether a Welch heteroscedastic analysis will change the conclusion, and state what would have to be true for it to matter a great deal.

Level 3: Calculate and Explain

Problem 11.9 — Reconstruct the pooled variance

With equal group sizes, the mean square within is the average of the three group variances. Use the standard deviations 8.101, 7.051 and 8.656 to approximate it, and explain why the group means play no part in this quantity.

Problem 11.10 — Recover the omnibus F

The analysis reports a mean square between of 2260.753 and a mean square within of 63.418. Compute the ratio and state what it represents. Then say why an F statistic cannot be interpreted without its two degrees of freedom.

Problem 11.11 — Compute a pairwise standard error and statistic

Use $SE = \sqrt{(MS_{\text{within}} \times (1/n_1 + 1/n_2))}$ with a mean square within of 63.418 and forty participants per group. Then divide the spaced-minus-interleaved difference of -5.943 by that standard error.

Problem 11.12 — Compute the planned contrast

The contrast weights are -1 for massed and 0.5 for each distributed schedule. Using means of 63.388, 72.378 and 78.320, compute the contrast estimate. Then use $SE = \sqrt{(MS_{\text{within}} \times \sum(w^2/n))}$ to approximate its standard error.

Level 4: Read jamovi Output

Problem 11.13 — Read the descriptives and the ANOVA table

The reconstructed jamovi output for the primary analysis is below.

Condition	N	Mean	SD	95% CI lower	95% CI upper
massed	40	63.388	8.101	60.894	65.881
spaced	40	72.378	7.051	69.884	74.871
interleaved	40	78.320	8.656	75.826	80.814

Source	Sum of Squares	df	Mean Square	F	p
spacing_condition	4521.506	2	2260.753	35.649	< .001
Residuals	7419.878	117	63.418	—	—

Write three sentences: the pattern of the means with their precision, the omnibus result, and the boundary the design imposes.

Problem 11.14 — Read the effect size with its interval

The analysis reports $\eta^2 = 0.379$, $\omega^2 = 0.366$, and a 95% interval for η^2 of [0.238, 0.484]. Interpret the estimate, explain why the interval is so wide with 120 participants, and say why ω^2 is slightly smaller.

Problem 11.15 — Read a familywise comparison table

The reconstructed Tukey output is below; the pooled standard error for each comparison is 1.781 and the studentized-range critical value is 3.357.

Comparison	Difference	Lower	Upper	t	p
massed – spaced	-8.990	-13.217	-4.763	-5.049	< .001
massed – interleaved	-14.933	-19.160	-10.705	-8.386	< .001
spaced – interleaved	-5.943	-10.170	-1.715	-3.337	.003

Summarize what the three intervals jointly support, and state which sentence a reader should not write from this table.

Problem 11.16 — Read the planned contrasts

The two prespecified contrasts are reported below.

Contrast	Estimate	SE	Lower	Upper	t	p
distributed – massed	11.961	1.542	8.907	15.015	7.756	< .001
interleaved – spaced	5.943	1.781	2.416	9.469	3.337	.001

Explain what each contrast answers, why the first has the smaller standard error, and how these two lines relate to the three lines in the previous table.

Level 5: Choose the Model

Problem 11.17 — Signature model-choice exercise

A colleague reports three separate two-group comparisons of Study 04 — massed versus spaced, massed versus interleaved, and spaced versus interleaved — each with its own standard error computed from only the two groups involved, and each with an unadjusted p value. Identify three distinct problems with this analysis, name the model that replaces it, and state the one situation in which comparing only two of the three conditions would be the correct analysis.

Problem 11.18 — Connect back to Unit 10

Unit 10 established that the decisive question is which observations are linked. Apply it here: (a) 120 different students, one schedule each; (b) 40 students who each work under all three schedules on different material; and (c) 120 students in twelve classrooms, with each classroom assigned one schedule. Choose a starting model for each and name the decisive cue.

Problem 11.19 — Choose the comparison set prospectively

Two teams analyze the same three-condition data. One prespecifies two contrasts; the other will examine all three pairwise gaps and report the largest. Explain which choice must be made before data inspection, and why a familywise adjustment is appropriate for one team and cannot rescue the other.

Problem 11.20 — Mixed three-group clinic

Choose a starting analysis for each case and state the decisive cue: (a) three independent groups with one quantitative outcome and markedly unequal variances and group sizes; (b) three ordered doses where the question is whether the outcome rises monotonically; (c) an outcome recorded as a rank position among three conditions; and (d) three conditions with a quantitative outcome and a strong pre-test measured on every participant.

Level 6: Report and Critique

Problem 11.21 — Repair a three-group causal claim

Critique: “The one-way ANOVA proved that spacing and interleaving cause better retention, $F = 35.649$, $p < .001$.” Identify at least four problems and write a defensible replacement using the verified Study 04 values.

Problem 11.22 — Repair omnibus-only reporting

Critique: “There was a significant difference among the three practice schedules.” Rewrite it using the group means, the omnibus test, the effect size with its interval, and at least one specific comparison.

Problem 11.23 — Report the prespecified contrasts

Write a concise results paragraph reporting the two planned contrasts. It must identify them as prespecified, give estimates and intervals in retention points, and keep the observational boundary visible.

Problem 11.24 — Expose selective comparison reporting

A report presents only the massed-versus-interleaved comparison, describes it as “the key comparison,” and does not mention that all three were examined. Explain why this is misleading and write a transparent replacement.

Worked Solutions

Each solution identifies the decisive cue, reasoning path, relevant evidence, rejected alternative, claim boundary, and a transfer cue.

Level 1 Solutions

Solution 11.1

A significant omnibus result establishes that the three represented condition means are not all equal. It leaves entirely open which conditions differ, in which direction, by how much, and whether any difference matters — and it says nothing about why the means differ. **Transfer cue:** an omnibus test locates a difference in the set, never in a pair.

Solution 11.2

Each comparison carries its own chance of a misleading result, and running three multiplies the chance that at least one appears convincing. More importantly, a reader who is shown one comparison and not told that three were run cannot judge the evidence at all. The count is part of the result, not part of the method section's housekeeping. **Rejected rule:** “each test was computed correctly, so the report is correct.”

Solution 11.3

The mean square within, 63.418, estimates the variance of retention scores among participants within the same condition — the background variability against which condition differences are judged. All 120 cases contribute, which is why it carries 117 degrees of freedom rather than the smaller number a single two-group comparison would use. Every later comparison uses it because pooling produces a more stable error estimate than re-deriving one from each pair.

Solution 11.4

No. Correct specification of the three-group model estimates differences; it does not identify their cause. Study 04's evidence record does not document how participants came to be in one schedule, so self-selection and pre-existing differences in motivation, prior knowledge, or available time remain live explanations. **Claim ceiling:** describe the pattern of differences among the represented participants; do not write that any schedule improved retention.

Study in the Larger Evidence Program: Practice Scheduling

- **What this study adds:** A reproducible estimate of how delayed retention differs across three represented practice schedules, with familywise and prespecified comparisons and an effect size interval.
- **What it cannot settle:** Whether scheduling causes the differences, since the assignment mechanism is undocumented, or whether the pattern holds for other material, delays, or learners.
- **Causal-support role:** Consistency evidence only; the ordered pattern is compatible with the literature but supplies no responsiveness evidence without documented assignment.
- **Next evidence priority:** A randomized schedule assignment with a recorded procedure, adherence monitoring, and a second retention delay.

Level 2 Solutions

Solution 11.5

The omnibus test should be clearly significant. The extreme means differ by almost fifteen points while the within-condition standard deviations are near eight, so the separation is close to two standard deviations across a design with forty participants per group. The quantity doing most of the work is the ratio of between-condition spread to within-condition spread, not the raw gap on its own. A fifteen-point gap would be unremarkable if scores varied by fifty points within a condition.

Solution 11.6

Every familywise interval becomes wider. What is bought is a joint guarantee: the stated coverage applies to the whole set of comparisons at once rather than to each one considered in isolation. Wider intervals are the price of being allowed to look at all three gaps and still make a calibrated statement about the one that turned out largest. **Transfer cue:** an adjustment is not a penalty for honesty; it is what makes several simultaneous statements mean something.

Solution 11.7

Smaller. The contrast averages two group means before comparing them with the third, and averaging two means of forty cases each produces a more precise quantity than either mean alone. The sum of squared weights divided by group sizes is smaller for the contrast — 1.5 divided by forty rather than 2 divided by forty — so its standard error is smaller than a pairwise difference's.

Solution 11.8

It should not change the conclusion. The group sizes are equal and the largest standard deviation, 8.656, is only about a quarter larger than the smallest, 7.051, so pooling is not doing damage here. Unequal variances matter a great deal when they are paired with unequal group sizes — particularly when the smallest group has the largest spread, which is exactly the situation Unit 10's Study 03A was built to show.

Level 3 Solutions

Solution 11.9

With equal group sizes the pooled variance is the mean of the three group variances: $(8.101^2 + 7.051^2 + 8.656^2) / 3 \approx 63.42$, which matches the recorded mean square within of 63.418 once the rounded standard deviations are allowed for. The group means play no part because this quantity measures variation of participants around their own condition's mean. Adding a constant to one condition would change the means and leave the pooled variance untouched. **Decisive cue:** the error term describes people within conditions, not conditions.

Solution 11.10

$F = 2260.753 / 63.418 = 35.649$, approximately. It is the ratio of variation among condition means to variation among participants within conditions, both expressed as mean squares. Its degrees of freedom, 2 and 117, are needed because the same numerical ratio means something quite different when the numerator summarizes three conditions rather than thirty, and when the denominator rests on 117 cases rather than a handful.

Solution 11.11

$SE = \sqrt{(63.418 \times (1/40 + 1/40))} = \sqrt{(3.171)} = 1.781$ points, approximately. Dividing the spaced-minus-interleaved difference of -5.943 by 1.781 gives approximately -3.337, matching the verified value. The same

standard error serves all three pairwise comparisons because the design is balanced and the error term is pooled.

Solution 11.12

The estimate is $0.5 \times 72.378 + 0.5 \times 78.320 - 63.388 = 11.961$ retention points. The standard error is $\sqrt{(63.418 \times (0.25/40 + 0.25/40 + 1/40))} = 1.542$ points, approximately. The contrast is a difference between an average of two conditions and a third, so it answers the "does distributing practice at all help" question in one number rather than in two overlapping pairwise comparisons.

Level 4 Solutions

Solution 11.13

Among the represented participants, mean delayed retention rose across the three schedules — massed 63.388 points, 95% CI [60.894, 65.881]; spaced 72.378, [69.884, 74.871]; interleaved 78.320, [75.826, 80.814] — and the three intervals do not overlap. The omnibus test indicates that the three condition means are not all equal, $F(2, 117) = 35.649$, $p < .001$, with a mean square within of 63.418 on 117 degrees of freedom. Because the evidence record does not document how participants came to be in a schedule, these statistics describe a pattern of differences among the represented participants and do not establish that scheduling produced them.

Solution 11.14

$\eta^2 = 0.379$ means that about 38 percent of the variation in retention scores in this sample is accounted for by condition membership. The interval [0.238, 0.484] is wide because an effect-size proportion is estimated far less precisely than the mean differences that generate it; 120 cases pin down a difference in points much better than they pin down a variance ratio. $\omega^2 = 0.366$ is slightly smaller because it corrects for the upward bias that η^2 carries as a sample quantity. **Transfer cue:** report an effect size with an interval, or do not report it as evidence.

Solution 11.15

All three familywise-adjusted intervals exclude zero, so the three conditions separate from one another jointly and not merely one pair at a time: massed scored -8.990 points below spaced, 95% CI [-13.217, -4.763]; -14.933 below interleaved, [-19.160, -10.705]; and spaced scored -5.943 below interleaved, [-10.170, -1.715]. Each interval used the pooled standard error of 1.781 and the studentized-range critical value of 3.357. The sentence a reader should not write is any sentence with a causal verb: "interleaving raised retention by fifteen points" is not licensed by an undocumented assignment mechanism, however cleanly the intervals separate.

Solution 11.16

The first contrast answers whether distributing practice at all is associated with higher retention: the two distributed schedules averaged 11.961 points above massed, 95% CI [8.907, 15.015], $t(117) = 7.756$, $p < .001$. The second asks whether mixing problem types adds anything beyond distributing them: interleaved exceeded spaced by 5.943 points, [2.416, 9.469], $t(117) = 3.337$, $p = .001$. The first has the smaller standard error, 1.542 against 1.781, because averaging two group means yields a more precise quantity than a single mean. These two lines are not a subset of the previous table: the second contrast is the same comparison as the third Tukey row, but the first has no pairwise counterpart, and its interval is narrower than any Tukey interval because it was planned rather than selected from three.

Study in the Larger Evidence Program: Comparison Selection

- **What this study adds:** A demonstration that two defensible reporting rules produce different intervals on identical data, and that the difference belongs to the procedure rather than to the participants.

- **What it cannot settle:** Anything causal; and it cannot tell a reader which comparisons were planned unless the report says so.
- **Causal-support role:** None directly; it protects the consistency evidence other studies contribute by keeping selective reporting out of the record.
- **Next evidence priority:** Preregister the comparison set for the next study and treat any comparison discovered here as a hypothesis rather than a finding.

Level 5 Solutions

Solution 11.17

First, each two-group comparison estimates the error term from only eighty cases instead of using the pooled estimate from all 120, so each is less stable than it needs to be. Second, three unadjusted comparisons make three simultaneous statements while reporting each as though it were the only one, so the stated coverage is not the coverage the set actually has. Third, the report never answers the omnibus half of the research question, and a reader cannot tell how many comparisons were run. The replacement is a one-way analysis of variance with a pooled error term, followed by the prespecified contrasts and a familywise-adjusted pairwise set. Comparing only two conditions is correct when the research question genuinely concerns only those two and that restriction was recorded before the data were seen — not when the third condition was dropped because its result was inconvenient.

Solution 11.18

(a) A one-way analysis of variance with independent groups; the decisive cue is that each student contributes one score to one condition. (b) A repeated-measures model; the decisive cue is that each student's three scores are linked through the student, exactly the dependence Unit 10 taught learners to look for, and treating 120 scores as independent would triple the apparent information. (c) A model that represents classroom-level assignment and within-classroom dependence; preserve the classroom identifier and the assignment record, because twelve classrooms rather than 120 students carry the assignment information. **Transfer cue:** the number of conditions changed, the dependence question did not.

Solution 11.19

The comparison set must be chosen before data inspection, because that choice determines what the reported intervals mean. For the first team a familywise adjustment is appropriate and sufficient: the set is fixed in advance, and the adjustment calibrates a known number of simultaneous statements. For the second team no adjustment repairs the procedure, because the reported quantity is the maximum of three noisy differences rather than any one of them, and an adjustment designed for three planned comparisons does not describe the sampling behaviour of a maximum chosen after the fact. **Rejected rule:** “we can always correct for it afterwards.”

Solution 11.20

(a) A Welch heteroscedastic one-way analysis, which does not pool variances; the decisive cue is unequal variances combined with unequal group sizes. (b) A prespecified trend contrast across the ordered doses, which asks the monotonic question directly instead of scattering it across three pairwise tests. (c) A rank-based procedure for three independent groups, because the outcome is ordinal and a mean of ranks is not the quantity of interest. (d) An analysis of covariance using the pre-test, chosen for precision — and note that it improves the estimate's precision without supplying the assignment mechanism a causal claim would need.

Level 6 Solutions

Solution 11.21

The statement uses “proved” and a causal verb where the assignment mechanism is undocumented; it reports an F statistic with no degrees of freedom; it gives no estimates, intervals, or effect size; and it collapses two distinct questions — whether distributing helps and whether interleaving adds more — into one unearned conclusion. A defensible replacement is: “Among the 120 represented participants, mean delayed retention differed across the three practice schedules, $F(2, 117) = 35.649$, $p < .001$, $\eta^2 = 0.379$, 95% CI [0.238, 0.484]. The two distributed schedules averaged 11.961 points above the massed schedule, 95% CI [8.907, 15.015], and interleaved exceeded spaced by 5.943 points, [2.416, 9.469]. Because the evidence record does not document how participants came to be in a schedule, these results describe differences among the represented participants and do not establish a causal effect of practice scheduling.”

Solution 11.22

“Mean delayed retention was 63.388 points under massed practice, 72.378 under spaced practice and 78.320 under interleaved practice. The three condition means were not all equal, $F(2, 117) = 35.649$, $p < .001$, $\eta^2 = 0.379$, 95% CI [0.238, 0.484]. The familywise-adjusted comparison of spaced with interleaved gave a difference of -5.943 points, 95% CI [-10.170, -1.715], $p = .003$. Assignment to schedule is not documented, so these are observed differences rather than effects.”

Solution 11.23

“Two contrasts were prespecified. Averaging the two distributed schedules and comparing them with massed practice gave a difference of 11.961 retention points, 95% CI [8.907, 15.015], $t(117) = 7.756$. Comparing interleaved with spaced practice gave 5.943 points, 95% CI [2.416, 9.469], $t(117) = 3.337$. All three pairwise differences were also examined with a familywise procedure and are reported in the table. Because the study does not document how participants came to be in a schedule, these differences describe the represented participants and support no causal conclusion.”

Solution 11.24

Calling one of three examined comparisons “the key comparison” after seeing the results converts a selected maximum into an apparently planned test, and omitting that all three were run removes the information a reader needs to discount it. The interval that is printed is also the wrong one: an interval calibrated for a single planned comparison does not describe a difference chosen for being the largest. A transparent replacement is: “All three pairwise comparisons were examined with a familywise procedure. Massed practice scored -14.933 retention points below interleaved practice, 95% CI [-19.160, -10.705], and the other two comparisons are reported in the table. Two contrasts were prespecified; the pairwise set was exploratory. Assignment to schedule is undocumented, so no causal interpretation is warranted.”

Unit Reflection Log

After attempting the unit, record:

- the problem number;
- your original answer and confidence;
- the cue you used and the cue you missed;
- the corrected decision rule;
- the claim ceiling you should have noticed;
- what the study adds to a larger evidence program; and
- the date on which you will attempt a new version.

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Version and Corrections

Version 1.0, September 2026. Corrections, questions, classroom adaptations, and suggestions for future versions are welcome through [GitHub Discussions](#).

About the Author

Nicholas Elliott Karlson is a statistician, educator, and author whose work emphasizes transparent reasoning from research question to reported result. His teaching approach treats software output as evidence to be checked rather than an answer to be copied.

Karlson holds a PhD in agricultural and applied economics and an MS in statistics. His professional experience includes statistical analysis and university instruction. Through NEKpress.ca, he develops proof-first guides and teaching resources that help readers build practical quantitative skills while keeping interpretation matched to design.